

ROBO Cylinder®

Slider Type

RCS4-SA4C/SA6C/SA7C/SA8C

SA4R/SA6R/SA7R/SA8R

RCS4CR-SA4C/SA6C/SA7C/SA8C

Instruction Manual

First Edition

ME3769-1H



Specifications

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Please Read Before Use

Thank you for purchasing our product.

This instruction manual explains the handling methods, structure and maintenance of this product, providing the information you need in order to use the product safely.

Before using the product, be sure to read this manual and fully understand the contents explained herein to ensure safe use of the product.

Please download the user's manual from our website.

You can download it free of charge. User registration is required for the first time downloading.

URL : www.iai-robot.co.jp/data_dl/CAD_MANUAL/

When using the product, print out of the necessary portions of the relevant manual, or please display it on your computer, tablet terminal, etc. so that you can check it immediately.

After reading the instruction manual, keep it in a convenient place so that whoever is handling the product can refer to it quickly when necessary.

[Important]

- This instruction manual is an original document dedicated for this product.
- This product cannot be used in ways not shown in this instruction manual. IAI shall not be liable for any result whatsoever arising from the use of the product in any other way than what is noted in the manual.
- The information contained in this instruction manual is subject to change without notice for the purpose of product improvement.
- If any issues arise regarding the information contained in this instruction manual, contact our customer center or the nearest sales office.
- Use or reproduction of this instruction manual in full or in part without permission is prohibited.
- The company names, names of products and trademarks of each company shown in the text are registered trademarks.

RCS4 Slider Type Instruction Manual Configuration

Product name	Instruction manual name	Control number
RCS4	First Step Guide	ME3775
RCS4 Slider Type	Instruction Manual (this document)	ME3769
RCON System	RCON System Instruction Manual	ME0384
RSEL System	RSEL System Instruction Manual	ME0392
SCON-CB/CFB Controller	SCON-CB/CFB Controller Instruction Manual	ME0340
SCON-CAL/CGAL Controller	SCON-CAL/CGAL Controller Instruction Manual	ME0243
MSCON-C Controller	MSCON-C Controller Instruction Manual	ME0306
SSEL-CS Controller	SSEL-CS Controller Instruction Manual	ME0157
XSEL-P/Q Controller	XSEL-P/Q Controller Instruction Manual	ME0148
XSEL-R/S Controller	XSEL-R/S Controller Instruction Manual	ME0313
XSEL-RA/SA Controller	XSEL-RA/SA Controller Instruction Manual	ME0359
PC Teaching Software for RC/EC	RCM-101-MW/RCM-101-USB Instruction Manual	ME0155
PC Teaching Software for XSEL	IA-101-X-MW/IA-101-X-USB Instruction Manual	ME0154
Touch Panel Teaching Pendant	TB-01/01D/01DR Applicable for Position Controller Instruction Manual	ME0324
Touch Panel Teaching Pendant	TB-02/02D Applicable for Position Controller Instruction Manual	ME0355
Touch Panel Teaching Pendant	TB-03 Applicable for Position Controller	ME0376
Touch Panel Teaching Pendant	TB-01/01D/01DR Applicable for Program Controller Instruction Manual	ME0325
Touch Panel Teaching Pendant	TB-02/02D Applicable for Program Controller Instruction Manual	ME0356
Touch Panel Teaching Pendant	TB-03 Applicable for Program Controller	ME0377

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Safety Guide

“Safety Guide” has been written to use the machine safely and so prevent personal injury or property damage beforehand. Make sure to read it before the operation of this product.

Safety Precautions for Our Products

The common safety precautions for the use of any of our robots in each operation.

No.	Operation Description	Description
1	Model Selection	- This product has not been planned and designed for the application where high level of safety is required, so the guarantee of the protection of human life is impossible. Accordingly, do not use it in any of the following applications. 1) Medical equipment used to maintain, control or otherwise affect human life or physical health. 2) Mechanisms and machinery designed for the purpose of moving or transporting people (For vehicle, railway facility or air navigation facility) 3) Important safety parts of machinery (Safety device, etc.) - Do not use the product outside the specifications. Failure to do so may considerably shorten the life of the product. - Do not use it in any of the following environments. 1) Location where there is any inflammable gas, inflammable object or explosive 2) Place with potential exposure to radiation 3) Location with the ambient temperature or relative humidity exceeding the specification range 4) Location where radiant heat is added from direct sunlight or other large heat source 5) Location where condensation occurs due to abrupt temperature changes 6) Location where there is any corrosive gas (sulfuric acid or hydrochloric acid) 7) Location exposed to significant amount of dust, salt or iron powder 8) Location subject to direct vibration or impact - For an actuator used in vertical orientation, select a model which is equipped with a brake. If selecting a model with no brake, the moving part may drop when the power is turned OFF and may cause an accident such as an injury or damage on the work piece.

No.	Operation Description	Description
2	Transportation	• When carrying a heavy object, do the work with two or more persons or utilize equipment such as crane. • When the work is carried out with 2 or more persons, make it clear who is to be the “leader” and who to be the “follower(s)” and communicate well with each other to ensure the safety of the workers. • When in transportation, consider well about the positions to hold, weight and weight balance and pay special attention to the carried object so it would not get hit or dropped. • Transport it using an appropriate transportation measure. The actuators available for transportation with a crane have eyebolts attached or there are tapped holes to attach bolts. Follow the instructions in the instruction manual for each model. • Do not step or sit on the package. • Do not put any heavy thing that can deform the package, on it. • When using a crane capable of 1t or more of weight, have an operator who has qualifications for crane operation and sling work. • When using a crane or equivalent equipments, make sure not to hang a load that weighs more than the equipment’s capability limit. • Use a hook that is suitable for the load. Consider the safety factor of the hook in such factors as shear strength. • Do not get on the load that is hung on a crane. • Do not leave a load hung up with a crane. • Do not stand under the load that is hung up with a crane.
3	Storage and Preservation	• The storage and preservation environment conforms to the installation environment. However, especially give consideration to the prevention of condensation. • Store the products with a consideration not to fall them over or drop due to an act of God such as earthquake.
4	Installation and Start	(1) Installation of Robot Main Body and Controller, etc. • Make sure to securely hold and fix the product (including the work part). A fall, drop or abnormal motion of the product may cause a damage or injury. Also, be equipped for a fall-over or drop due to an act of God such as earthquake. • Do not get on or put anything on the product. Failure to do so may cause an accidental fall, injury or damage to the product due to a drop of anything, malfunction of the product, performance degradation, or shortening of its life. • When using the product in any of the places specified below, provide a sufficient shield. 1) Location where electric noise is generated 2) Location where high electrical or magnetic field is present 3) Location with the mains or power lines passing nearby 4) Location where the product may come in contact with water, oil or chemical droplets

No.	Operation Description	Description
4	Installation and Start	(2) Cable Wiring • Use our company's genuine cables for connecting between the actuator and controller, and for the teaching tool. • Do not scratch on the cable. Do not bend it forcibly. Do not pull it. Do not coil it around. Do not insert it. Do not put any heavy thing on it. Failure to do so may cause a fire, electric shock or malfunction due to leakage or continuity error. • Perform the wiring for the product, after turning OFF the power to the unit, so that there is no wiring error. • When the direct current power (+24V) is connected, take the great care of the directions of positive and negative poles. If the connection direction is not correct, it might cause a fire, product breakdown or malfunction. • Connect the cable connector securely so that there is no disconnection or looseness. Failure to do so may cause a fire, electric shock or malfunction of the product. • Never cut and/or reconnect the cables supplied with the product for the purpose of extending or shortening the cable length. Failure to do so may cause the product to malfunction or cause fire. (3) Grounding • The grounding operation should be performed to prevent an electric shock or electrostatic charge, enhance the noise-resistance ability and control the unnecessary electromagnetic radiation. • For the ground terminal on the AC power cable of the controller and the grounding plate in the control panel, make sure for grounding work. For security grounding, it is necessary to select an appropriate wire thickness suitable for the load. Perform wiring that satisfies the specifications (electrical equipment technical standards). For detail, follow the description in an instruction manual of each controller or controller built-in actuator. • Perform Class D Grounding (former Class 3 Grounding with ground resistance 100Ω or below).

No.	Operation Description	Description
4	Installation and Start	(4) Safety Measures • When the work is carried out with 2 or more persons, make it clear who is to be the “leader” and who to be the “follower(s)” and communicate well with each other to ensure the safety of the workers. • When the product is under operation or in the ready mode, take the safety measures (such as the installation of safety and protection fence) so that nobody can enter the area within the robot’s movable range. When the robot under operation is touched, it may result in death or serious injury. • Make sure to install the emergency stop circuit so that the unit can be stopped immediately in an emergency during the unit operation. • Take the safety measure not to start up the unit only with the power turning ON. Failure to do so may start up the machine suddenly and cause an injury or damage to the product. • Take the safety measure not to start up the machine only with the emergency stop cancellation or recovery after the power failure. Failure to do so may result in an electric shock or injury due to unexpected power input. • When the installation or adjustment operation is to be performed, give clear warnings such as “Under Operation; Do not turn ON the power!” etc. Sudden power input may cause an electric shock or injury. • Take the measure so that the work part is not dropped in power failure or emergency stop. • Wear protection gloves, goggle or safety shoes, as necessary, to secure safety. • Do not insert a finger or object in the openings in the product. Failure to do so may cause an injury, electric shock, damage to the product or fire. • When releasing the brake on a vertically oriented actuator, exercise precaution not to pinch your hand or damage the work parts with the actuator dropped by gravity.
5	Teaching	• When the work is carried out with 2 or more persons, make it clear who is to be the “leader” and who to be the “follower(s)” and communicate well with each other to ensure the safety of the workers. • Perform the teaching operation from outside the safety protection fence, if possible. In the case that the operation is to be performed unavoidably inside the safety protection fence, prepare the “Stipulations for the Operation” and make sure that all the workers acknowledge and understand them well. • When the operation is to be performed inside the safety protection fence, the worker should have an emergency stop switch at hand with him so that the unit can be stopped any time in an emergency. • When the operation is to be performed inside the safety protection fence, in addition to the workers, arrange a watchman so that the machine can be stopped any time in an emergency. Also, keep watch on the operation so that any third person can not operate the switches carelessly. • Place a sign “Under Operation” at the position easy to see. • When releasing the brake on a vertically oriented actuator, exercise precaution not to pinch your hand or damage the work parts with the actuator dropped by gravity. * Safety protection Fence : In the case that there is no safety protection fence, the movable range should be indicated.

No.	Operation Description	Description
6	Trial Operation	• When the work is carried out with 2 or more persons, make it clear who is to be the “leader” and who to be the “follower(s)” and communicate well with each other to ensure the safety of the workers. • After the teaching or programming operation, perform the check operation one step by one step and then shift to the automatic operation. • When the check operation is to be performed inside the safety protection fence, perform the check operation using the previously specified work procedure like the teaching operation. • Make sure to perform the programmed operation check at the safety speed. Failure to do so may result in an accident due to unexpected motion caused by a program error, etc. • Do not touch the terminal block or any of the various setting switches in the power ON mode. Failure to do so may result in an electric shock or malfunction.
7	Automatic Operation	• Check before starting the automatic operation or rebooting after operation stop that there is nobody in the safety protection fence. • Before starting automatic operation, make sure that all peripheral equipment is in an automatic-operation-ready state and there is no alarm indication. • Make sure to operate automatic operation start from outside of the safety protection fence. • In the case that there is any abnormal heating, smoke, offensive smell, or abnormal noise in the product, immediately stop the machine and turn OFF the power switch. Failure to do so may result in a fire or damage to the product. • When a power failure occurs, turn OFF the power switch. Failure to do so may cause an injury or damage to the product, due to a sudden motion of the product in the recovery operation from the power failure.

No.	Operation Description	Description
8	Maintenance and Inspection	• When the work is carried out with 2 or more persons, make it clear who is to be the “leader” and who to be the “follower(s)” and communicate well with each other to ensure the safety of the workers. • Perform the work out of the safety protection fence, if possible. In the case that the operation is to be performed unavoidably inside the safety protection fence, prepare the “Stipulations for the Operation” and make sure that all the workers acknowledge and understand them well. • When the work is to be performed inside the safety protection fence, basically turn OFF the power switch. • When the operation is to be performed inside the safety protection fence, the worker should have an emergency stop switch at hand with him so that the unit can be stopped any time in an emergency. • When the operation is to be performed inside the safety protection fence, in addition to the workers, arrange a watchman so that the machine can be stopped any time in an emergency. Also, keep watch on the operation so that any third person can not operate the switches carelessly. • Place a sign “Under Operation” at the position easy to see. • For the grease for the guide or ball screw, use appropriate grease according to the instruction manual for each model. • Do not perform the dielectric strength test. Failure to do so may result in a damage to the product. • When releasing the brake on a vertically oriented actuator, exercise precaution not to pinch your hand or damage the work parts with the actuator dropped by gravity. • The slider or rod may get misaligned OFF the stop position if the servo is turned OFF. Be careful not to get injured or damaged due to an unnecessary operation. • Pay attention not to lose the removed cover or screws, and make sure to put the product back to the original condition after maintenance and inspection works. Use in incomplete condition may cause damage to the product or an injury. * Safety protection Fence : In the case that there is no safety protection fence, the movable range should be indicated.
9	Modification and Dismantle	• Do not modify, disassemble, assemble or use of maintenance parts not specified based at your own discretion.
10	Disposal	• When the product becomes no longer usable or necessary, dispose of it properly as an industrial waste. • When removing the actuator for disposal, pay attention to drop of components when detaching screws. • Do not put the product in a fire when disposing of it. The product may burst or generate toxic gases.
11	Other	• Do not come close to the product or the harnesses if you are a person who requires a support of medical devices such as a pacemaker. Doing so may affect the performance of your medical device. • See Overseas Specifications Compliance Manual to check whether complies if necessary. • For the handling of actuators and controllers, follow the dedicated instruction manual of each unit to ensure the safety.

Alert Indication

The safety precautions are divided into "Danger", "Warning", "Caution" and "Notice" according to the warning level, as follows, and described in the instruction manual for each model.

Level	Degree of Danger and Damage	Symbol
Danger	This indicates an imminently hazardous situation which, if the product is not handled correctly, will result in death or serious injury.	 Danger
Warning	This indicates a potentially hazardous situation which, if the product is not handled correctly, could result in death or serious injury.	 Warning
Caution	This indicates a potentially hazardous situation which, if the product is not handled correctly, may result in minor injury or property damage.	 Caution
Notice	This indicates lower possibility for the injury, but should be kept to use this product properly.	 Notice

Precautions for Handling

1. The Safety Guide attached with the product is intended to permit safe use of the product and thus to prevent risks and property damage. Be sure to read it before handling the product.

2. Do not attempt any handling or operation that is not indicated in this instruction manual.

3. Make sure to secure the actuator properly in accordance with this instruction manual.

If the actuator is not securely fixed, this may lead to abnormal noise, vibration, breakdown or shortened product life.

4. Make sure to observe the usage conditions and environment of the product.

Operation outside the warranty could cause decreased performance or product breakdown.
Use within the allowable range for each item.

Item	Cautions for use	Problems or breakdowns which may occur if the allowable range is exceeded
Speed and acceleration/deceleration	Use within the allowable range	May lead to abnormal noise, vibration, breakdown, or shortened product life.
Allowable load moment	Use within the allowable range	May lead to abnormal noise, vibration, breakdown, or shortened product life. In extreme cases, flaking may occur on the guide or ball screw.
Overhang load length		Mounting a load with an overhang length greater than the allowable values may lead to vibration or abnormal noise.

5. If return operations are continued over a short distance, they may rapidly degrade the film of grease.

Continuous return operation within a distance less than 30mm may cause the grease film to degrade rapidly.

As a guideline, in every 5,000 to 10,000 cycles, have approximately 5 cycles of return operation over a 50mm distance or more to regenerate the oil film. Continued use of the actuator in that state may lead to breakdown.

In extreme cases, flaking may occur on the guide or ball screw.

6. Do not attempt to have sliders collide with an obstacle at high speed.

This may damage the coupling or other mechanical parts.

7. For Double Slider Specification (Option Code: W) in Reversed-home Specification (Option Code: NM), make sure to have the home-return operation conducted after joining the driver slider and free slider.

8. In some conditions of environment of use, postures of installation and conditions of operation, the base oil separated from the grease may come out of ROBO Cylinder.

It is recommended to have a protection in case the peripheral devices could get influence of the base oil.

International Standard Compliance

The ROBO Cylinder complies with the following overseas standards.

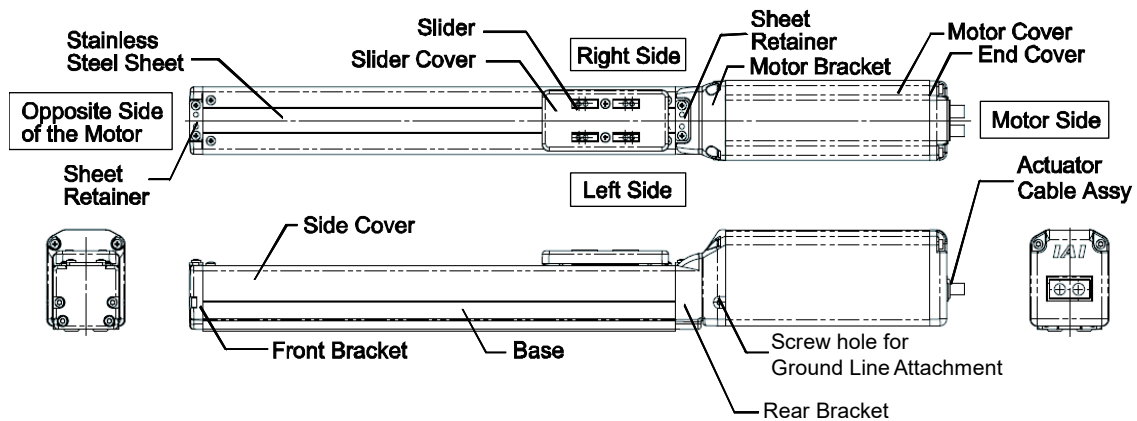
Refer to the Overseas Standard Compliance Manual (ME0287) for more detailed information.

CE Marking	RoHS3 Directive
○	○

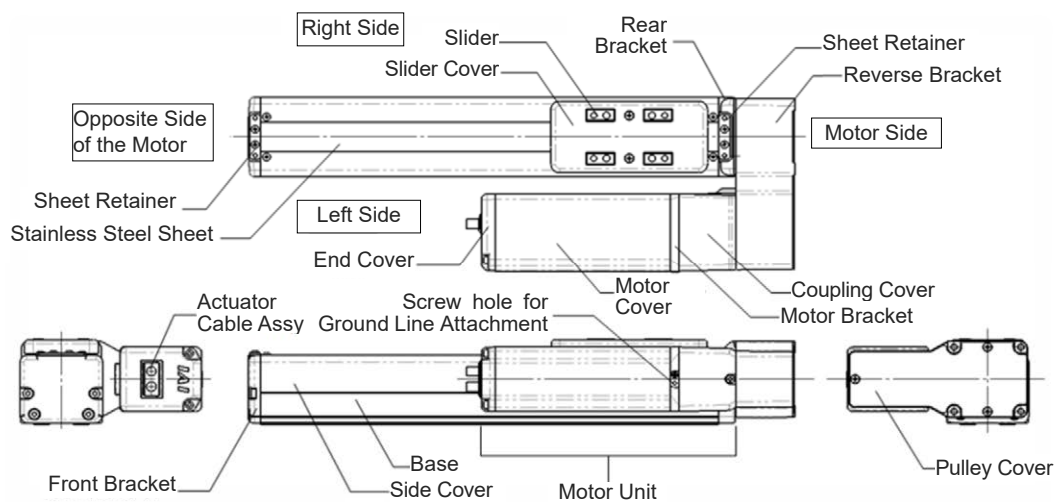
Names of the Parts

In this manual, the actuator left/right sides and motor/opposite sides are shown as in the figure below.

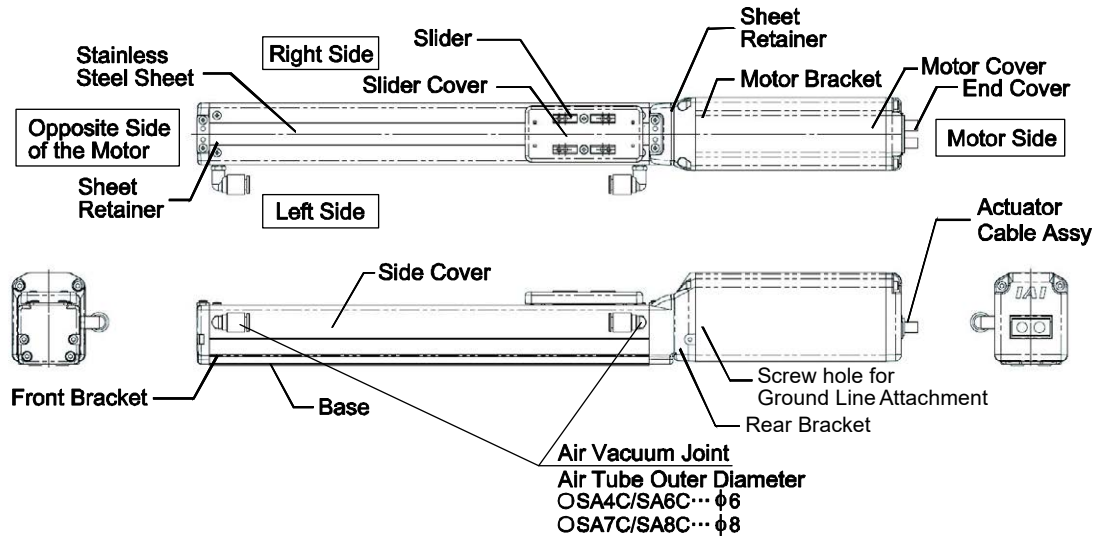
Standard Type Motor Straight Type



Standard Type Motor Reversing Type



Cleanroom Type Motor Straight Type



ROBO Cylinder

Chapter 1

Specifications

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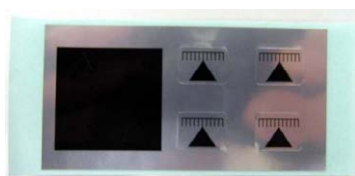
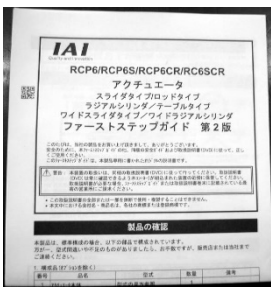
1.1 Checking the Product

1.1 Checking the Product

1.1.1 Parts

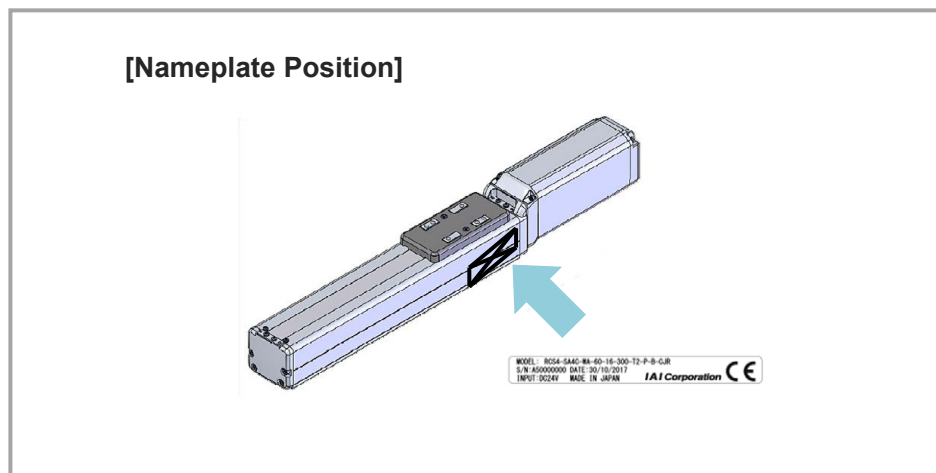
The following table shows the product configuration for the standard specification.

See the component list for the details of the enclosed components. If you find any fault or missing parts, contact your local IAI distributor.

Body	Accessories	
<u>Actuator Quantity 1</u>	<u>Motor Cable Quantity 1</u>	<u>Encoder Cable Quantity 1</u>
		
Accessories		
<u>In-house Made Seals Quantity 1</u>	<u>Cross Recessed Pan Head Screw with Captive Washer M3×6 Quantity 2</u>	<u>Zip ties Quantity 2</u>
	For affixing ground cable 	For clamping connector cover 
* Refer to International Application Manual (ME0287) for how to use it.		
Accessories		
<u>Safety Guide Quantity 1</u>	<u>First Step Guide Quantity 1</u>	
		

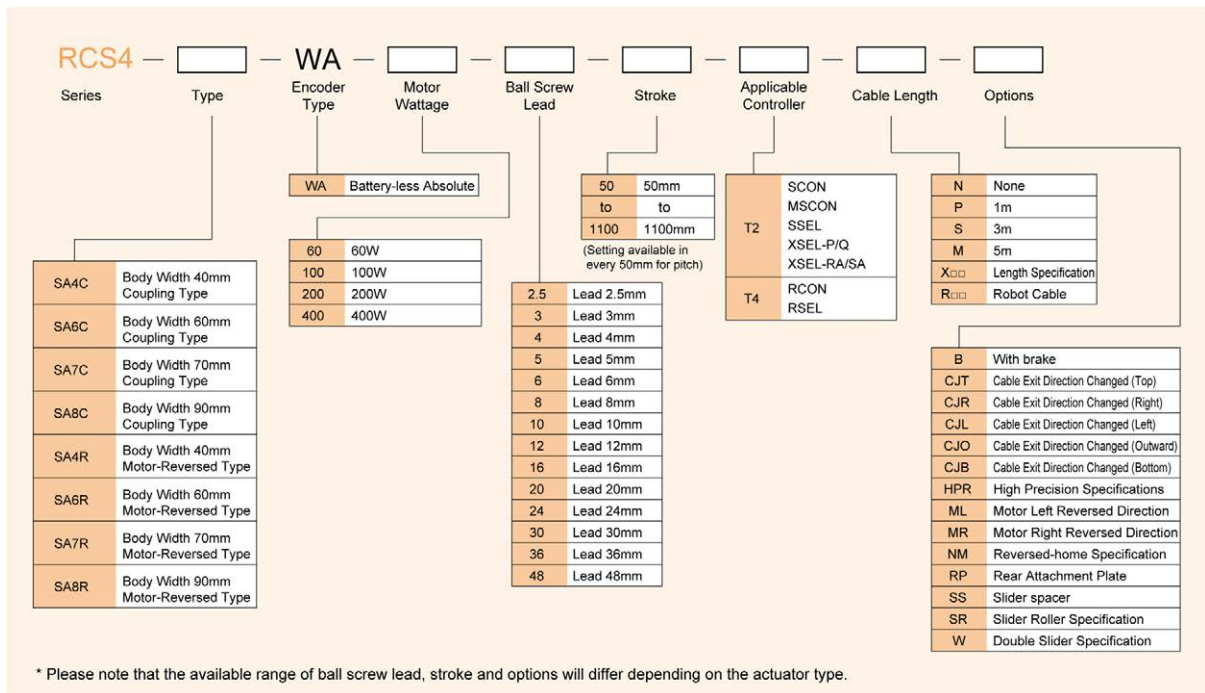
1.1.2 How to Read the Model Nameplate

Model→ MODEL: RCS4-SA4C-WA-60-16-300-T2-P-B-CJR
Serial number→ S/N: A50000000 DATE: 30/10/2017
INPUT: DC24V MADE IN JAPAN **IAI Corporation** **CE**

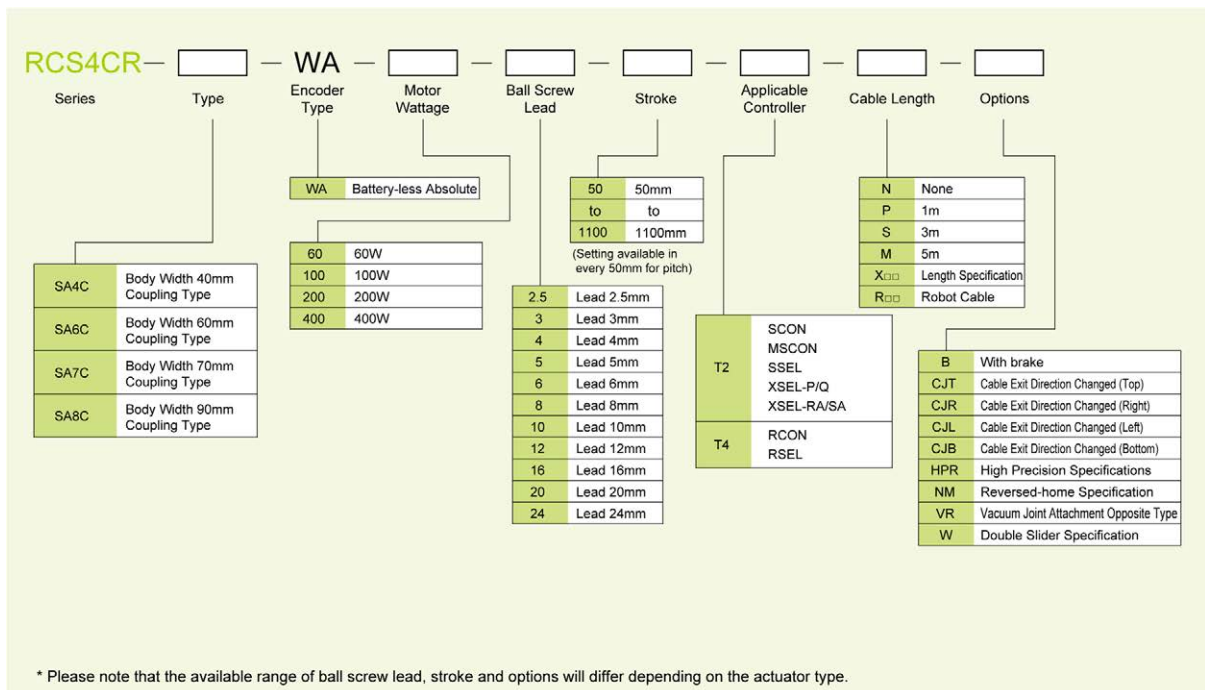


1.1.3 How to Read the Model Number

[Standard Type]



[Cleanroom Type]



1.1.4 Product list

[Standard Type]

Category	Type	Appearance	Body Width (mm)	Motor Wattage (W)	Lead (mm)	Positioning Repeatability (mm)	Stroke (mm)	Max. Speed (mm/s)	Rated Thrust (N)	Max. Payload (kg)	
										Horizontal	Vertical
Motor Straight Type	SA4C			60	16	±0.01 [±0.005]	50 to 500 (Every 50 st)	960	53	10	3
					10			600	85	14	5
					5			300	170	17	8
					2.5			150	340	20	12
	SA6C			100	30	±0.01 [±0.005]	50 to 800 (Every 50 st)	1600	57	11	3.5
					20			1200	85	18	6
					12			720	142	30	11
					6			360	283	45	15
	SA7C			200	3	±0.01 [±0.005]	50 to 800 (Every 50 st)	180	566	45	15
					36			1800	95	7	4
					24			1500	142	30	7
					16			1000	214	40	12
	SA8C			400	8	±0.01 [±0.005]	50 to 1100 (Every 50 st)	500	427	45	20
					4			240	855	50	25
					48			2200	141	8	-
					30			1800	226	30	12
Motor Reversing Type	SA4R			60	20	±0.01	50 to 500 (Every 50 st)	1200	339	60	20
					10			600	678	80	35
					5			300	1357	90	45
					2.5			150	340	20	12
	SA6R			100	30	±0.01	50 to 800 (Every 50 st)	1600	57	11	3
					20			1200	85	18	5
					12			720	142	30	9
					6			360	283	45	15
	SA7R			200	3	±0.01	50 to 800 (Every 50 st)	180	566	45	15
					36			1800	95	7	4
					24			1500	142	30	6
					16			1000	214	38	12
	SA8R			400	8	±0.01	50 to 1100 (Every 50 st)	500	427	45	18
					4			240	855	50	25
					48			2100	141	8	-
					30			1800	226	30	12

Values in brackets [] show specifications for high-precision type

[Cleanroom Type]

Category	Type	Appearance	Body Width (mm)	Motor Wattage (W)	Lead (mm)	Positioning Repeatability (mm)	Stroke (mm)	Max. Speed (mm/s)	Rated Thrust (N)	Max. Payload (kg)		Cleanliness
										Horizontal	Vertical	
Motor Straight Type	SA4C			60	16	±0.01 [±0.005]	50 to 500 (Every 50 st)	960	53	10	3	Class 10 (Fed.Std.209D Standards) Class 2.5 Equivalent (ISO14644-1 Standards)
					10			600	85	14	5	
					5			300	170	17	8	
					2.5			150	340	20	12	
	SA6C			100	20	±0.01 [±0.005]	50 to 800 (Every 50 st)	1200	85	18	6	
					12			720	142	30	11	
					6			360	283	45	15	
					3			180	566	45	15	
	SA7C			200	24	±0.01 [±0.005]	50 to 800 (Every 50 st)	1500	142	30	7	
					16			1000	214	40	12	
					8			500	427	45	20	
					4			240	855	50	25	
	SA8C			400	20	±0.01 [±0.005]	50 to 1100 (Every 50 st)	1200	339	60	20	
					10			600	678	80	35	
					5			300	1357	90	45	

Values in brackets [] show specifications for high-precision type

1.2 Specifications

1.2.1 Specifications

[1] RCS4(CR)-SA4C

[Lead and Payload]

Lead (mm)	Max. Payload		Rated Thrust (N)
	Horizontal(kg)	Vertical(kg)	
16	10	3	53
10	14	5	85
5	17	8	170
2.5	20	12	340

[Stroke and Max Speed]

Unit: mm/s

Lead (mm)	50 to 450 (Every 50mm)	500 (mm)
16	960	875
10	600	555
5	300	275
2.5	150	135



Caution

- The maximum speed may not be achieved when the stroke or movement distance is short or when acceleration/deceleration is set low.
- Setting at or below the minimum speed may lead to abnormal noise or unstable speeds. Do not attempt to set below the minimum speed.
- The minimum speed can be calculated with the equation below.

$$\text{Minimum speed (mm/s)} = \text{Ball screw lead (mm/r)} \div 16384 \text{ (p/r)} \times 1000 \text{ (1/s)}$$

(mm/r: movement per 1 motor revolution, p/r: pulse per 1 motor revolution)

[Payload by Acceleration]

At low load capacity, the acceleration/deceleration can be increased.

Lead 16

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
10	8	6	6	4	3	3	1.5	1.5	1

Lead 10

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
14	14	12	10	6	5	5	3	3	2

Lead 5

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
17	17	15	15	—	8	8	6	6	—

Lead 2.5

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
20	20	20	—	—	12	12	10	—	—

**Caution**

- Do not attempt to configure settings for acceleration/deceleration above the specifications. This may lead to vibration, breakdown, or shortened product life.

[Payload and Acceleration/Deceleration at Offboard Tuning]

(Note) RCS4CR Cleanroom Type is not applicable for the offboard tuning.

Have the tuning conducted with the offboard tuning in the PC software.

Operation with the acceleration/deceleration described in the table below should be performed.

Make sure not to operate with a payload higher than what is described in the table.

The unit for payload is kg. blanked boxes represent operation not available.

	Lead	Payload for Each Acceleration/Deceleration (kg)																	Max. Speed (mm/s)	
		0.3G	0.5G	0.7G	1.0G	1.2G	1.5G	2.0G	2.2G	2.5G	2.7G	3.0G	3.2G	3.5G	3.7G	4.0G	4.2G	4.5G	Standard	After Tuning
Horizontal	16	10	8	6	6	4	3.5	3	2.75	2.5	2.25	2	2	1.75	1.5	1.25	1	1	960	960
	10	14	14	12	10	6	5	4.5	4	3.5	3	2.5	2.25	2	1.75	1.5	1.25	1	600	600
	5	17	17	15	15	14.5	14	11	10	9	8	7	6.5	6	5.5	5	4.5	4	300	300
	2.5	20	20	20	20	17	11												150	150
Vertical	16	3	3	1.5	1.5	1	1	1	1	1	0.75	0.75							960	960
	10	5	5	3	3	2	2	2	1.5	1.5	1	1							600	600
	5	8	8	6	6	5.75	5.5	5.25	5	4.75	4.5	4.25							300	300
	2.5	12	12	10	8	7	6												150	150

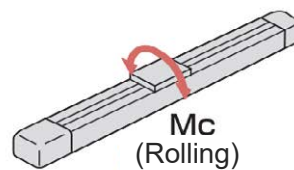
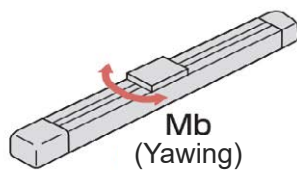
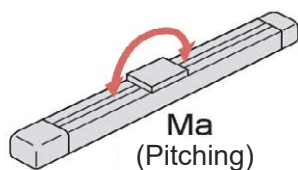
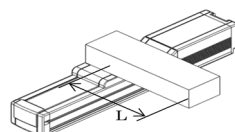
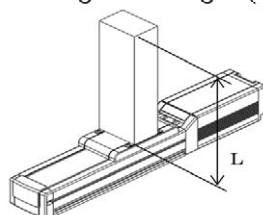
[Actuator Specifications]

Item	Content
Drive System	Ball Screw $\phi 8\text{mm}$, Rolled C10
Positioning Repeatability ^{(*)1}	$\pm 0.01\text{mm}$ [$\pm 0.005\text{mm}$]
Lost Motion	0.1mm or less
Base	Material: Aluminum, White Anodized
Allowable static moment	Ma direction: 13.0 N•m, Mb direction: 18.6 N•m, Mc direction: 25.3 N•m
Allowable dynamic moment ^{(*)2}	Ma direction: 5.0 N•m, Mb direction: 7.1 N•m, Mc direction: 9.7 N•m
No. of Encoder Pulses	16384
Ambient operating temperature/humidity	0 to 40°C, 85%RH or less (Non-condensing)

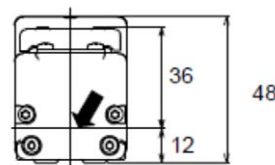
- Overhang load length guideline: 150mm or less

(*)1 Number in brackets [] show the specification for high precision type.

(*)2 For reference rated life of 5000km. The running life differs according to operation conditions and mounting status.

Slider type moment direction**Slider type overhang load length (L = 150mm or less)**

- * For calculation of Ma and Mc moments, take the position pointed with an arrow as the datum.

**Caution**

- If the actuator is used with excessive allowable moment and overhang load, it may not only lead to abnormal noise and vibration but also significantly reduce the life of the actuator.

[2] RCS4(CR)-SA6C

* There is no Lead 30 prepared for Cleanroom Type RCS4CR-SA6C.

[Lead and Payload]

Lead (mm)	Max. Payload		Rated Thrust (N)
	Horizontal(kg)	Vertical(kg)	
30	11	3.5	57
20	18	6	85
12	30	11	142
6	45	15	283
3	45	15	566

[Stroke and Max Speed]

Unit: mm/s

Lead (mm)	50 to 500 (Every 50mm)	550 (mm)	600 (mm)	650 (mm)	700 (mm)	750 (mm)	800 (mm)
30	1600		1450	1260	1100	970	860
20	1200	1130	970	840	735	650	575
12	720	620	535	460	405	355	315
6	360	305	265	230	200	175	155
3	180	150	130	115	100	85	75

**Caution**

- The maximum speed may not be achieved when the stroke or movement distance is short or when acceleration/deceleration is set low.
- Setting at or below the minimum speed may lead to abnormal noise or unstable speeds. Do not attempt to set below the minimum speed.
- The minimum speed can be calculated with the equation below.

$$\text{Minimum speed (mm/s)} = \text{Ball screw lead (mm/r)} \div 16384 \text{ (p/r)} \times 1000 \text{ (1/s)}$$
 (mm/r: movement per 1 motor revolution, p/r: pulse per 1 motor revolution)

[Payload by Acceleration]

At low load capacity, the acceleration/deceleration can be increased.

Lead30

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
11	6	6	4	3	3.5	2	2	1.5	1.5

Lead20

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
18	15	10	8	6	6	5	4	4	2

Lead12

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
30	24	20	15	15	11	10	10	8	8

Lead6

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
45	30	20	20	—	15	15	15	15	—

Lead3

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
45	30	20	—	—	15	14	10	—	—

**Caution**

- Do not attempt to configure settings for acceleration/deceleration above the specifications. This may lead to vibration, breakdown, or shortened product life.

[Payload and Acceleration/Deceleration at Offboard Tuning]

(Note) RCS4CR Cleanroom Type is not applicable for the offboard tuning.

Have the tuning conducted with the offboard tuning in the PC software.

Operation with the acceleration/deceleration described in the table below should be performed.

Make sure not to operate with a payload higher than what is described in the table.

The unit for payload is kg. blanked boxes represent operation not available.

	Lead	Payload for Each Acceleration/Deceleration (kg)																	Max. Speed (mm/s)	
		0.3G	0.5G	0.7G	1.0G	1.2G	1.5G	2.0G	2.2G	2.5G	2.7G	3.0G	3.2G	3.5G	3.7G	4.0G	4.2G	4.5G	Standard	After Tuning
Horizontal	30	11	6	6	4	3	2.5	2	1.5	1.5	1.5	1.5	1.5	1.25	1.25	1	1	1	1600	1800
	20	18	15	10	8	6	5	4.5	4	3.5	2.5	2	2	2	2	2	2	1.75	1200	1200
	12	30	24	20	15	15	9	6	5.5	5	4	3.5	3.5	3	3	2.5	2.5	2	720	720
	6	45	30	20	20	18	16	11	9	7.5	7	5.75	5	4	3.5	3	2.5	2	360	360
	3	45	30	20	12	7.5	2.5												180	180
Vertical	30	3.5	2	2	1.5	1.5	1.5	1.5	1.5	1	1	1							1600	1800
	20	6	5	4	4	2	2	2	2	2	2	2							1200	1200
	12	11	10	10	8	8	7	6	5	4	3.5	3							720	720
	6	15	15	15	15	11	9	6.5	6	5	4	4							360	360
	3	15	14	10	6	4	1												180	180

[Actuator Specifications]

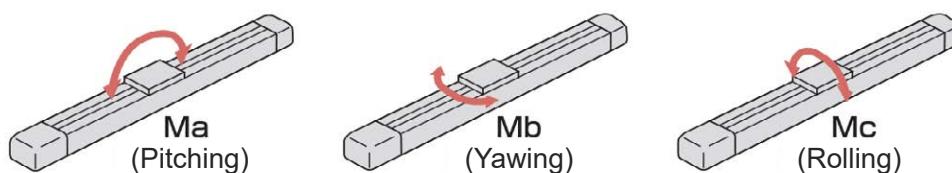
Item	Content
Drive System	Ball Screw ϕ 10mm, Rolled C10
Positioning Repeatability ^(*1)	$\pm 0.01\text{mm}$ [$\pm 0.005\text{mm}$]
Lost Motion	0.1mm or less
Base	Material: Aluminum White Anodized
Allowable static moment	Ma direction: 48.5 N•m, Mb direction: 69.3 N•m, Mc direction: 103 N•m
Allowable dynamic moment ^(*2)	Ma direction: 11.6 N•m, Mb direction: 16.6 N•m, Mc direction: 24.6 N•m
No. of Encoder Pulses	16384
Ambient operating temperature/humidity	0 to 40°C, 85%RH or less (Non-condensing)

- Overhang load length guideline: 220mm or less

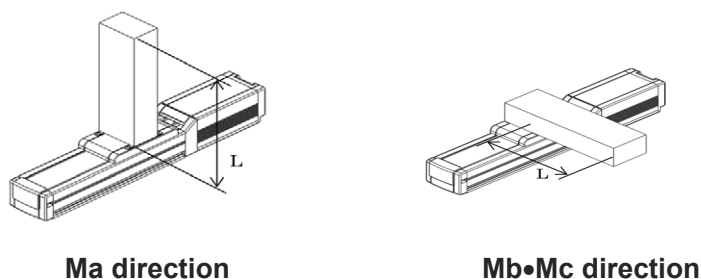
(*1) Number in brackets [] show the specification for high precision type.

(*2) For reference rated life of 5000km. The running life differs according to operation conditions and mounting status.

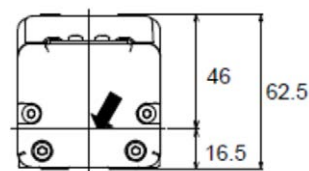
Slider type moment direction



Slider type overhang load length (L = 220mm or less)



- * For calculation of Ma and Mc moments, take the position pointed with an arrow as the datum.

**Caution**

- If the actuator is used with excessive allowable moment and overhang load, it may not only lead to abnormal noise and vibration but also significantly reduce the life of the actuator.

[3] RCS4(CR)-SA7C

* There is no Lead 36 prepared for Cleanroom Type RCS4CR-SA7C.

[Lead and Payload]

Lead (mm)	Max. Payload		Rated Thrust (N)
	Horizontal(kg)	Vertical(kg)	
36	7	4	95
24	30	7	142
16	40	12	214
8	45	20	427
4	50	25	855

[Stroke and Max Speed]

Unit: mm/s

Lead (mm)	50 to 500 (Every 50mm)	550 (mm)	600 (mm)	650 (mm)	700 (mm)	750 (mm)	800 (mm)
36	1800			1620	1420	1260	1120
24	1500	1440	1240	1095	965	850	760
16	1000	965	830	720	635	560	500
8	500	475	410	355	315	275	245
4	240	235	205	175	155	135	120

**Caution**

- The maximum speed may not be achieved when the stroke or movement distance is short or when acceleration/deceleration is set low.
- Setting at or below the minimum speed may lead to abnormal noise or unstable speeds. Do not attempt to set below the minimum speed.
- The minimum speed can be calculated with the equation below.

$$\text{Minimum speed (mm/s)} = \text{Ball screw lead (mm/r)} \div 16384 \text{ (p/r)} \times 1000 \text{ (1/s)}$$

(mm/r: movement per 1 motor revolution, p/r: pulse per 1 motor revolution)

[Payload by Acceleration]

At low load capacity, the acceleration/deceleration can be increased.

Lead 36

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
7	7	6	4	3	4	4	3	2	2

Lead 24

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
30	12	10	6	5	7	7	6	5	4

Lead 16

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
40	30	15	15	12	12	12	10	8	8

Lead 8

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
45	40	40	35	—	20	20	20	18	—

Lead 4

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
50	50	40	—	—	25	25	20	—	—

**Caution**

- Do not attempt to configure settings for acceleration/deceleration above the specifications. This may lead to vibration, breakdown, or shortened product life.

[Payload and Acceleration/Deceleration at Offboard Tuning]

(Note) RCS4CR Cleanroom Type is not applicable for the offboard tuning.

Have the tuning conducted with the offboard tuning in the PC software.

Operation with the acceleration/deceleration described in the table below should be performed.

Make sure not to operate with a payload higher than what is described in the table.

The unit for payload is kg. blanked boxes represent operation not available.

	Lead	Payload for Each Acceleration/Deceleration (kg)																	Max. Speed (mm/s)	
		0.3G	0.5G	0.7G	1.0G	1.2G	1.5G	2.0G	2.2G	2.5G	2.7G	3.0G	3.2G	3.5G	3.7G	4.0G	4.2G	4.5G	Standard	After Tuning
Horizontal	36	7	7	6	4	3	3	2.5	2.5	2	2	1.5	1.5	1.5	1.5	1.5	1.5	1	1800	2150
	24	30	12	10	6	5	5	5	5	4	4	3	3	3	3	3	3	3	1500	1500
	16	40	30	15	15	12	12	10	10	9	8	7	6	6	5.5	5	4	3.5	1000	1000
	8	45	40	40	35	30	25	18	15	10	9	9	9	8	7	6	5	4	500	500
	4	50	50	40	25	15	5												240	240
Vertical	36	4	4	3	2	2	2	2	2	1.5	1.5	1.5							1800	2150
	24	7	7	6	5	4	4	4	3.5	3.5	3	3							1500	1500
	16	12	12	10	8	8	8	5	5	5	5	5							1000	1000
	8	20	20	20	18	15	12	10	10	8	7	7							500	500
	4	25	25	20	12	8	3												240	240

[Actuator Specifications]

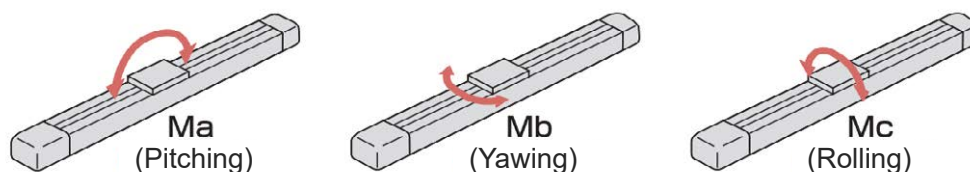
Item	Content
Drive System	Ball Screw $\phi 12\text{mm}$, Rolled C10
Positioning Repeatability ^(*1)	$\pm 0.01\text{mm}[\pm 0.005\text{mm}]$
Lost Motion	0.1 mm or less
Base	Material: Aluminum, White Anodized
Allowable static moment	Ma direction:115 N•m,Mb direction:115 N•m,Mc direction:229 N•m
Allowable dynamic moment ^(*2)	Ma direction:44.7 N•m,Mb direction:44.7 N•m,Mc direction:89.1 N•m
No. of Encoder Pulses	16384
Ambient operating temperature/humidity	0 to 40°C, 85%RH or less (Non-condensing)

- Overhang load length guideline:300mm or less

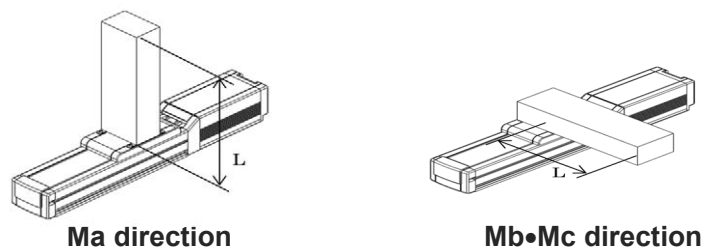
(*1) Number in brackets [] show the specification for high precision type.

(*2) For reference rated life of 5000km. The running life differs according to operation conditions and mounting status.

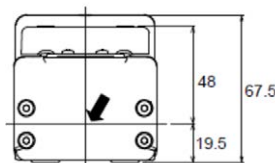
Slider type moment direction



Slider type overhang load length (L = 300mm or less)



* For calculation of Ma and Mc moments, take the position pointed with an arrow as the datum.

**Caution**

- If the actuator is used with excessive allowable moment and overhang load, it may not only lead to abnormal noise and vibration but also significantly reduce the life of the actuator.

[4] RCS4(CR)-SA8C

* There is no Lead 30 and Lead 48 prepared for Cleanroom Type RCS4CR-SA8C.

[Lead and Payload]

Lead (mm)	Max. Payload		Rated Thrust (N)
	Horizontal(kg)	Vertical(kg)	
48	8	—	141
30	30	12	226
20	60	20	339
10	80	35	678
5	90	45	1357

[Stroke and Max Speed]

Unit: mm/s

Lead (mm)	50 to 600 (Every 50mm)	650 (mm)	700 (mm)	750 (mm)	800 (mm)	850 (mm)	900 (mm)	950 (mm)	1000 (mm)	1050 (mm)	1100 (mm)
48	2200			2180	1950	1760	1590	1450	1320	1210	1110
30	1800	1640	1440	1280	1155	1040	940	855	780	715	660
20	1200	1090	960	860	770	695	630	570	520	480	440
10	600	540	480	430	385	345	310	285	260	235	220
5	300	270	240	215	190	175	155	140	130	120	110

**Caution**

- The maximum speed may not be achieved when the stroke or movement distance is short or when acceleration/deceleration is set low.
- Setting at or below the minimum speed may lead to abnormal noise or unstable speeds. Do not attempt to set below the minimum speed.
- The minimum speed can be calculated with the equation below.

$$\text{Minimum speed (mm/s)} = \text{Ball screw lead (mm/r)} \div 16384 \text{ (p/r)} \times 1000 \text{ (1/s)}$$

(mm/r: movement per 1 motor revolution, p/r: pulse per 1 motor revolution)

[Payload by Acceleration]

At low load capacity, the acceleration/deceleration can be increased.

Lead 48

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
8	8	6	5	4	—	—	—	—	—

Lead 30

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
30	25	20	15	10	12	12	10	8	6

Lead 20

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
60	40	30	20	15	20	20	18	15	12

Lead 10

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
80	80	70	60	—	35	35	35	30	—

Lead 5

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
90	90	70	—	—	45	45	35	—	—

**Caution**

- Do not attempt to configure settings for acceleration/deceleration above the specifications. This may lead to vibration, breakdown, or shortened product life.

[Payload and Acceleration/Deceleration at Offboard Tuning]

(Note) RCS4CR Cleanroom Type is not applicable for the offboard tuning.

Have the tuning conducted with the offboard tuning in the PC software.

Operation with the acceleration/deceleration described in the table below should be performed.

Make sure not to operate with a payload higher than what is described in the table.

The unit for payload is kg. blanked boxes represent operation not available.

	Lead	Payload for Each Acceleration/Deceleration (kg)																	Max. Speed (mm/s)	
		0.3G	0.5G	0.7G	1.0G	1.2G	1.5G	2.0G	2.2G	2.5G	2.7G	3.0G	3.2G	3.5G	3.7G	4.0G	4.2G	4.5G	Standard	After Tuning
Horizontal	48	8	8	6	5	4	3.5	3	2.5	2	1.5	1	1	1	0.5	0.5	0.5	0.5	2200	2500
	30	30	25	20	15	10	9	8	7	6	5	4	4	4	3.75	3.75	3	3	1800	1800
	20	60	40	30	20	15	13	10	9	8	7	6	5	5	4	4	3	3	1200	1200
	10	80	80	70	60	58	55	43	35	30	25	20	18	15	13	12	10	8	600	600
	5	90	90	70	50	40	25												300	300
Vertical	48																			
	30	12	12	10	8	6	5.5	5	4.5	3.5	3.5	3							1800	1800
	20	20	20	18	15	12	10	8	7	6	5	4							1200	1200
	10	35	35	35	30	28	25	23	20	18	16	14							600	600
	5	45	45	35	30	22	15												300	300

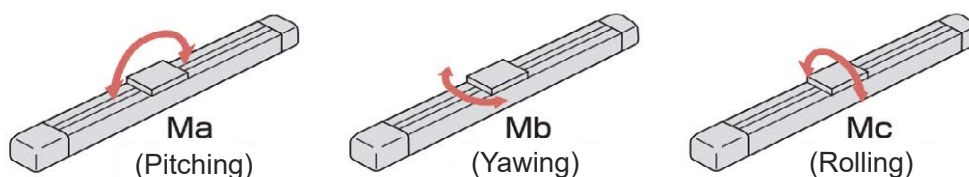
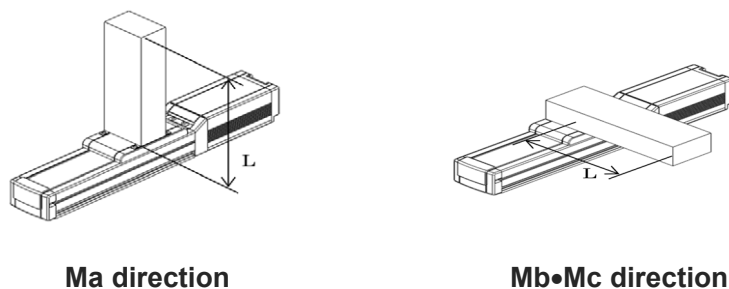
[Actuator Specifications]

Item	Content
Drive System	Ball Screw $\phi 16\text{mm}$, Rolled C10
Positioning Repeatability ^(*1)	$\pm 0.01\text{mm}[\pm 0.005\text{mm}]$
Lost Motion	0.1mm or less
Base	Material: Aluminum, White Anodized
Allowable static moment	Ma direction: 219 N•m, Mb direction: 219 N•m, Mc direction: 414 N•m
Allowable dynamic moment ^(*2)	Ma direction: 77.0 N•m, Mb direction: 77.0 N•m, Mc direction: 146 N•m
No. of Encoder Pulses	16384
Ambient operating temperature/humidity	0 to 40°C, 85%RH or less (Non-condensing)

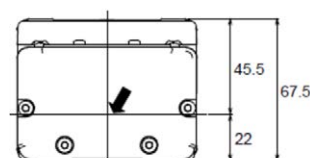
- Overhang load length guideline: 400mm or less

(*1) Number in brackets [] show the specification for high precision type.

(*2) For reference rated life of 5000km. The running life differs according to operation conditions and mounting status.

Slider type moment direction**Slider type overhang load length (L = 400mm or less)**

*For calculation of Ma and Mc moments, take the position pointed with an arrow as the datum.

**Caution**

- If the actuator is used with excessive allowable moment and overhang load, it may not only lead to abnormal noise and vibration but also significantly reduce the life of the actuator.

[5] RCS4-SA4R**[Lead and Payload]**

Lead (mm)	Max. Payload		Rated Thrust (N)
	Horizontal(kg)	Vertical(kg)	
16	10	2.5	53
10	14	4.5	85
5	17	8	170
2.5	20	12	340

[Stroke and Max Speed]

Unit: mm/s

Lead (mm)	50 to 450 (Every 50mm)	500 (mm)
16	960	875
10	600	555
5	300	275
2.5	150	135

**Caution**

- The maximum speed may not be achieved when the stroke or movement distance is short or when acceleration/deceleration is set low.
- Setting at or below the minimum speed may lead to abnormal noise or unstable speeds. Do not attempt to set below the minimum speed.
- The minimum speed can be calculated with the equation below.

Minimum speed (mm/s) = Ball screw lead (mm/r) ÷ 16384 (p/r) × 1000 (1/s)
 (mm/r: movement per 1 motor revolution, p/r: pulse per 1 motor revolution)

[Payload by Acceleration]

At low load capacity, the acceleration/deceleration can be increased.

Lead 16

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
10	8	6	6	—	2.5	2.5	1.5	1.5	—

Lead 10

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
14	14	12	10	—	4.5	4.5	3	3	—

Lead 5

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
17	17	15	13	—	8	8	6	6	—

Lead 2.5

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
20	20	20	—	—	12	12	10	—	—

**Caution**

- Do not attempt to configure settings for acceleration/deceleration above the specifications. This may lead to vibration, breakdown, or shortened product life.

[Payload and Acceleration/Deceleration at Offboard Tuning]

Have the tuning conducted with the offboard tuning in the PC software.

Operation with the acceleration/deceleration described in the table below should be performed.

Make sure not to operate with a payload higher than what is described in the table.

The unit for payload is kg. blanked boxes represent operation not available.

	Lead	Payload for Each Acceleration/Deceleration (kg)																		Max. Speed (mm/s)	
		0.3G	0.5G	0.7G	1.0G	1.2G	1.5G	2.0G	2.2G	2.5G	2.7G	3.0G	3.2G	3.5G	3.7G	4.0G	4.2G	4.5G	Standard	After Tuning	
Horizontal	16	10	8	6	6	4	3.5	3	2.75	2.5	2.25	2	2	1.75	1.5	1.25	1	1	960	960	
	10	14	14	12	10	6	5	4.5	4	3.5	3	2.5	2.25	2	1.75	1.5	1.25	1	600	600	
	5	17	17	15	13	12.5	12	11	10	9	8	7	6.5	6	5.5	5	4.5	4	300	300	
	2.5	20	20	20	20	17	11												150	150	
Vertical	16	2.5	2.5	1.5	1.5	1	1	1	1	1	0.75	0.75							960	960	
	10	4.5	4.5	3	3	2	2	2	1.5	1.5	1	1							600	600	
	5	8	8	6	6	5.75	5.5	5.25	5	4.75	4.5	4.25							300	300	
	2.5	12	12	10	8	7	6												150	150	

[Actuator Specifications]

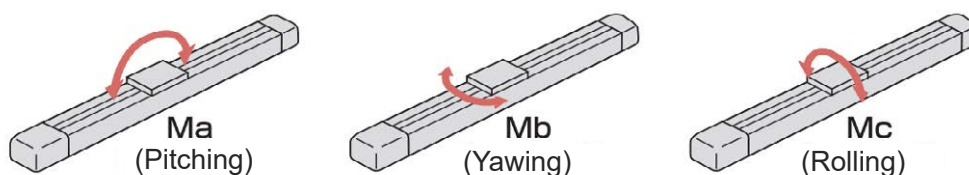
Item	Content
Drive System	Ball Screw $\phi 8\text{mm}$, Rolled C10
Positioning Repeatability ^{(*)1}	$\pm 0.01\text{mm}[\pm 0.005\text{mm}]$
Lost Motion	0.1mm or less
Base	Material: Aluminum, White Anodized
Allowable static moment	Ma direction: 13.0 N•m, Mb direction: 18.6 N•m, Mc direction: 25.3 N•m
Allowable dynamic moment ^{(*)2}	Ma direction: 5.0 N•m, Mb direction: 7.1 N•m, Mc direction: 9.7 N•m
No. of Encoder Pulses	16384
Ambient operating temperature/humidity	0 to 40°C, 85%RH or less (Non-condensing)

- Overhang load length guideline: 150mm or less

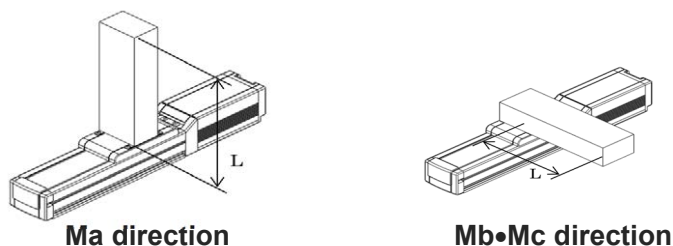
(*)1 Number in brackets [] show the specification for high precision type.

(*)2 For reference rated life of 5000km. The running life differs according to operation conditions and mounting status.

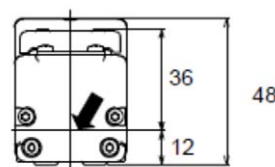
Slider type moment direction



Slider type overhang load length (L = 150mm or less)



- * For calculation of Ma and Mc moments, take the position pointed with an arrow as the datum.

**Caution**

- If the actuator is used with excessive allowable moment and overhang load, it may not only lead to abnormal noise and vibration but also significantly reduce the life of the actuator.

[6] RCS4-SA6R**[Lead and Payload]**

Lead (mm)	Max. Payload		Rated Thrust (N)
	Horizontal(kg)	Vertical(kg)	
30	11	3	57
20	18	5	85
12	30	9	142
6	45	15	283
3	45	15	566

[Stroke and Max Speed]

Unit: mm/s

Lead (mm)	50 to 500 (Every 50mm)	550 (mm)	600 (mm)	650 (mm)	700 (mm)	750 (mm)	800 (mm)
30	1600		1450	1260	1100	970	860
20	1200	1130	970	840	735	650	575
12	720	620	535	460	405	355	315
6	360	305	265	230	200	175	155
3	180	150	130	115	100	85	75

**Caution**

- The maximum speed may not be achieved when the stroke or movement distance is short or when acceleration/deceleration is set low.
- Setting at or below the minimum speed may lead to abnormal noise or unstable speeds. Do not attempt to set below the minimum speed.
- The minimum speed can be calculated with the equation below.

$$\text{Minimum speed (mm/s)} = \text{Ball screw lead (mm/r)} \div 16384 \text{ (p/r)} \times 1000 \text{ (1/s)}$$
 (mm/r: movement per 1 motor revolution, p/r: pulse per 1 motor revolution)

[Payload by Acceleration]

At low load capacity, the acceleration/deceleration can be increased.

Lead 30

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
11	6	6	4	—	3	2	2	1.5	—

Lead 20

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
18	15	10	8	—	5	5	4	4	—

Lead 12

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
30	24	20	15	—	9	9	9	8	—

Lead 6

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
45	30	20	20	—	15	15	15	15	—

Lead 3

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
45	30	20	—	—	15	14	10	—	—

**Caution**

- Do not attempt to configure settings for acceleration/deceleration above the specifications. This may lead to vibration, breakdown, or shortened product life.

[Payload and Acceleration/Deceleration at Offboard Tuning]

Have the tuning conducted with the offboard tuning in the PC software.

Operation with the acceleration/deceleration described in the table below should be performed.

Make sure not to operate with a payload higher than what is described in the table.

The unit for payload is kg. blanked boxes represent operation not available.

	Lead	Payload for Each Acceleration/Deceleration (kg)																	Max. Speed (mm/s)	
		0.3G	0.5G	0.7G	1.0G	1.2G	1.5G	2.0G	2.2G	2.5G	2.7G	3.0G	3.2G	3.5G	3.7G	4.0G	4.2G	4.5G	Standard	After Tuning
Horizontal	30	11	6	6	4	3	2.5	2	1.5	1.5	1.5	1.25	1.25	1.25	1.25	1	1	1	1600	1800
	20	18	15	10	8	6	5	4.5	4	3.5	2.5	2	2	2	2	1.75	1.75	1.75	1200	1200
	12	30	24	20	15	15	9	6	5.5	5	4	3.5	3.5	3	3	2.5	2.5	2	720	720
	6	45	30	20	20	18	16	11	9	7.5	7	5.75	5	4	3.5	3	2.5	2	360	360
	3	45	30	20	12	7.5	2.5												180	180
Vertical	30	3	2	2	1.5	1.5	1.5	1.5	1.5	1	1	1							1600	1800
	20	5	5	4	4	3	2	2	2	2	2	2							1200	1200
	12	9	9	9	8	8	7	6	5	4	3.5	3							720	720
	6	15	15	15	15	11	9	6.5	6	5	4	4							360	360
	3	15	14	10	6	4	1												180	180

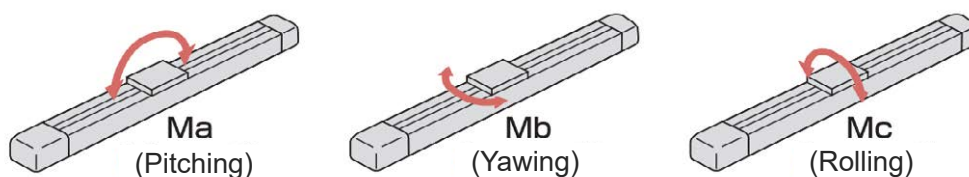
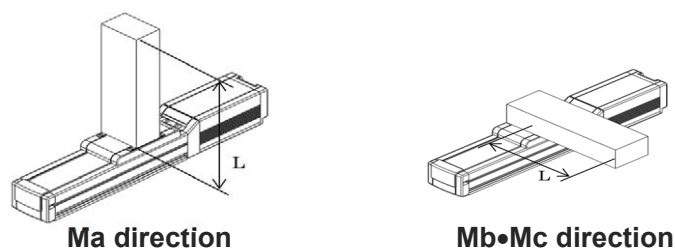
[Actuator Specifications]

Item	Content
Drive System	Ball Screw $\phi 10\text{mm}$, Rolled C10
Positioning Repeatability ^(*1)	$\pm 0.01\text{mm}[\pm 0.005\text{mm}]$
Lost Motion	0.1 mm or less
Base	Material: Aluminum, White Anodized
Allowable static moment	Ma direction: 48.5 N•m, Mb direction: 69.3 N•m, Mc direction: 103 N•m
Allowable dynamic moment ^(*2)	Ma direction: 11.6 N•m, Mb direction: 16.6 N•m, Mc direction: 24.6 N•m
No. of Encoder Pulses	16384
Ambient operating temperature/humidity	0 to 40°C, 85%RH or less (Non-condensing)

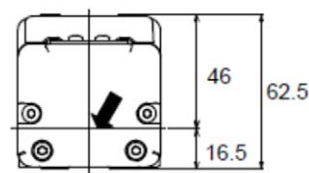
- Overhang load length guideline: 220mm or less

(*1) Number in brackets [] show the specification for high precision type.

(*2) For reference rated life of 5000km. The running life differs according to operation conditions and mounting status.

Slider type moment direction**Slider type overhang load length (L = 220mm or less)**

- * For calculation of Ma and Mc moments, take the position pointed with an arrow as the datum.

**Caution**

- If the actuator is used with excessive allowable moment and overhang load, it may not only lead to abnormal noise and vibration but also significantly reduce the life of the actuator.

[7] RCS4-SA7R**[Lead and Payload]**

Lead (mm)	Max. Payload		Rated Thrust (N)
	Horizontal(kg)	Vertical(kg)	
36	7	4	95
24	30	6	142
16	38	12	214
8	45	18	427
4	50	25	855

[Stroke and Max Speed]

Unit: mm/s

Lead (mm)	50 to 500 (Every 50mm)	550 (mm)	600 (mm)	650 (mm)	700 (mm)	750 (mm)	800 (mm)
36	1800			1620	1420	1260	1120
24	1500	1440	1240	1095	965	850	760
16	1000	965	830	720	635	560	500
8	500	475	410	355	315	275	245
4	240	235	205	175	155	135	120

**Caution**

- The maximum speed may not be achieved when the stroke or movement distance is short or when acceleration/deceleration is set low.
- Setting at or below the minimum speed may lead to abnormal noise or unstable speeds. Do not attempt to set below the minimum speed.
- The minimum speed can be calculated with the equation below.

Minimum speed (mm/s) = Ball screw lead (mm/r) ÷ 16384 (p/r) × 1000 (1/s)
 (mm/r: movement per 1 motor revolution, p/r: pulse per 1 motor revolution)

[Payload by Acceleration]

At low load capacity, the acceleration/deceleration can be increased.

Lead 36

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
7	7	6	4	—	4	4	3	2	—

Lead 24

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
30	12	10	6	—	6	6	5	5	—

Lead 16

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
38	30	15	15	—	12	12	10	8	—

Lead 8

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
45	40	40	35	—	18	18	18	16	—

Lead 4

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
50	50	40	—	—	25	25	20	—	—

**Caution**

- Do not attempt to configure settings for acceleration/deceleration above the specifications. This may lead to vibration, breakdown, or shortened product life.

[Payload and Acceleration/Deceleration at Offboard Tuning]

Have the tuning conducted with the offboard tuning in the PC software.

Operation with the acceleration/deceleration described in the table below should be performed.

Make sure not to operate with a payload higher than what is described in the table.

The unit for payload is kg. blanked boxes represent operation not available.

	Lead	Payload for Each Acceleration/Deceleration (kg)																	Max. Speed (mm/s)	
		0.3G	0.5G	0.7G	1.0G	1.2G	1.5G	2.0G	2.2G	2.5G	2.7G	3.0G	3.2G	3.5G	3.7G	4.0G	4.2G	4.5G	Standard	After Tuning
Horizontal	36	7	7	6	4	3	3	2.5	2.5	2	2	1.5	1.5	1.5	1.5	1.5	1.5	1	1800	2150
	24	30	12	10	6	5	5	5	5	4	4	3	3	3	3	3	3	3	1500	1500
	16	38	30	15	15	12	12	10	10	9	8	7	6	6	5.5	5	4	3.5	1000	1000
	8	45	40	40	35	30	25	18	15	10	9	9	9	8	7	6	5	4	500	500
	4	50	50	40	25	15	5												240	240
Vertical	36	4	4	3	2	2	2	2	2	1.5	1.5	1.5							1800	2150
	24	6	6	5	5	4	4	4	3.5	3.5	3	3							1500	1500
	16	12	12	10	8	8	8	5	5	5	5	5							1000	1000
	8	18	18	18	16	15	12	10	10	8	7	7							500	500
	4	25	25	20	12	8	3												240	240

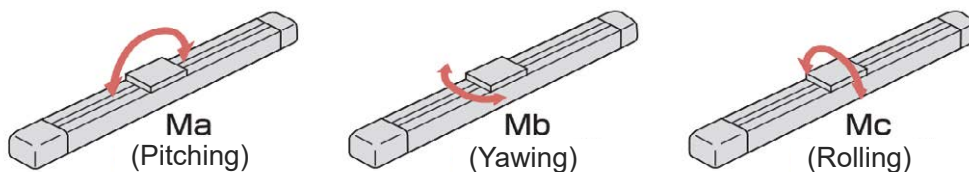
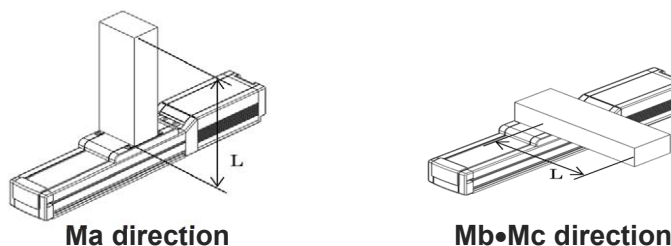
[Actuator Specifications]

Item	Content
Drive System	Ball Screw $\phi 12\text{mm}$, Rolled C10
Positioning Repeatability ^(*1)	$\pm 0.01\text{mm}[\pm 0.005\text{mm}]$
Lost Motion	0.1mm or less
Base	Material: Aluminum, White Anodized
Allowable static moment	Ma direction:115 N•m, Mb direction:115 N•m, Mc direction:229 N•m
Allowable dynamic moment ^(*2)	Ma direction:44.7 N•m, Mb direction:44.7 N•m, Mc direction:89.1 N•m
No. of Encoder Pulses	16384
Ambient operating temperature/humidity	0 to 40°C 85%RH or less (Non-condensing)

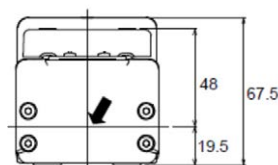
- Overhang load length guideline:300mm or less

(*1) Number in brackets [] show the specification for high precision type.

(*2) For reference rated life of 5000km. The running life differs according to operation conditions and mounting status.

Slider type moment direction**Slider type overhang load length (L = 300mm or less)**

- * For calculation of Ma and Mc moments, take the position pointed with an arrow as the datum.

**Caution**

- If the actuator is used with excessive allowable moment and overhang load, it may not only lead to abnormal noise and vibration but also significantly reduce the life of the actuator.

[8] RCS4-SA8R**[Lead and Payload]**

Lead (mm)	Max. Payload		Rated Thrust (N)
	Horizontal(kg)	Horizontal(kg)	
48	8	-	141
30	30	12	226
20	60	20	339
10	80	35	678
5	90	45	1357

[Stroke and Max Speed]

Unit: mm/s

Lead (mm)	50 to 600 (Every 50mm)	650 (mm)	700 (mm)	750 (mm)	800 (mm)	850 (mm)	900 (mm)	950 (mm)	1000 (mm)	1050 (mm)	1100 (mm)
48	2100				1950	1760	1590	1450	1320	1210	1110
30	1800	1640	1440	1280	1155	1040	940	855	780	715	660
20	1200	1090	960	860	770	695	630	570	520	480	440
10	600	540	480	430	385	345	310	285	260	235	220
5	300	270	240	215	190	175	155	140	130	120	110

**Caution**

- The maximum speed may not be achieved when the stroke or movement distance is short or when acceleration/deceleration is set low.
- Setting at or below the minimum speed may lead to abnormal noise or unstable speeds. Do not attempt to set below the minimum speed.
- The minimum speed can be calculated with the equation below.

Minimum speed (mm/s) = Ball screw lead (mm/r) ÷ 16384 (p/r) × 1000 (1/s)
 (mm/r: movement per 1 motor revolution, p/r: pulse per 1 motor revolution)

[Payload by Acceleration]

At low load capacity, the acceleration/deceleration can be increased.

Lead 48

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
8	8	6	5	—	—	—	—	—	—

Lead 30

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
30	25	20	15	—	12	12	10	8	—

Lead 20

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
60	40	30	20	—	20	20	18	15	—

Lead 10

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
80	80	70	60	—	35	35	35	30	—

Lead 5

Horizontal					Vertical				
0.3	0.5	0.7	1.0	1.2	0.3	0.5	0.7	1.0	1.2
90	90	70	—	—	45	45	35	—	—

**Caution**

- Do not attempt to configure settings for acceleration/deceleration above the specifications. This may lead to vibration, breakdown, or shortened product life.

[Payload and Acceleration/Deceleration at Offboard Tuning]

Have the tuning conducted with the offboard tuning in the PC software.

Operation with the acceleration/deceleration described in the table below should be performed.

Make sure not to operate with a payload higher than what is described in the table.

The unit for payload is kg. blanked boxes represent operation not available.

	Lead	Payload for Each Acceleration/Deceleration (kg)																	Max. Speed (mm/s)	
		0.3G	0.5G	0.7G	1.0G	1.2G	1.5G	2.0G	2.2G	2.5G	2.7G	3.0G	3.2G	3.5G	3.7G	4.0G	4.2G	4.5G	Standard	After Tuning
Horizontal	48	8	8	6	5	4	3.5	3	2.5	2	1.5	1	1	1	0.5	0.5	0.5	0.5	2100	2500
	30	30	25	20	15	10	9	8	7	6	5	4	4	4	3.75	3.75	3.5	3.5	1800	1800
	20	60	40	30	20	15	13	10	9	8	7	6	6	6	6	6	5	5	1200	1200
	10	80	80	70	60	58	55	43	35	30	25	20	18	15	13	12	10	8	600	600
	5	90	90	70	50	40	25												300	300
Vertical	48																			
	30	12	12	10	8	6	5.5	5	4.5	4	3.5	3							1800	1800
	20	20	20	18	15	12	10	8	7	6	5	4							1200	1200
	10	35	35	35	30	28	25	23	20	18	16	14							600	600
	5	45	45	35	30	22	15												300	300

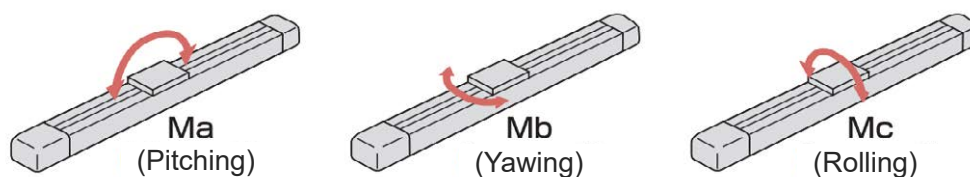
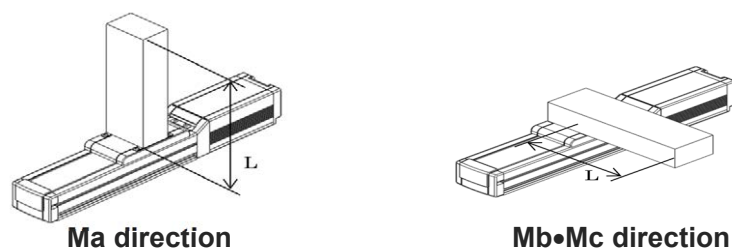
[Actuator Specifications]

Item	Content
Drive System	Ball Screw $\phi 16\text{mm}$, Rolled C10
Positioning Repeatability ^{(*)1}	$\pm 0.01\text{mm}[\pm 0.005\text{mm}]$
Lost Motion	0.1mm or less
Base	Material: Aluminum, White Anodized
Allowable static moment	Ma direction: 219 N•m, Mb direction: 219 N•m, Mc direction: 414 N•m
Allowable dynamic moment ^{(*)2}	Ma direction: 77.0 N•m, Mb direction: 77.0 N•m, Mc direction: 146 N•m
No. of Encoder Pulses	16384
Ambient operating temperature/humidity	0 to 40°C, 85%RH or less (Non-condensing)

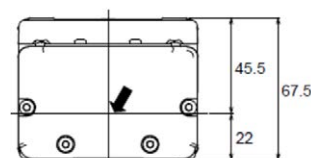
- Overhang load length guideline: 400mm or less

(*)1 Number in brackets [] show the specification for high precision type.

(*)2 For reference rated life of 5000km. The running life differs according to operation conditions and mounting status.

Slider type moment direction**Slider type overhang load length (L = 400mm or less)**

- * For calculation of Ma and Mc moments, take the position pointed with an arrow as the datum.

**Caution**

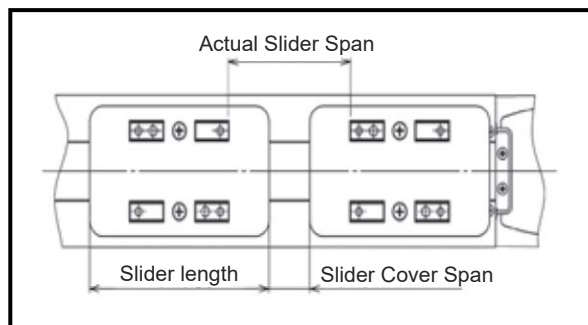
- If the actuator is used with excessive allowable moment and overhang load, it may not only lead to abnormal noise and vibration but also significantly reduce the life of the actuator.

1.2.2 Double Slider (Option)

[Table for Double Slider Type]

Name of Type	Standard rated life (km)	Slider Span (mm)		Allowable dynamic moment			Allowable static moment			Overhang load length (mm)	Slider Mass (kg)	Slider length (mm)	Min. Effective Stroke for Double Slider (mm)
		Actual Slider Span	Slider Cover Span	Ma direction (N·m)	Mb direction (N·m)	Mc direction (N·m)	Ma direction (N·m)	Mb direction (N·m)	Mc direction (N·m)				
SA4	5000	60	24	44.6	63.6	15.7	143	204	50.5	420	1	76	50
SA6	5000	90	40	106	152	40	474	678	178	630	2	110	50
SA7	5000	70	20	285	285	145	711	711	361	810	2	130	50
SA8	5000	120	35	565	565	237	1980	1980	828	1200	2.5	165	50

[Figure of Double Slider]



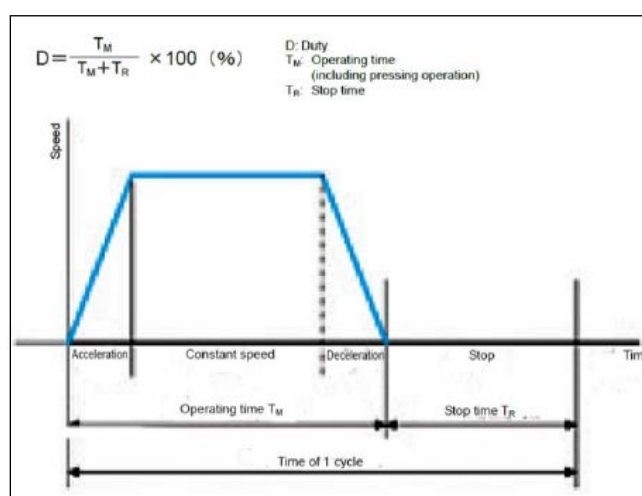
- If double slider (option) is ordered, consider the available stroke should be the value that the length of slider and slider cover span are subtracted from the stroke shown in the model code. Therefore, add the following number to the necessary stroke when indicate the stroke at order. Also, make sure that the “necessary stroke” should be the minimum effective stroke of the double slider type or above.

Stroke Length to Order (Nominal Stroke)
“Necessary Stroke” + “Slider Cover Span” + “Slider Length” or more

- The payload of the double slider should consider the weight of the slider itself subtracted from the performance in the specification for the maximum.
- When you consider the maximum velocity, consider the stroke length to order (nominal stroke) as the value of stroke in [Stroke and Max Speed].

1.2.3 Duty ratio

The duty ratio is the operating rate, shown in %, of the actuator operating time within one cycle.



As the reference for duty available to use may differ depending on the operation conditions (payload, acceleration / deceleration, etc.), it is necessary to figure out the load factor LF and acceleration / deceleration time ratio t_{od} using the calculation formulae below and find it out from the graph.

1) Figure out the load factor LF using the calculation formulae below.

Maximum payload at the acceleration 0.3G is described in 1.2 Specifications.

[When indicated acceleration / deceleration is at acceleration / deceleration 0.3G or below]

$$\text{Load Factor: LF} = \frac{M \times \alpha}{M_r \times 0.3} \quad [\%]$$

Max. Payload at Acceleration 0.3G : M_r [kg]

Acceleration / Deceleration 0.3G : 0.3 [G]

Payload during Operation : M [kg]

Acceleration during Operation : α [G]

[When indicated acceleration / deceleration is at acceleration / deceleration 0.3G or above]

$$\text{Load Factor: LF} = \frac{M \times \alpha}{M_d \times \alpha} = \frac{M}{M_d} \quad [\%]$$

Payload at Indicated Acceleration : M_d [kg]

Payload during Operation : M [kg]

Acceleration during Operation : α [G]

2) Figure out the acceleration / deceleration time ratio t_{od} using the calculation formulae below.

Acceleration / Deceleration Time Ratio t_{od} =

$$\frac{\text{Acceleration Time during Operation} + \text{Deceleration Time during Operation}}{\text{Duration of Operation}} \quad [\%]$$

$$\text{Acceleration Time} = \frac{\text{Velocity during Operation [mm/s]}}{\text{Acceleration during Operation [mm/s}^2\text{]} [\text{sec}]}$$

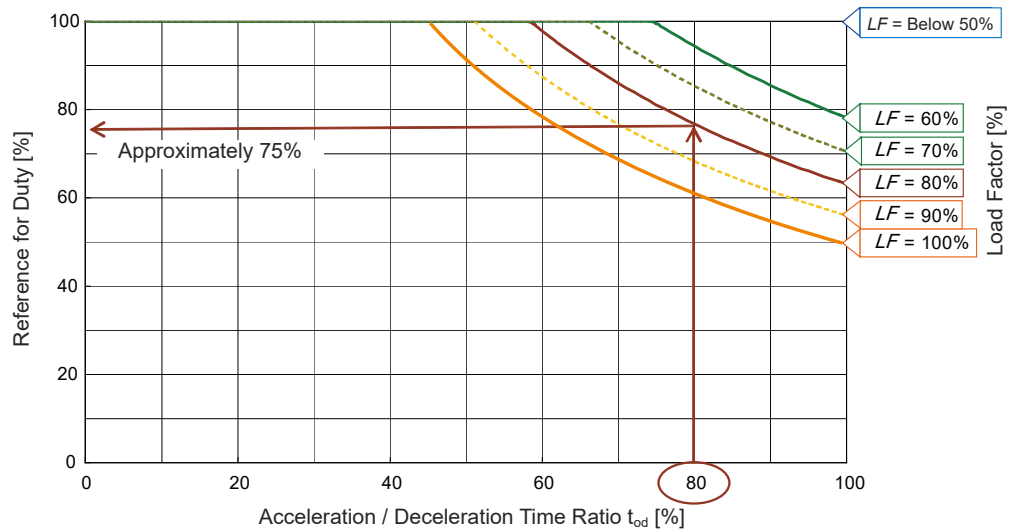
$$\text{Acceleration [mm/s}^2\text{]} = \text{Acceleration [G]} \times 9,800 \text{ mm/s}^2$$

$$\text{Deceleration Time} = \frac{\text{Velocity during Operation [mm/s]}}{\text{Deceleration during Operation [mm/s}^2\text{]} [\text{sec}]}$$

$$\text{Deceleration [mm/s}^2\text{]} = \text{Deceleration [G]} \times 9,800 \text{ mm/s}^2$$

3) Read a reference for duty with the figured out “Load Factor” and “Acceleration / Deceleration Time Ratio”.

e.g.) The reference for duty when the load factor LF is 80% and the acceleration / deceleration time ratio t_{od} is 80% should be approximately 75%.



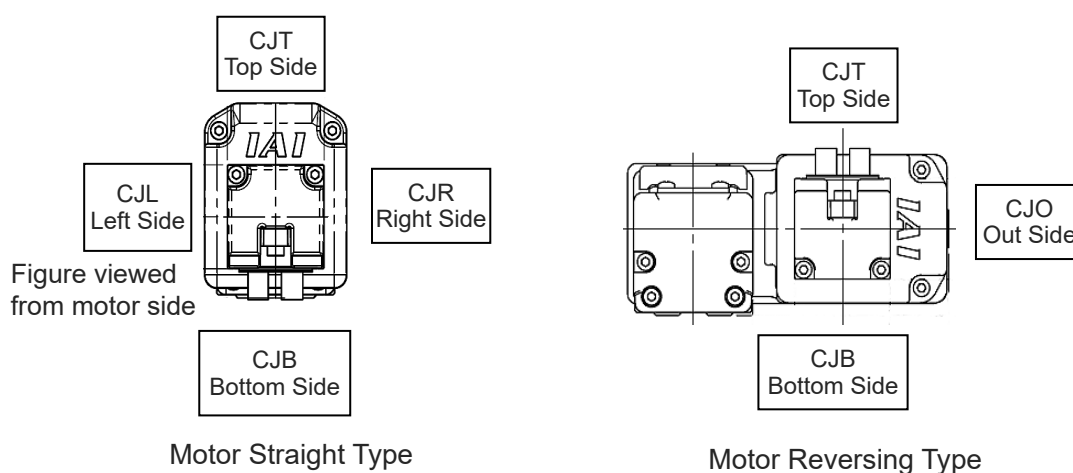
1.3 Options

1.3.1 With brake (Model Code: B)

This is used to prevent the slider from moving during power outages or when the servo is OFF. It can also be used to prevent the slider from falling when mounted vertically.

1.3.2 Cable Exit Direction Changed (Model Code: CJT, CJR, CJL, CJB and CJO)

The orientation of the motor / encoder cable to be installed on the actuator unit can be changed to top/bottom/right/left.



1.3.3 High Precision Specifications (Model Code: HPR)

Standard positioning repeatability is $\pm 0.01\text{mm}$. When this option is specified, positioning repeatability will become $\pm 0.005\text{mm}$. It should be available to order only for lead length 20mm or less.

1.3.4 Motor Reversed Direction (Model Code: ML, MR)

It is the code to indicate the direction of motor reversed when selecting the motor reversed type. ML shows reversed to left and MR to right.



1.3.5 Reversed-home Specification (Model Code: NM)

The standard home position is on the motor side.

However, the opposite side specification is selected if the home position direction is reversed in accordance with equipment layout or assembly direction.

1.3.6 Slider Roller Specification (Model Code: SR)

The structure of the slider in the standard type can be changed to the roller structure same as the one in the cleanroom type.

If it gets chosen to have the slider roller type, the appearance and dimensions of the slider cover should become the same as the cleanroom type.

1.3.7 Vacuum Joint Attachment Opposite Type (Model Code: VR)

The vacuum joint should be attached on the left of the main unit in the view from the motor end in standard, which can be changed to the right (opposite) in option.

1.3.8 Double Slider Specification (Model Code: W)

This is an option that adds a free slider that is not connected to the ball screw.

Doubling the slider makes it possible to increase the moment and the overhang load length.

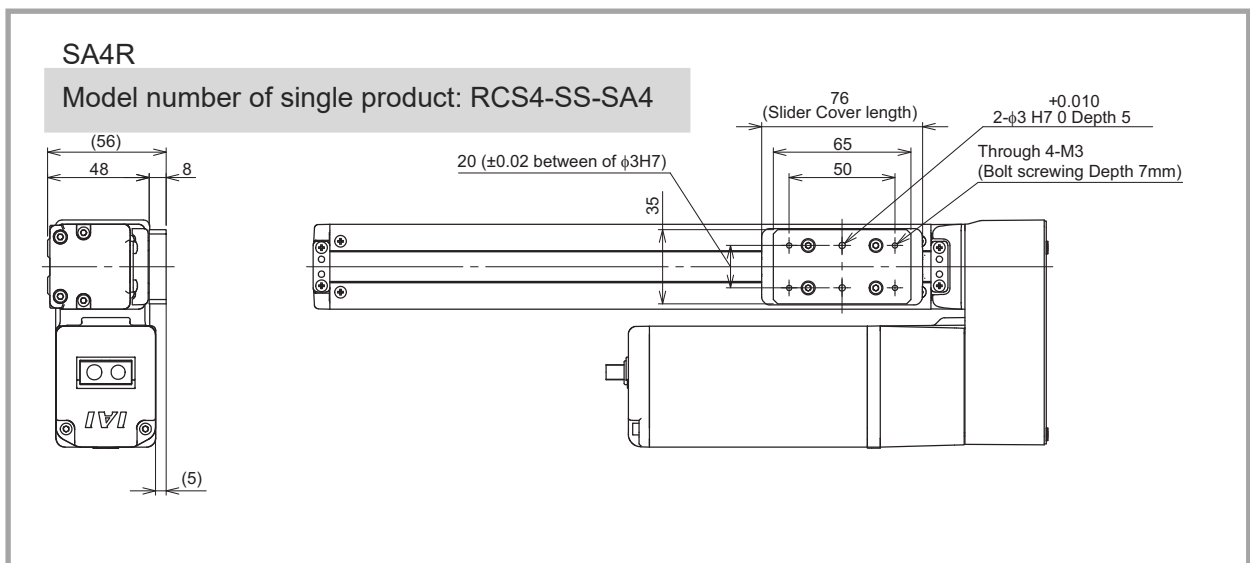
* Refer to [1.2.2 Double Slider (Option)] for details.

1.3.9 Rear Attachment Plate (Model Code: RP)

It is a bracket (plate) to affix the motor reversed type (SA4R) at the back to a device.

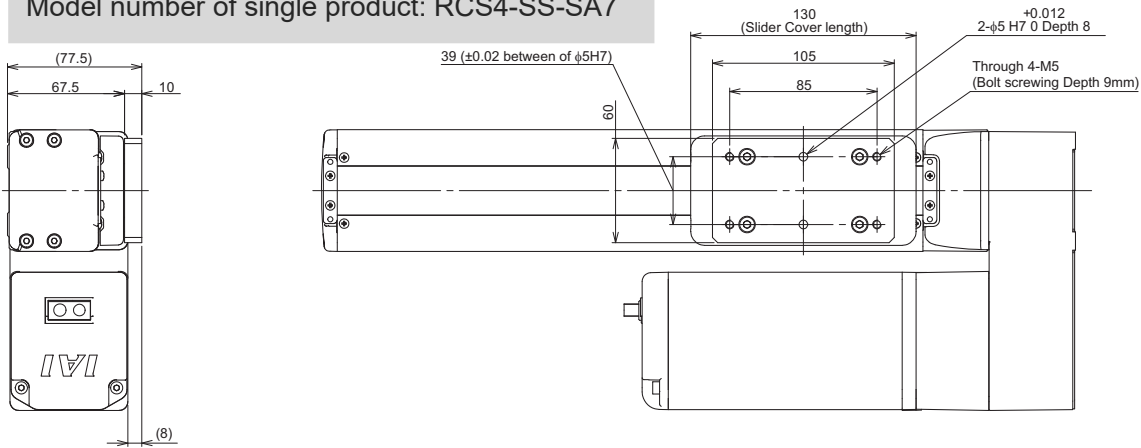
1.3.10 Slider spacer (Model Code: SS)

It is a spacer to set the top of the slider higher than the top of the motor cover.



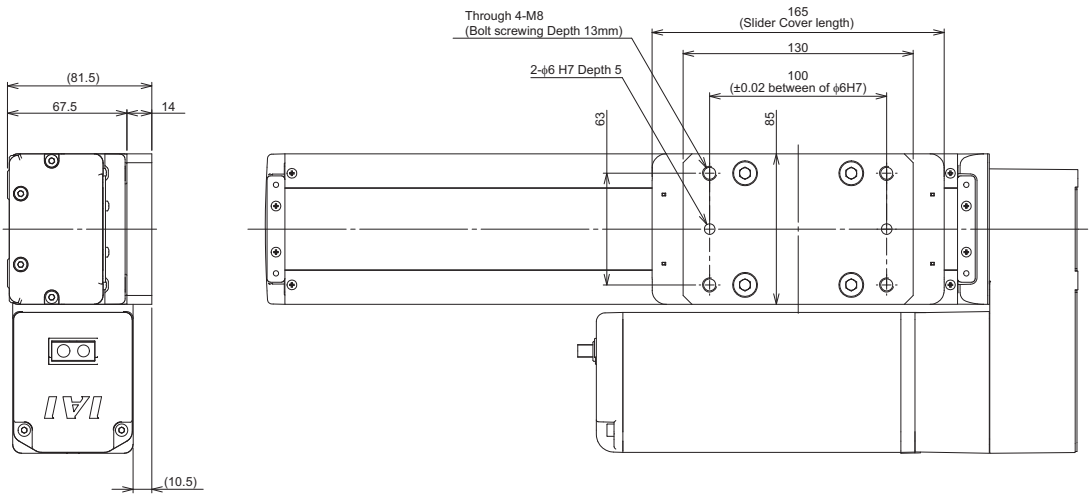
SA7R

Model number of single product: RCS4-SS-SA7



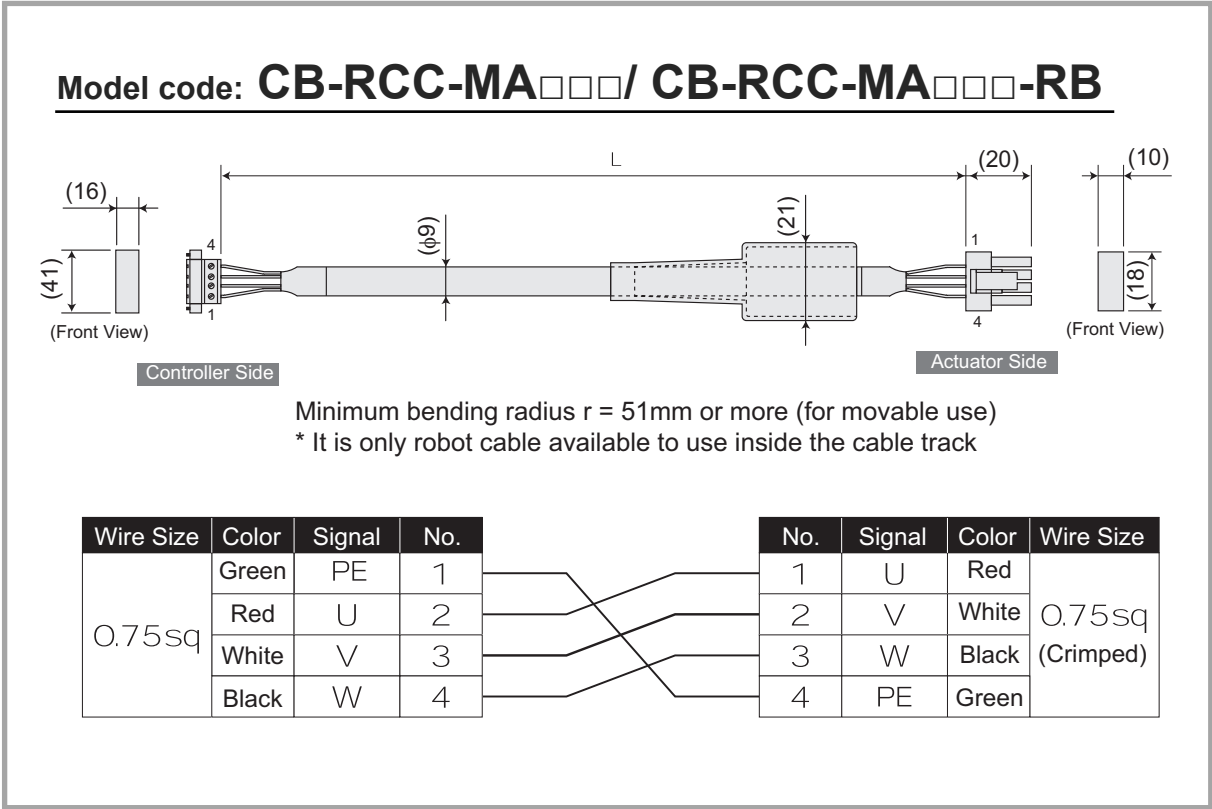
SA8R

Model number of single product: RCS4-SS-SA8



1.4 Accessories

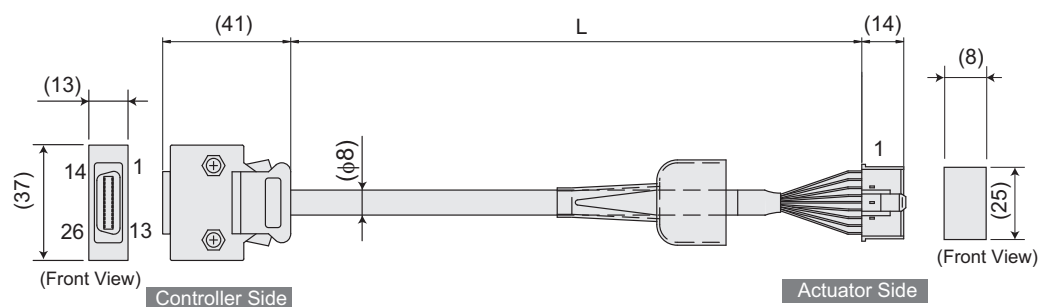
1.4.1 Motor Cable



- The cable length (L) should be 1m at minimum and 20m at maximum.
Order can be make in unit of 1m long.
- An example for model codes is as shown below.
Cable length 1m → CB-RCC-MA010 (-RB)
Cable length 3m → CB-RCC-MA030 (-RB)
Cable length 10m → CB-RCC-MA100 (-RB)

1.4.2 Encoder Cable

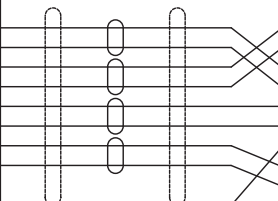
Model code: **CB-X1-PA**□□□



Minimum bending radius $r = 44\text{mm}$ or more (for movable use)

*Robot cable is standard for this model.

Wire Size	Color	Signal	No.
AWG26 (Soldered)	—	—	10
	—	—	11
	—	E24V	12
	—	OV	13
	—	LS	26
	—	CREEP	25
	—	OT	24
	—	RSV	23
	—	—	9
	—	—	18
	—	—	19
	—	A+	1
	—	A-	2
	—	B+	3
	—	B-	4
	—	Z+	5
	—	Z-	6
	Orange	SRD+	7
	Green	SRD-	8
	Purple	BAT+	14
	Gray	BAT-	15
	Red	VCC	16
Black	GND	17	
Blue	BKR-	20	
Yellow	BKR+	21	
—	—	22	



No.	Signal	Color	Wire Size
1	B A T +	Purple	AWG26 (Crimped)
2	B A T -	Gray	
3	SD	Orange	
4	SD	Green	
5	V C C	Red	
6	G N D	Black	
7	F G	Drain	
8	B K -	Blue	
9	B K +	Yellow	

Clamp the shield to the hood.

Ground wire or braided shield wire

- The cable length (L) should be 1m at minimum and 20m at maximum.
Order can be made in unit of 1m long.
- An example for model codes is as shown below.

Cable length 1m → CB-X1-PA010

Cable length 3m → CB-X1-PA030

Cable length 10m → CB-X1-PA100

ROBO Cylinder

Chapter 2

Installation

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2.2	Installation and storage/preservation environment	2-3
2.2.1	Installation Environment	2-3
2.2.2	Storage/preservation environment	2-4
2.3	How to Install	2-5
2.3.1	Mounting Orientation Type: RCS4 –SA4/SA6/SA7/SA8	2-5
2.3.2	Precautions regarding stainless steel sheet	2-6
2.3.3	Installation surface	2-7
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2.3.5	Mounting transported objects	2-15
2.3.6	About Suction of the Cleanroom Type	2-17

2.1 Precautions for transportation

[Handling the package]

- Do not damage or drop the package.

The package is not specially designed to withstand dropping or shock due to collision.

- Keep the unit in horizontal orientation for stationary positioning or transportation.
- Do not climb onto the package.
- Do not put anything that could deform the package on it.



[Handling after unpacking]

- Hold the base part when you carry the unit.
- Do not carry the unit by its motor cover or stainless steel sheet.
- Do not damage or drop the package during transportation.
- Do not apply excessive force to any part of the actuator.

→For the names of each part, refer to [Names of the Parts] on page Intro-11, 12.

**[Handling when assembled into machinery (system)]**

- Fix the slider so they would not accidentally move during transportation.
- If the actuator or any moving part is overhanging, fix it appropriately to avoid large wobbles due to external vibration. When transporting without fixing the tip, do not apply impact of 0.3G or more.
- When suspending machinery (system) with ropes, be careful not to catch the rope on the actuator or cable.

2.2 Installation and storage/preservation environment

Usage is possible in environments of pollution degree 2 or equivalent.

Pollution degree 2: Environment in which generally only nonconductive pollution occurs, but temporary conductive pollution may occur due to condensation (IEC 60664-1)

2.2.1 Installation Environment

The actuator should be installed in a location other than those specified below.

In general, the installation environment should be one in which an operator can work without protective gear.

- Where the actuator receives radiant heat from strong heat sources such as heat treatment furnaces
- Where the ambient temperature exceeds the range of 0 to 40°C
- Where the temperature changes rapidly and condensation occurs
- Where the relative humidity exceeds 85% RH
- Where the actuator receives direct sunlight
- Where the actuator is exposed to corrosive or combustible gases
- Where the ambient air contains a large amount of powder dust, salt or iron (at level exceeding what is normally expected in an assembly plant)
- Where the actuator is subject to splashed water, oil (including oil mist or cutting fluid) or chemical solutions
- Where the actuator receives impact or vibration
- Where the altitude is more than 2,000m

Also, provide sufficient work space for the following maintenance and inspection:

- Space to replenish grease
- Space to replace the motor

If the actuator is used in any of the following locations, provide sufficient shielding measures:

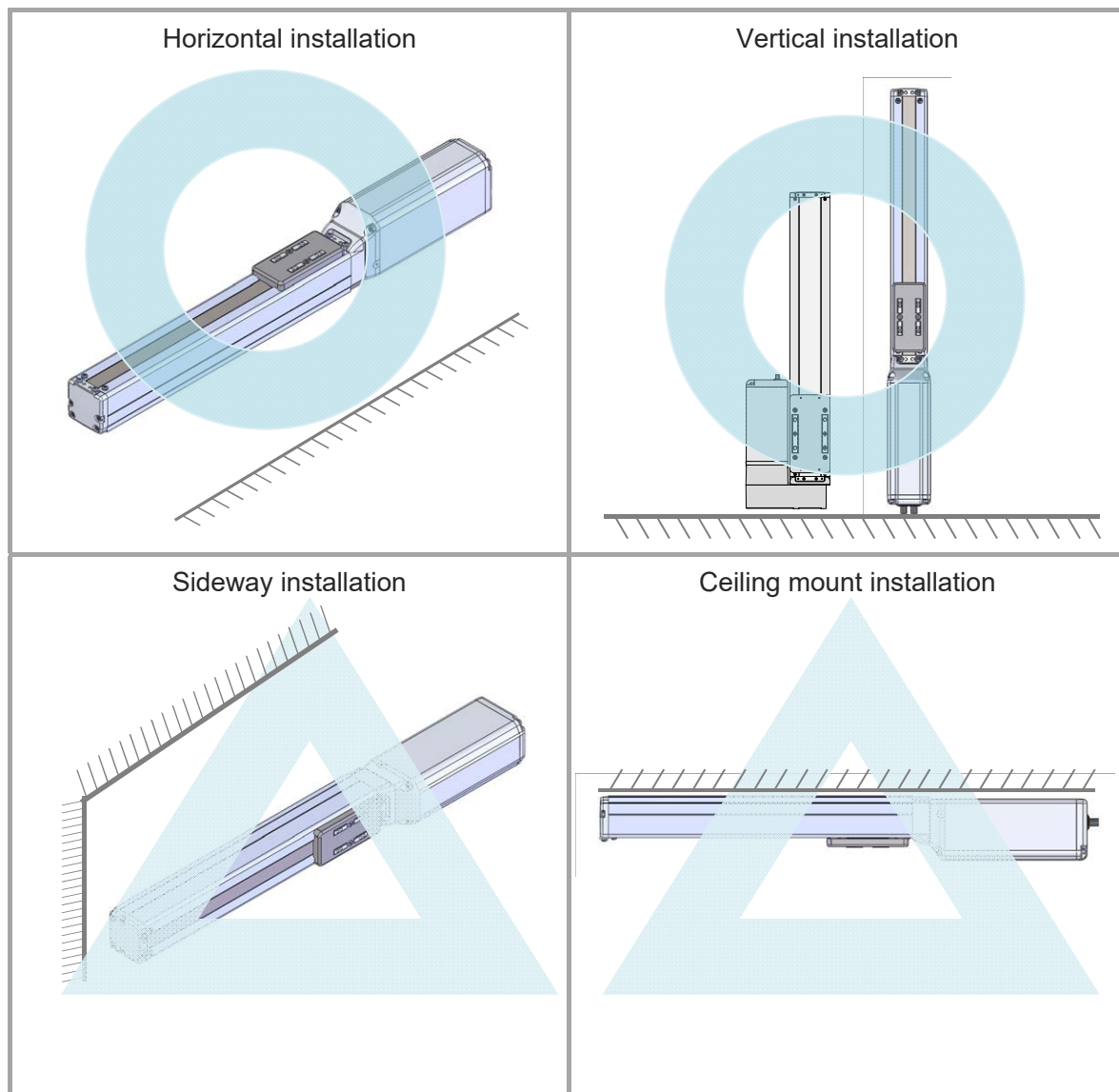
- Where noise generates due to static electricity, etc.
- Where the actuator is subject to a strong electric or magnetic field
- Where the actuator is subject to ultraviolet ray or radiation

2.2.2 Storage/preservation environment

- For the storage and preservation environment, see the installation environment.
However, give especial consideration to the prevention of condensation during long-term storage/preservation.
- Unless especially specified, desiccant is not included in the package at shipping.
If the product is to be stored/preserved in an environment where condensation is anticipated, take condensation preventive measures.
- For short-term storage, it can be stored at 60°C or below.
For storage of one month or more, make sure that the temperature does not exceed 50°C.
- The product should be placed horizontally for storage and preservation.
If storing in the packaged condition, observe the conditions, if any, regarding storage orientation.

2.3 How to Install

2.3.1 Mounting Orientation Type: RCS4 –SA4/SA6/SA7/SA8



Caution

- Install the unit with the motor on the top side when it is installed vertically for Motor Straight Type.
- When the motor is installed at the bottom, the grease may get separated and the base oil could flow into the motor, which could cause malfunction of the controller, motor or encoder.

Therefore it is not recommended to install it with the motor on the bottom side.

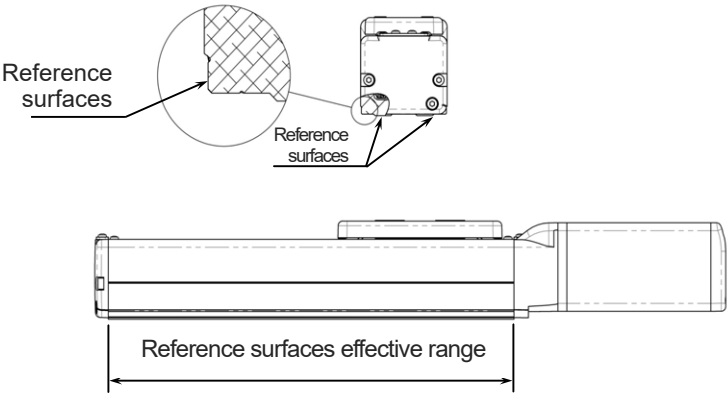
Consider the motor reversed type when it is necessary to install the motor at the bottom.

2.3.2 Precautions regarding stainless steel sheet

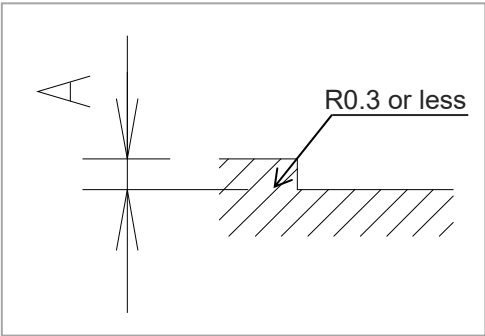
- During installation and transport, do not grasp or press on the stainless steel sheet. Otherwise, this may lead to stainless steel sheet damage.
- Although lateral or ceiling installation is possible, it may cause slackness or misalignment of the stainless steel sheet.
Continued usage in this orientation may lead to damage to the stainless steel sheet.
- Adjust the stainless steel sheet if slackness or misalignment occurs.
- Keep adhesive, paint, and other viscous material off the stainless steel sheet.
Such material adhering to the stainless steel sheet can lead to defective slider operation or sheet damage.
Avoid installation in environments of this kind, or take safety measures to prevent adhesion.

2.3.3 Installation surface

- The body mounting surface should be a machined surface or a plane with similar accuracy, with flatness within 0.05mm/m.
 - The mounting frame should have a structure rigid enough to prevent the generation of vibration, etc.
 - Also consider enough space necessary for maintenance work such as actuator replacement and inspection.
 - The bottom surface and both side surfaces of the body base serve as the reference surfaces for slider running accuracy.
- Use these surfaces as the reference surfaces for mounting when running accuracy is required.



When mounting using the side reference surface, machine the installation surface according to the figure below.



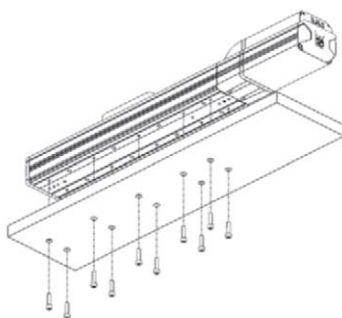
A dimension	
SA4, SA6, SA7	2 to 4mm
SA8	3 to 5mm

2.3.4 Installation of the Main Unit

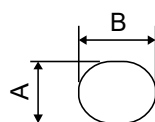
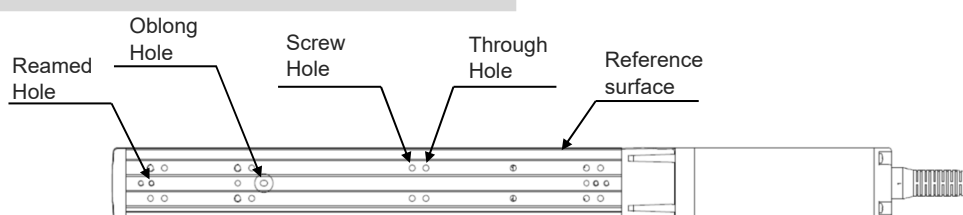
[When using the screw holes on the bottom of the base]

There are screw holes for attachment at the back of the base that allow fixation from the back.

→For details regarding the position and dimensions, refer to [Chapter 5 External Dimensions].



Dimensions for Screw Hole Positions



Oblong Hole

Type	Screw Hole Size	Screw Holes Depth	Reamed Hole (mm)	Oblong Hole
SA4C SA4R	M4	Through (screwing depth should be 6mm or less)	$\phi 3H7$ Depth 4	A: $3^{+0.010}_0$ B: 4 Depth 4mm or less
SA6C SA6R	M5	Through (screwing depth should be 10mm or less)	$\phi 4H7$ Depth 5.5	A: $4^{+0.012}_0$ B: 5 Depth 5.5mm or less
SA7C SA7R	M5	Through (screwing depth should be 10mm or less)	$\phi 4H7$ Depth 6	A: $4^{+0.012}_0$ B: 5 Depth 6mm or less
SA8C SA8R	M6	Through (screwing depth should be 12mm or less)	$\phi 6H7$ Depth 6.5	A: $6^{+0.012}_0$ B: 7 Depth 6.5mm or less

[Tightening Torque]

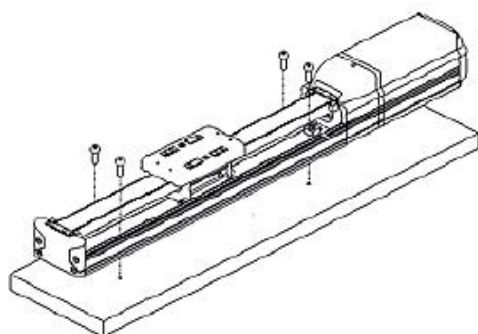
Type	Screw Hole Size	Tightening Torque	Reamed Hole (mm)
		In the case that steel is used for the bolt seating surface:	In the case that aluminum is used for the bolt seating surface:
SA4C SA4R	M4	3.59N•m	1.76N•m
SA6C SA6R	M5	7.27N•m	3.42N•m
SA7C SA7R	M5	7.27N•m	3.42N•m
SA8C SA8R	M6	12.3N•m	5.4N•m

**Notice**

- The use of high-strength bolts of ISO-10.9 or higher is recommended.
- Make sure the internal thread and bolt effective engagement length is approximately 1.8 times the nominal diameter or more.

[Using the Through Holes on the Top of the Base]

There are through holes for attachment on the base for SA4, SA6 and SA7 that allow fixation from the top when the side covers on both ends are taken off.



For mounting bolts, use hexagonal socket head bolts indicated in the following table depending on the platform material.

Type	Through Hole	Mounting Bolt	Tightening Torque
SA4C SA4R	φ3.4 drilled hole, 6.5 counter boring depth 3.5	M3	0.83N•m
SA6C SA6R	φ4.5 drilled hole, 8 counter boring depth 4.5	M4	1.76N•m
SA7C SA7R	φ6 drilled hole, 9.5 counter boring depth 5.5	M5	3.42N•m

Detaching Side Cover

- Remove the 4 pieces of attachment screws with using a Philips screwdriver.
- The slider cannot be driven only with ROBO Cylinder itself if it is equipped with a brake. Connect a controller and have JOG operation to move the slider to perform installation.



Caution

- Attempt not to drop the bolts, tools, etc. on the stainless steel sheet when tightening the bolts. Please avoid making a dent mark or scratches.



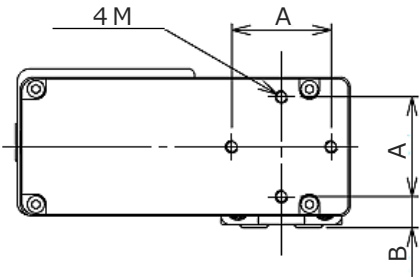
Notice

- The use of high-strength bolts of ISO-10.9 or higher is recommended.
- Make sure the internal thread and bolt effective engagement length is approximately 1.8 times the nominal diameter or more.

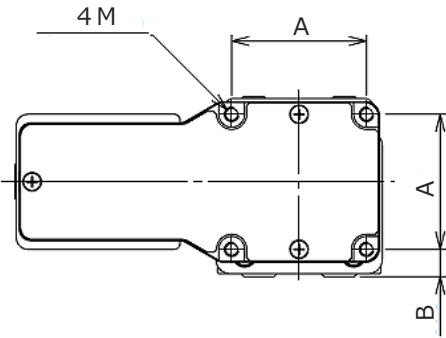
[When Using Attachment Holes on Bracket in Motor Reversing Type]

As there are screw holes on the reversing bracket, it is available to screw in from the back. It should be an option (Model: RP) for SA4R.

- For SA4R (Option)

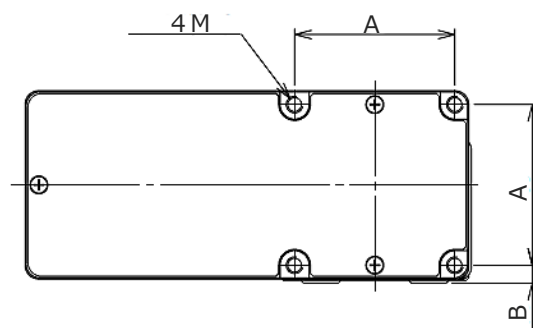


- For SA6R



Type	A	B	Attachment Hole M	Tightening Torque
SA4R	32	10	M4 Depth 8	1.76N•m
SA6R	47	9.5	M5 Depth 10	3.42N•m

- For SA7R and SA8R



Type	A	B	Attachment Hole M	Tightening Torque
SA7R	58	6.5	M6 Depth 12	5.4N•m
SA8R	58	8	M6 Depth 12	5.4N•m

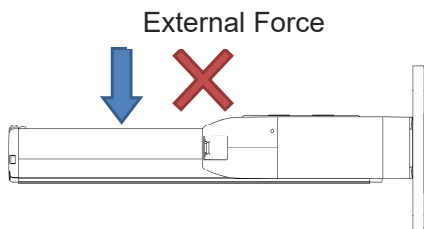


Notice

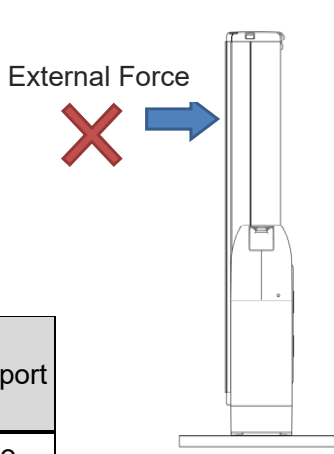
- The use of high-strength bolts of ISO-10.9 or higher is recommended.
- Make sure the internal thread and bolt effective engagement length is approximately 1.8 times the nominal diameter or more.

[Precautions when using reversing brackets]

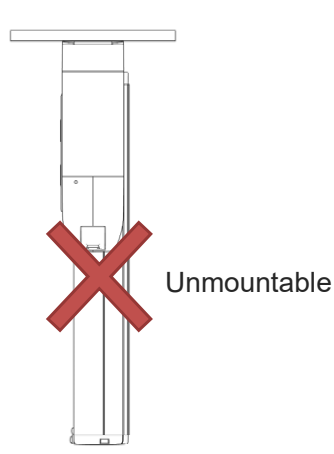
- Availability of installation for each installation posture is as shown below:



Horizontal



Vertical



Ceiling Mount

Installation Posture			Support
Horizontal	Vertical	Ceiling Mount	
×	×	×	No
○	○	×	Yes

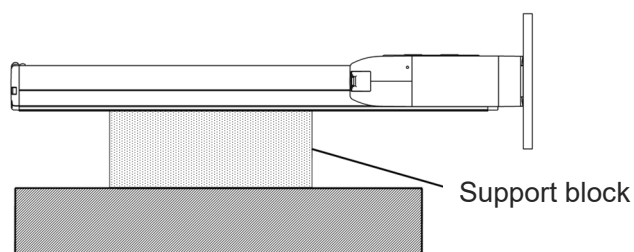


Caution

- Do not attempt to affix the unit only with the screw holes on the side-mounted bracket.
 - Do not apply external force to the actuator body after installation.
- External force may cause malfunctions or damage to parts.

[Precautions for horizontal mounting using reversing brackets]

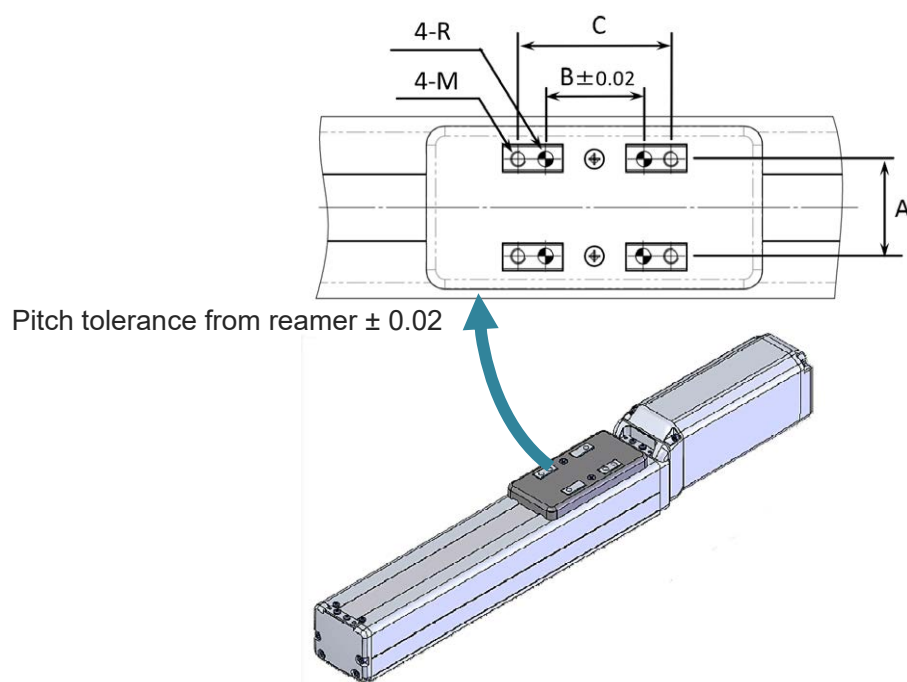
- Prepare a support block for the body, as shown in the figure below.



2.3.5 Mounting transported objects

[For SA4C/R, SA6C/R, SA7C/R]

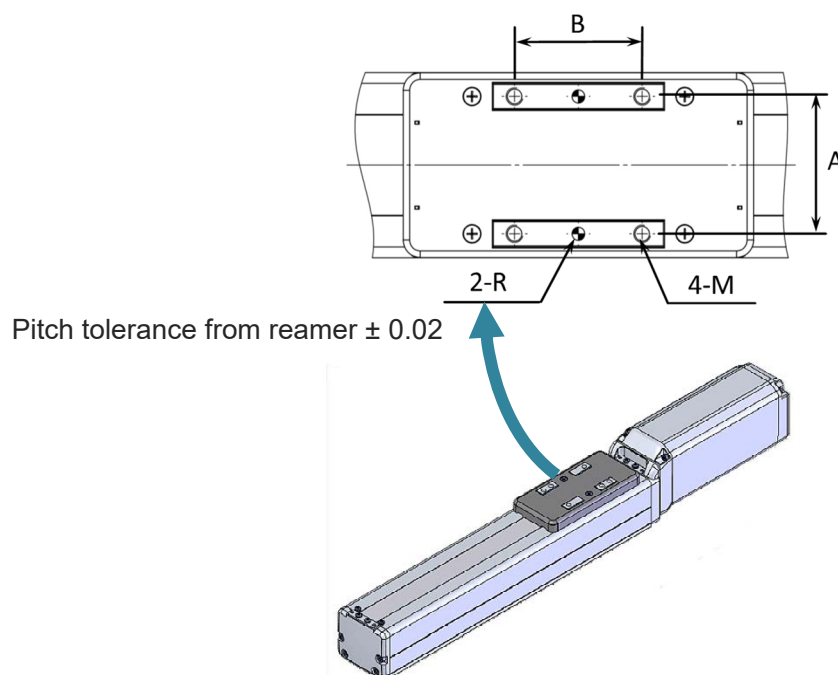
- Use the screw mounting holes on the top surface of the slider to fix transported objects.
- There are 4 reamed holes on the top surface of the slider.
Use these reamed holes if repeated attaching and detaching is required.
- If fine-tuning of perpendicularity, etc. is required, use one of the reamed holes for this adjustment.
- Refer to the figure below for the screw-in depth and reamed hole depth.



Type	A	B	C	Reamed Hole R	Screw Hole M	Mounting Bolt	
						Bolt Nominal Diameter	Tightening Torque
SA4C, SA4R	20	24	32	$\phi 3H7$ Depth 6	M3 Depth 7	M3	0.83N•m
SA6C, SA6R	31	32	50	$\phi 5H7$ Depth 6	M5 Depth 10	M5	3.42N•m
SA7C, SA7R	39	42	65	$\phi 5H7$ Depth 10	M5 Depth 10	M5	3.42N•m

[For SA8C/R]

- Use the screw mounting holes on the top surface of the slider to fix transported objects.
- There are 2 reamed holes on the top surface of the slider.
Use these reamed holes if repeated attaching and detaching is required.
- If fine-tuning of perpendicularity, etc. is required, use one of the reamed holes for this adjustment.
- Refer to the figure below for the screw-in depth and reamed hole depth.



Type	A	B	C	Reamed Hole R	Screw Hole M	Mounting Bolt	
						Bolt Nominal Diameter	Tightening Torque
SA8C, SA8R	63	60		$\phi 6H7$ Depth 10	M8 Through (screwing depth should be 12mm max.)	M8	11.5N•m

**Caution**

- Be careful with regard to the length of the mounting bolt and positioning pin.
- The use of screw-in depth greater than that of the screw or reamed mounting holes may damage the screw hole or reduce the mounting strength of the transported object, leading to decreased accuracy or unexpected accidents.

**Notice**

- The mounting bolts are to be prepared by the customer.
- The use of high-strength bolts of ISO-10.9 or higher is recommended.

2.3.6 About Suction of the Cleanroom Type

- Cleanroom type actuators are able to exert performance corresponding to cleanroom class 10 (0.1μm, Fed.Std.209D) and class 2.5 equivalent (ISO 14644-1) by aspirating air from the vacuum joint.

The guidelines of the suction amounts for the rated speeds of each model are shown in the following table.

[Suction amount guideline]

Single slider specification

Type	Lead [mm]	Suction Amount NI/min (L/min)
SA4C	16	60 (65)
	10	40 (43)
	5	20 (22)
	2.5	10 (11)
SA6C	20	100 (108)
	12	70 (76)
	6	30 (32)
	3	15 (16)
SA7C	24	90 (97)
	16	70 (76)
	8	40 (43)
	4	30 (32)
SA8C	20	160 (173)
	10	80 (86)
	5	40 (43)

[Suction amount guideline]

Double slider specification (option model W)

Type	Lead [mm]	Suction Amount NI/min (L/min)
SA4C	10	60 (65)
	5	30 (32)
	2.5	20 (22)
SA6C	12	110 (119)
	6	60 (65)
	3	35 (38)
SA7C	16	100 (108)
	8	50 (54)
	4	40 (43)
SA8C	10	120 (130)
	5	50 (54)

**Notice**

- Please perform suction of air from the two vacuum joints located on the sides of the actuator according to the suction amounts (total amount of the two locations) shown in the table below.
Also, please manage the joints so that the suction amounts of the two locations are equivalent.
- Please use the actuator in a state in which all of the base attachment holes on the bottom of the actuator are covered. When there is a hole that passes through the actuator, the level of cleanliness decreases.

ROBO Cylinder

Chapter 3

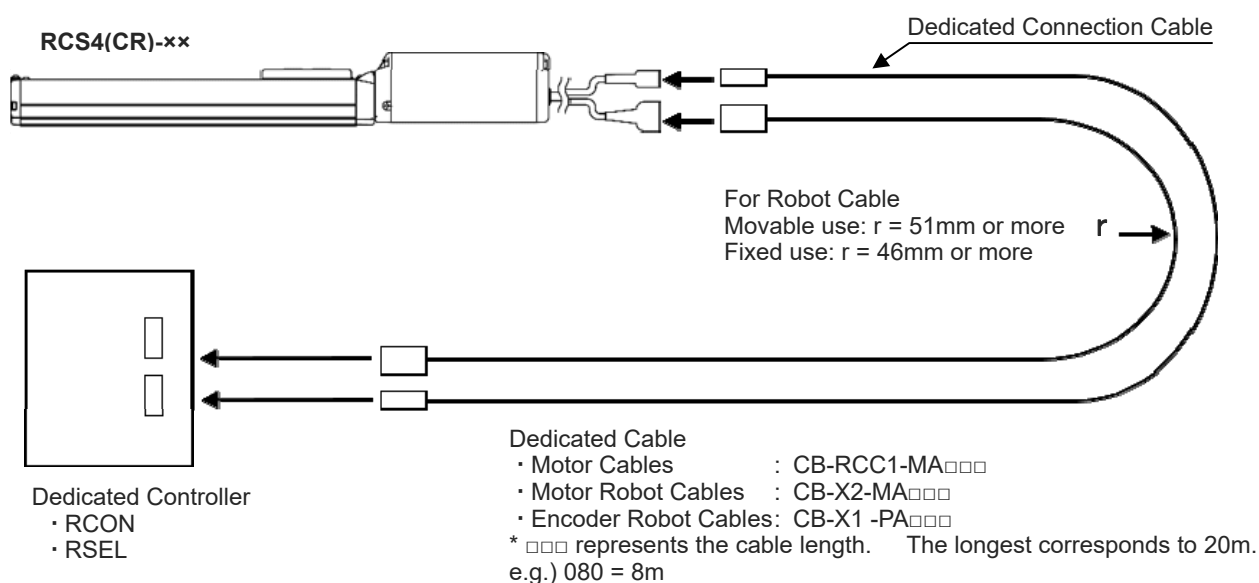
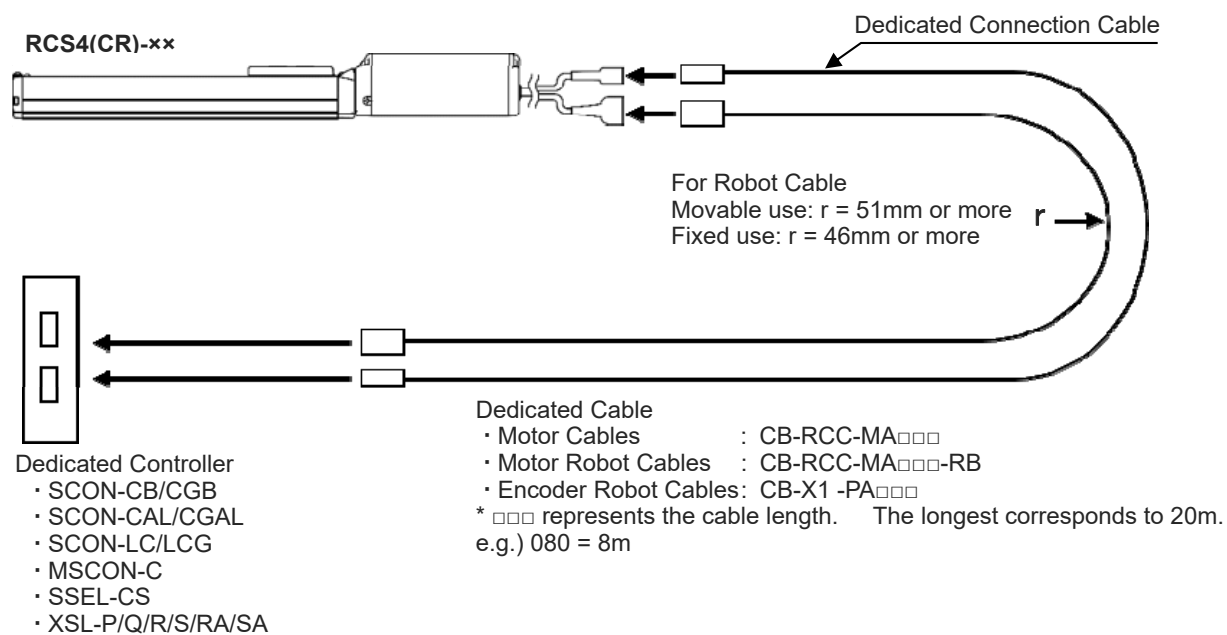
Connecting with the Controller

3.1	Connecting with the Controller.....	3-1
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3.1 Connecting with the Controller

As the connection cable for the controller and the actuator, use the IAI-dedicated connection cable.

Please consult with IAI if you require a different kind of cable than the one supplied.



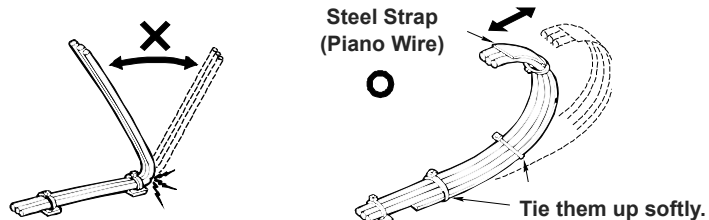
**Caution**

- For wiring, please follow the warnings stated below. When constructing a system as the machinery equipment, pay attention to the wiring and connection of each cable so they are conducted properly. Not following them may cause not only a malfunction such as cable breakage or connection failure, or an operation error, but also electric shock or electric leakage, or may even cause a fire.
 - Use dedicated cables of IAI indicated in this instruction manual. Contact us if you wish to have a change to the specifications of the dedicated cables.
 - Make sure to turn the power off in the process of power line or cable connection or disconnection.
 - Do not attempt to cut a dedicated cable with connectors on both ends to extend, shorten or re-joint it.
 - Hold the dedicated cable to avoid mechanical force being applied to the terminals and connectors.
 - Use a cable pipe or duct to have an appropriate protection when there is a possibility of mechanical damage on a dedicated cable.
 - In case a dedicated cable is to be used at a moving part, make sure to lay out the cable without applying any force to pull the connector or extreme bend on the cable. Do not attempt to use the cable with a bending radius below the allowable value.
 - Make certain that the connectors are plugged properly. Insufficient connection may cause an operation error, thus it is extremely risky.
 - Do not lay out the cables to where the machine runs over them.
 - Pay attention to the cable layout so it would not hit peripherals during an operation. In case it does, have an appropriate protection such as a cable track.
 - When a cable is used hanging on the ceiling, prevent an environment that the cable swings with acceleration or wind velocity.
 - Make sure there is not too much friction inside the cable storage equipment.
 - Do not apply radiated heat to power line or cables.
-

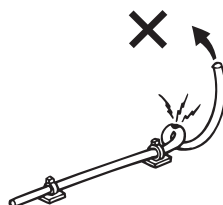


Caution

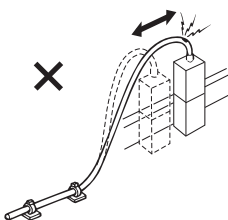
- Have a sufficient radius for bending, and avoid a bend concentrating on one point.



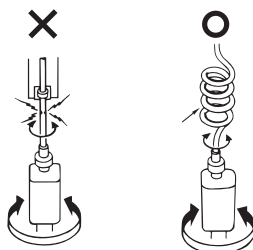
- Do not let the cable bend, kink or twist.



- Do not pull the cable with a strong force.



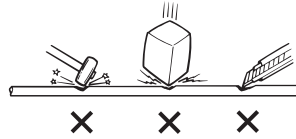
- Pay attention not to concentrate the twisting force to one point on a cable.



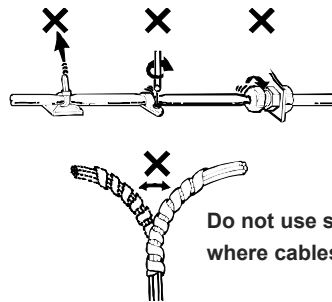


Caution

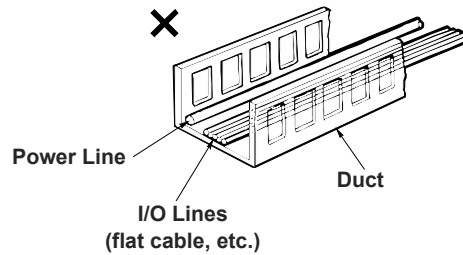
- Do not pinch, drop a heavy object onto or cut the cable.



- When a cable is fastened to affix, make sure to have an appropriate force and do not tighten too much.



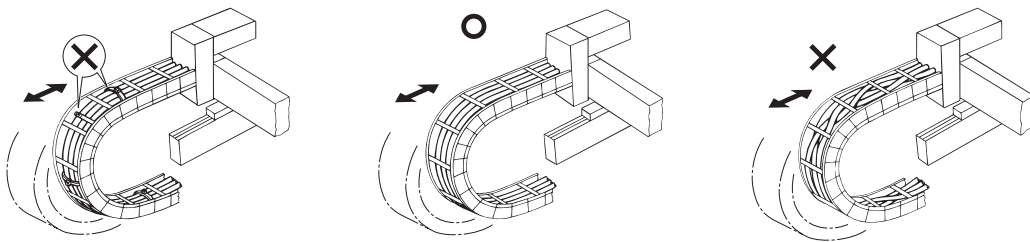
- PIO line, communication line, power and driving lines are to be put separately from each other and do not tie them together. Arrange so that such lines are independently routed in the duct.





Caution

- Follow the instructions below when using a cable track.
- If there is an indication to the cable for the space factor in a cable track, refer to the wiring instruction given by the supplier when storing the cable in the cable track.
- Avoid the cables to get twined or twisted in the cable track, and also to have the cables move freely and do not tie them up. (Avoid tension being applied when the cables are bent.)
- Do not pile up cables. It may cause faster abrasion of the sheaths or cable breakage.



ROBO Cylinder

Chapter 4

Maintenance and Inspection

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4.1 Precautions for maintenance and inspection work

Make sure to read the following precautions before conducting any maintenance or inspection work.



Caution

- Do not climb on or put anything on the actuator.
Otherwise, this may lead to accidental falling, injury or damage to the product due to falling objects, product loss of function or performance degradation, or shortening of product life.
- Before releasing the brake, make sure to check there is nothing that will interfere with moving parts within the operation range.
- The slider may fall, possibly injuring the operator or people nearby and damaging the actuator, workpiece or equipment.
- Check that the power to the actuator is OFF before conducting any maintenance or inspections.
- Be careful not to lose the cover or any removed screws.
Be sure to return the product to the original condition after maintenance and inspection work.
Mounting in an incomplete state may cause injuries or damage to the product.
- Do not modify, disassemble/assemble, or use maintenance parts not specified on your own discretion under any circumstances.
- The grease film may run out if the actuator performs return operation continuously over a distance of 30mm or less. As a guideline, every 5,000 to 10,000 cycles, have approximately 5 cycles of return operation over a 50mm distance or more to regenerate the oil film.
The ball screw or guide may be damaged if the oil film runs out.

**Notice**

- First, be sure to wipe off the old grease, and then supply new grease.
- The degradation speed of grease may differ depending on the operating environment (temperature, humidity and ambient atmosphere).

It is recommended to shorten the grease supply period if the actuator is used under poor environmental conditions such as high temperatures, high humidity or dusty atmospheres.

- Also, it is recommended to improve the environmental conditions in case the grease changes color notably due to poor operating conditions.
- Base oil may separate from the grease due to the mounting orientation or operating conditions.

Base oil may also leak from the inside of the actuator to the exterior through gaps.
Check visually for oil drips when supplying grease.

- An actuator stored for 6 months or more may suffer from grease degradation.
Supply grease before the start of use.
→For details, refer to [4.5 Greasing method].
-

4.2 Inspection schedule

Follow the maintenance inspection schedule below.

It is assumed that the equipment is operating 8 hours per day.

If the equipment is running continuously night and day or otherwise running at a high operating rate, inspect more often as needed.

4.2.1 Inspection items and schedule *Stainless steel sheet life guideline: 5,000km

Inspection period	External inspection	Internal inspection	Greasing	
			Ball screw	Guide
Start of work inspection	○	—	—	—
1-month inspection	○	—	—	—
3-month inspection	—	—	○	○
3-month inspection on	—	—	Grease supply timing (Guideline) dependent	
6-month inspection	○	○		
Every 6 months thereafter	○	○		

4.2.2 Grease supply timing (Guideline)

Maximum operating speed (mm/s)	Grease supply timing (Guideline)	
	Operation distance	Months
0 to 750 or less	1,250km	12 months
750 to 1500 or less	2,500km	
1500 to 2200	5,000km	

4.3 Visual inspection items

Refer to [4.6 How to replace components] for detailed information about specific component replacement and adjustment methods.

4.3.1 External visual inspection

Inspection items	Maintenance work
Is abnormal noise or vibration generated?	Take an action by referring to [Troubleshooting in Controller Instruction Manual].
Are actuator mounting bolts loose?	Tighten them further.
Is the cable scratched?	Replace if the damage is severe.
Is the connector loose?	Re-insert correctly.
Is grease dripping out? (especially if vertically mounted)	Clean up any drips. Replenish the grease.
Is the stainless steel sheet scratched?	Replace the stainless steel sheet.
Is the stainless steel sheet misaligned or slack?	Adjust the alignment and hang.

Refer to [4.6 How to replace components] for detailed information about specific component replacement and adjustment methods.

4.3.2 Internal visual inspection

Inspection items	Maintenance work
Backlash/play or wear debris found?	The ball screw guide may be damaged. Contact IAI.
Has foreign matter penetrated inside?	Remove the foreign matter, clean, and inspect the interior for any damage.
Is the ball screw or guide grease not lubricating well? (Even if the grease is brown, lubrication is adequate if the running surface is shiny)	Wipe away the old grease, then replenish with new grease.
Is dust or foreign matter adhered to the ball screw or guide?	Replenish with new grease after cleaning.

Internal inspection method

- 1) Loosen the screws on the side cover and detach the side cover.
- 2) Check inside.
- 3) After finishing the inspection, assemble back in the reverse order.

If you touch the edge of the stainless steel sheet in the attaching the side cover process, the sheet may get damaged or wavy which result in shortening life or earlier wear-out. To avoid touching the edge of the sheet, insert a spacer (approximately 0.1 to 0.2mm) between the sheet and cover to push up the sheet, and then push in the cover. When affixing the side cover, tighten the screws with the tightening torque described below.

Screw Diameter	Tightening Torque
M3	0.41N·m

- 4) Make sure that the slider and the side cover do not interfere, moving the slider backwards and forwards between the strokes.



Caution

- Do not damage the stainless steel sheet by bending it forcefully during internal inspection. Do not tug on the stainless steel sheet or in any way attempt to reposition it. Change in the attached condition may cause the sheet to be mounted unevenly or impact the product life. Keep in mind that the edges of the stainless steel sheet can cause injuries. Always wear gloves when working on it.

4.4 Cleaning

4.4.1 External cleaning

- Clean exterior surfaces as necessary.
- Use a soft cloth to wipe away dirt and buildup.
- Do not blow too hard with compressed air as it may cause dust to get in through the gaps.
- Do not use oil-based solvents as they can harm lacquered and painted surfaces.
- To remove severe buildup, wipe gently with a soft cloth soaked in a neutral detergent or alcohol.

4.4.2 Internal cleaning

- Use a soft cloth to wipe away dirt and buildup.
- Do not blow too hard with compressed air as it may cause dust to get in through the gaps.
- Do not use oil-based solvents, neutral detergent or alcohol.

4.5 Greasing method

[1] Grease used: Use an equivalent product

Type	Oil Supply Point	During Maintenance (recommended product)	Default (reference)
SA4: All Leads SA6: Lead 3/6/12/20 SA7: Lead 4/8/16/24	Ball screw	Kyodo Yushi / Multemp LRL No.3	Kyodo Yushi / Multemp LRL No.3
	Guide		
SA8: Lead 5/10/20/30	Ball screw	Kyodo Yushi / Multemp PS No.2	Kyodo Yushi / Multemp PS No.2
	Guide		
SA6: Lead 30 SA7: Lead 36 SA8: Lead 48	Ball screw	Kyodo Yushi / Multemp ET-R No.2	Kyodo Yushi / Multemp ET-R No.2
	Guide		
Cleanroom	Ball screw	Kuroda Precision Industries / Kuroda C Grease	Kuroda Precision Industries / Kuroda C Grease
	Guide		



Caution

[Standard Type]

- Do not attempt to apply any grease other than lithium grease.
Mixing with grease types other than lithium grease not only reduces the performance of the grease, but it may even cause damage to the actuator.

[Cleanroom Type]

- For grease to be used with an actuator that has a cleanroom setting, please use a type of grease that emits a low amount of dust and is described for cleanroom use. When standard grease is used, dust may be emitted.
- Do not attempt to apply any grease other than urea based grease.
Mixing with grease types other than urea based grease not only reduces the performance of the grease, but it may even cause damage to the actuator.

[2] Greasing method: Ball screw/guide (supply to both through grease fitting)

◆ Preparation

Type	Amount of Grease Supply (Reference)	Grease Nipple Diameter
SA4	0.5cc to 1.0cc	φ3.5
SA6	1.5cc to 2.0cc	
SA7	2.0cc to 2.5cc	
SA8	3.5cc to 4.0cc	

Grease Gun	Nozzle	Supplier of nozzle
Grease gun of mounting screw R1/8 (Example) GC-57K (Yamada Corporation)	NZ3	NSK

◆ Greasing method

- 1) Remove the screw and detach the side cover on either side, right or left.

The grease nipple will appear.

(Note) When the side cover on the other side has been detached, the other grease nipple also appears.

Side Cover Attachment Screws



Opposite Side

Side Cover Attachment Screws



Motor Side



Side Cover

After removal of the side cover on either side



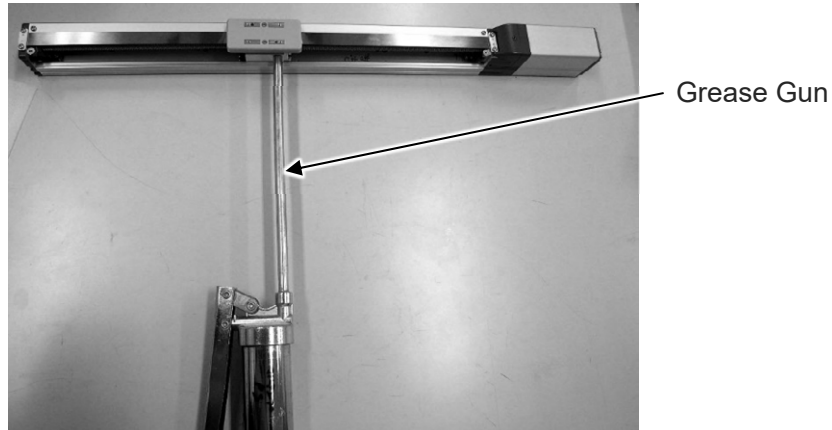
Grease Nipple
(There is one
on the other side.)

- 2) Supply grease from the grease nipple on either side, using the grease gun.

Wipe off the grease before supplying new in case it is extremely dirty.

(Note) Grease can also be applied from the grease nipple on the other side.

Grease can be supplied on either of the two grease nipples to the ball screw and the guides on both sides.



Move the slider back and forth in its full stroke after the oil supply so the oil can be spread evenly.

- Confirm that the ball tracks on the ball screw and guide look glossy with oil or grease.
Supply grease again if it is not spread enough.
- Wipe off excess grease.

- 3) Attach the side covers after grease supply is finished.

If you touch the edge of the stainless steel sheet in the attaching process, the sheet may get damaged or wavy which result in shortening life or earlier wear-out.

To avoid touching the edge of the sheet, insert a spacer (approximately 0.1 to 0.2mm) between the sheet and cover to push up the sheet, and then push in the cover.

When affixing the side cover, tighten the screws with the tightening torque described below.

Screw Diameter	Tightening Torque
M3	0.41N·m

- 4) Move the slider back and forth in its full stroke to make sure there is no interference of the slider to the side covers.



Caution

- Supplying too much grease may increase sliding resistance and load to the motor, resulting in a drop of performance.

Also, excess grease on the ball screw may be splashed around in the ambience.

- Do not damage the stainless steel sheet by bending it forcefully during work.
- Keep in mind that the edges of the stainless steel sheet can cause injuries. Always wear gloves when working on it.
- The front bracket is supporting the ball screw. Do not detach it.
If the front bracket loses its tuned condition, it may cause an increase of driving resistance, shortened life of each component or abnormal noise due to the center of the axis being off the right position.
- In case the grease got into your eye, wash it with clean water for 15 minutes and immediately go see the doctor to get appropriate care.
- After finishing the grease supply work, wash your hands carefully with water and soap to rinse the grease OFF.

4.6 How to replace components

4.6.1 Stainless steel sheet replacement/adjustment

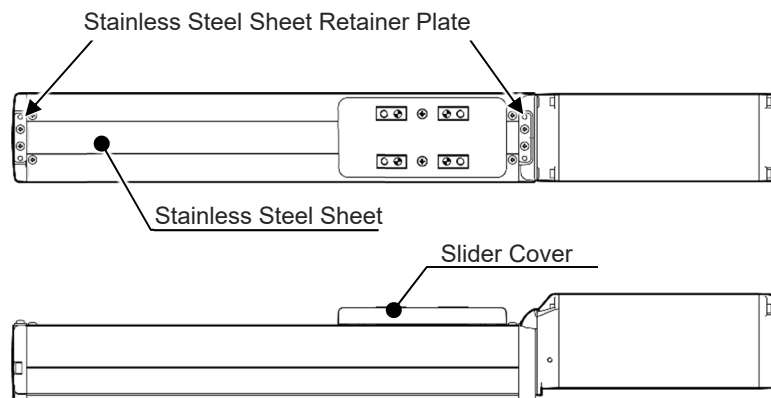
◆ Preparation

(1) Items required for replacing

- Phillips screwdriver
- Scale
- Adhesive tape
- Replacement stainless steel sheet

Type	Stainless Steel Sheet Model Code
RCS4-SA4C/SA4R, RCS4CR-SA4C	ST-6A4- (Stroke)
RCS4-SA6C/SA6R, RCS4CR-SA6C	ST-6A6- (Stroke)
RCS4-SA7C/SA7R, RCS4CR-SA7C	ST-6A7- (Stroke)
RCS4-SA8C/SA8R, RCS4CR-SA8C	ST-6A8- (Stroke)

◆ Names of the Parts



Caution

- Degradation and wear-out of the stainless steel sheet relies on its tension of attachment. If the stainless steel sheet is pulled with a huge force and the gap between the sheet and the slider cover is large, there is a risk of metal fatigue. On the other hand, if the tensile strength is too low, the stainless steel sheet would interfere with the back side of the slider cover and make dust to be generated.

For SA7 and SA8 have a replacement and tuning in Procedure 1.

For SA4 and SA6 have a replacement and tuning in either Procedure 1 or Procedure 2.

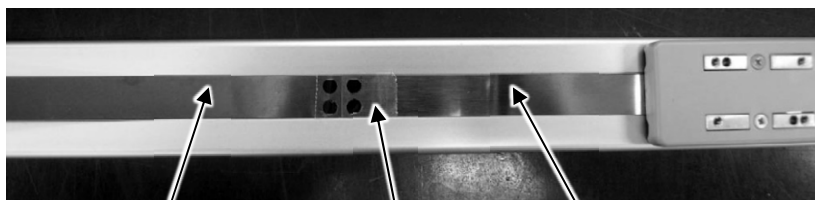
◆ Stainless steel sheet replacement method Procedure 1

- 1) Remove the screws (4 places) holding the stainless steel sheet and stainless steel sheet retainer plates (2 pieces) with using a Phillips screwdriver.
- 2) Pull the old stainless steel sheet till in front of the slider.
Make sure not to pull it out completely off the slider.



In front of Slider

- 3) Join a new stainless steel sheet to the old one with using adhesive tape.



New Stainless Steel Sheet

Old Stainless Steel Sheet

Join with Adhesive Tape

- 4) Pull the old stainless steel sheet so the new one goes through below the slider.
Pull it till the new stainless steel sheet comes to the end of the actuator.



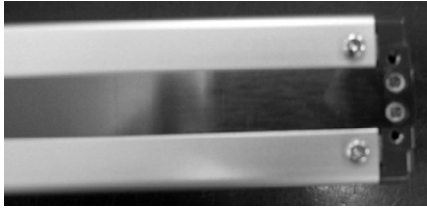
New Stainless Steel Sheet

Old Stainless Steel Sheet



End of Actuator

- 5) Remove the old stainless steel sheet.



- 6) Fix the two stainless steel sheet retainer plates using the screws (four locations).
Use a Phillips screwdriver to attach it.



There are two grooves for stainless steel sheet attachment guides. Align so that the stainless steel sheet is positioned at the center between the two grooves. Attach the stainless steel sheet so that it is not bent.

Tightening Torque

0.41N·m

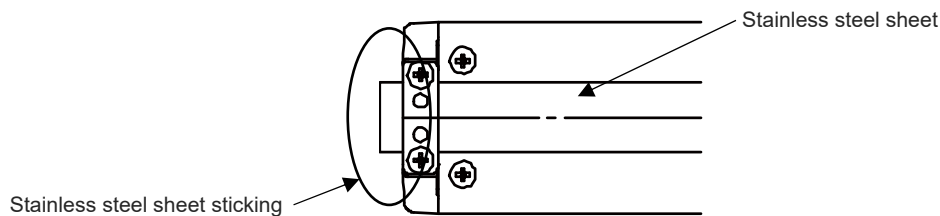
- 7) After the stainless steel sheet retainer plates have been fixed, move the slider from one side to the other side (full stroke) with your hand to check that there is no rise or slackness in the stainless steel sheet.
If there is any problem, repeat from step 6).



Caution

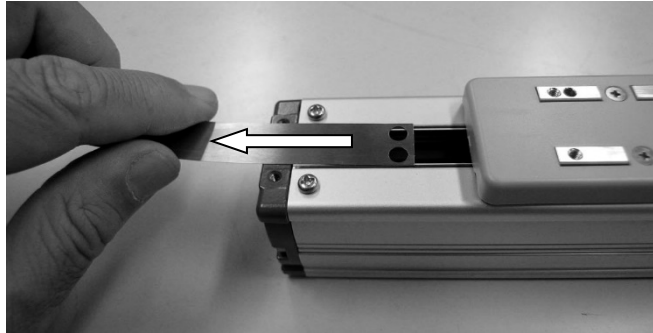
- Ensure that the stainless steel sheet does not stick out of the end of the actuator when it is mounted.

It may lead to a risk of cutting your finger on the stainless steel sheet.

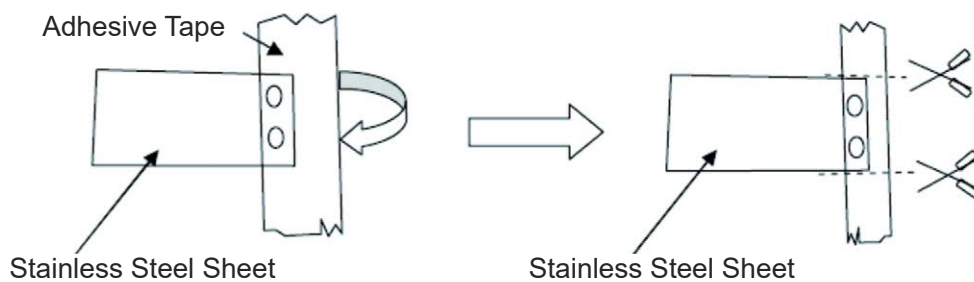


◆ Stainless steel sheet replacement method Procedure 2

- 1) Remove the screws (4 places) holding the stainless steel sheet and stainless steel sheet retainer plates (2 pieces) with using a Phillips screwdriver.
- 2) Pull out the old stainless steel sheet.

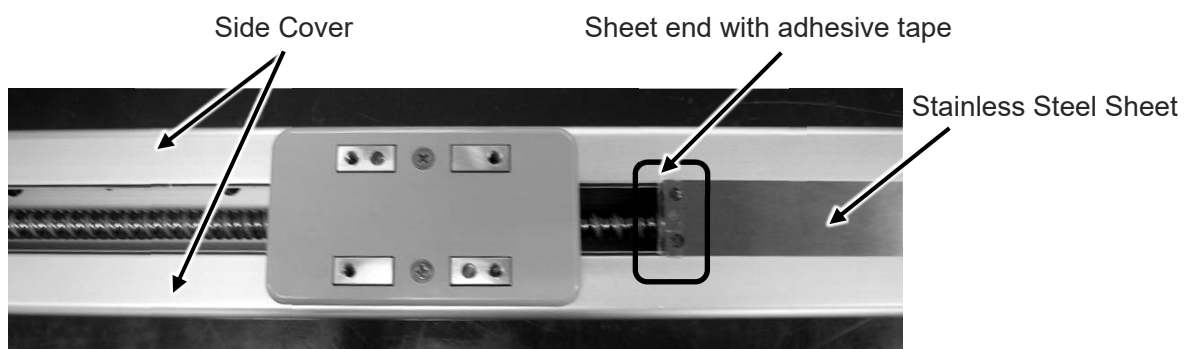


- 3) Attach adhesive tape on one side of new stainless steel sheet.



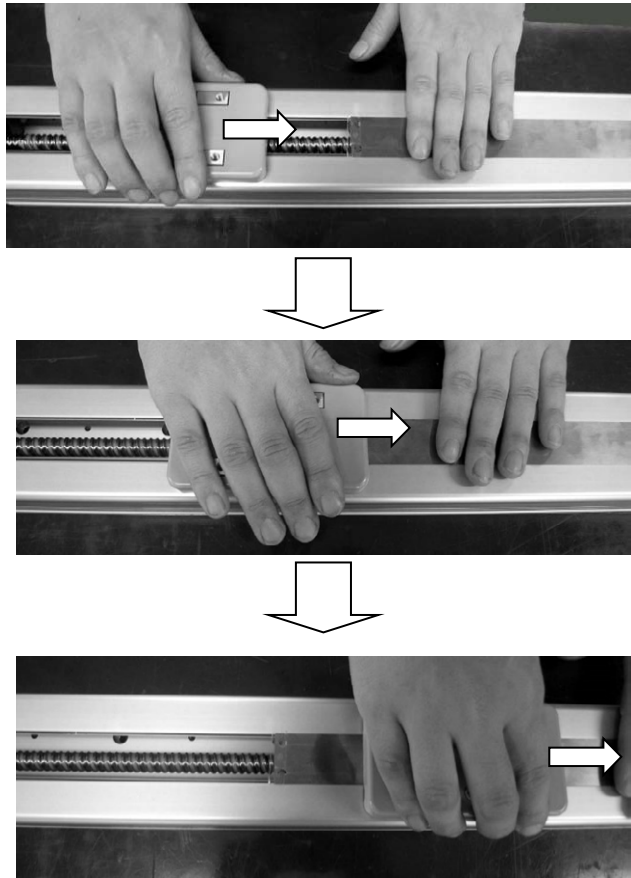
Attach adhesive tape so that it wraps around the end side of the stainless steel sheet. At that time, arrange it so that about 3mm of the end of the tape projects from the stainless steel sheet. Cut any tape excess.

- 4) Put the stainless steel sheet with an adhesive tape on the side cover.

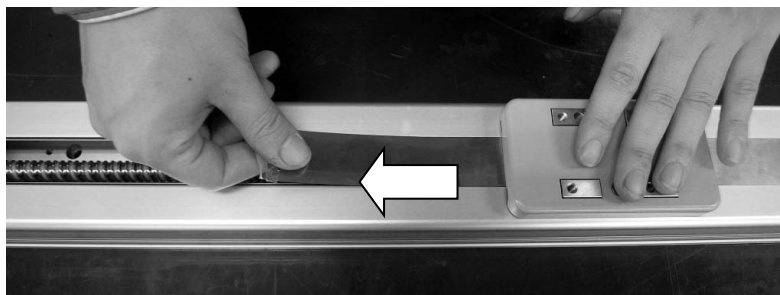


Take care not to drop the stainless steel sheet into the hole.

- 5) With the stainless steel sheet being held, move the slider to put through the stainless steel sheet.



- 6) After the stainless steel sheet passes through the slider, pull one end of the stainless steel sheet out to the position of the stainless steel retainer plate. Take off the adhesive tape once positioning is finished.



- 7) Fix the stainless steel sheet retainer plates using the screws (four locations).

Use a Phillips screwdriver to attach it.



There are two grooves for stainless steel sheet attachment guides. Align so that the stainless steel sheet is positioned at the center between the two grooves. Attach the stainless steel sheet so that it is not bent.

Tightening Torque

0.41N·m

- 8) After the stainless steel sheet retainer plates have been fixed, move the slider from one side to the other side (full stroke) with your hand to check that there is no rise or slackness in the stainless steel sheet.

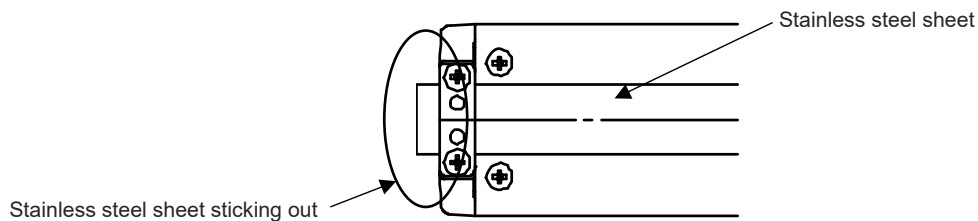
If there is any problem, repeat from step 7).



Caution

- Ensure that the stainless steel sheet does not stick out of the end of the actuator when it is mounted.

It may lead to a risk of cutting your finger on the stainless steel sheet.



- Make sure that there is no misalignment or slack in the stainless steel sheet. This may lead to damage to the stainless steel sheet.
- When installing the sheet clamp, do not apply excessive force to the stainless steel sheet.



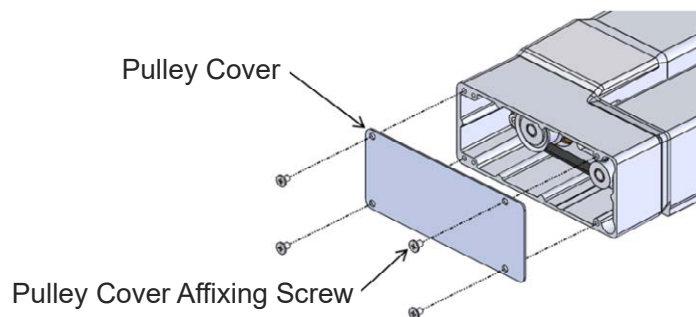
Notice

- As a guideline, the stainless steel sheet should last for about 5,000km of motion. However, depending on the usage or environment, replace the stainless steel sheet appropriately.
- It is available for customers to replace the stainless steel sheet. However, if you feel anxious about replacing it, it is recommended you ask us to do the replacement work for you. Either you bring in our product to our service office or our service staff visits your site to do the replacement work.

4.6.2 Timing belt replacement/adjustment

[Timing Belt Inspection]

- 1) Detach the pulley cover affixing screws and take off the pulley cover.



- 2) Check the condition of the timing belt visually.

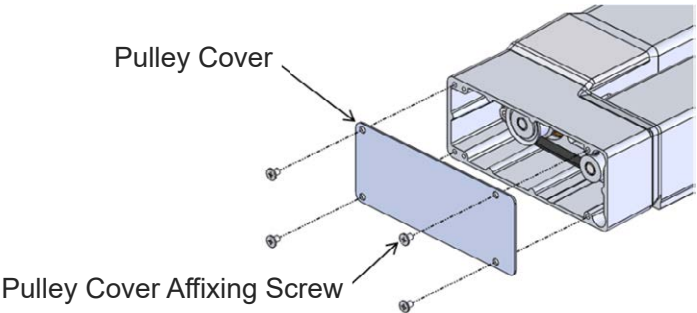
- In generally speaking, it possesses bending life of several million times. However, the period of replacement for the belt cannot be clearly defined as the durability of it is impacted so much by the operational conditions.
- The timing belt gets worn away as the time passes, and it is necessary to have replacement at regular intervals with the following conditions as reference.
 - When the gear and belt area show obvious friction.
 - When swelling occurs as a result of oil adhesion.
 - When damages such as a crack occurs on the belt gear and back side.
- For the toothed belt, it is recommended to set the interval of regular replacement cycle when in use under high wire fatigue condition in high acceleration and deceleration because it is difficult to judge the right timing for replacement by checking appearance or looseness of the wires strengthening the belt.

[Timing Belt Model Code]

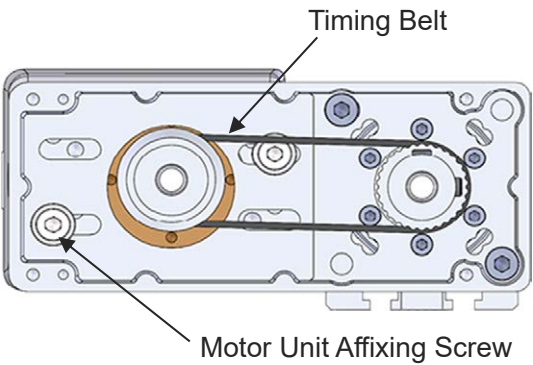
Type	Model Code
SA4R	TB-RCS4-STRA4R
SA6R	TB-RCS4-STRA6R
SA7R	TB-RCS4-STRA7R
SA8R	TB-RCS4-SA8R

[Timing Belt Replacement]

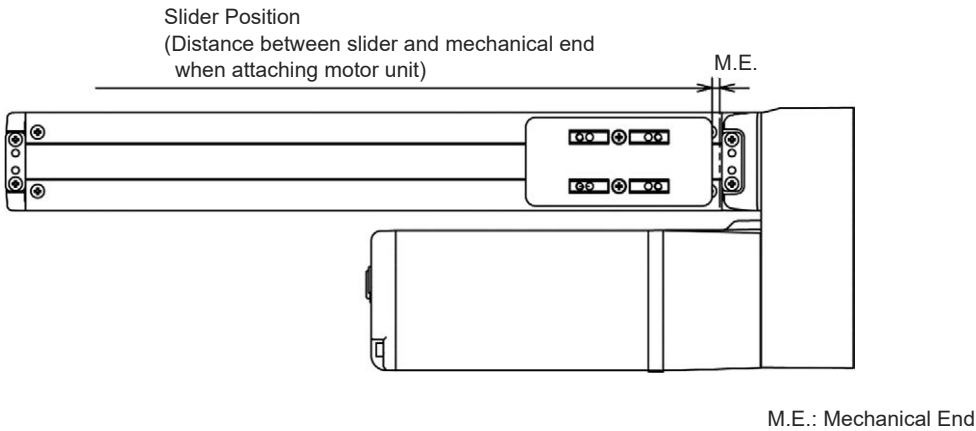
- 1) Detach the pulley cover affixing screws and take off the pulley cover.



- 2) Loosen the motor unit affixing screw and take off the timing belt.

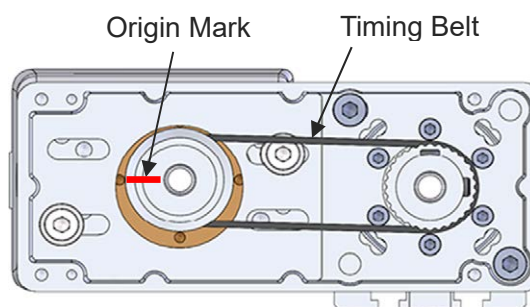


- 3) Keep the slider at the distance shown in the table below from the mechanical end.

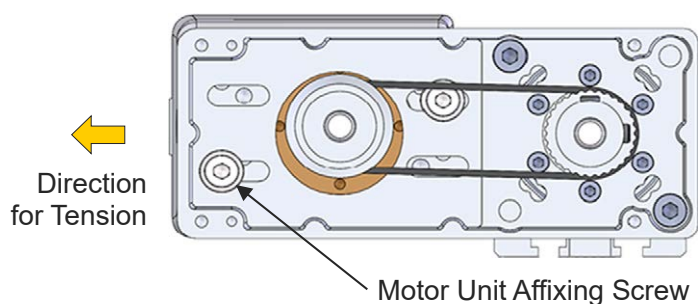


Slider Position	
Type	Slider Position (mm)
SA4R, SA6R, SA7R, SA8R	2

- 4) With the origin mark marked on the motor end pulley facing outwards, hang the belt on the pulleys.



- 5) Apply tension in the force shown in the table below to the motor unit, and tighten the motor unit affixing screw in the tightening torque shown in the table below.



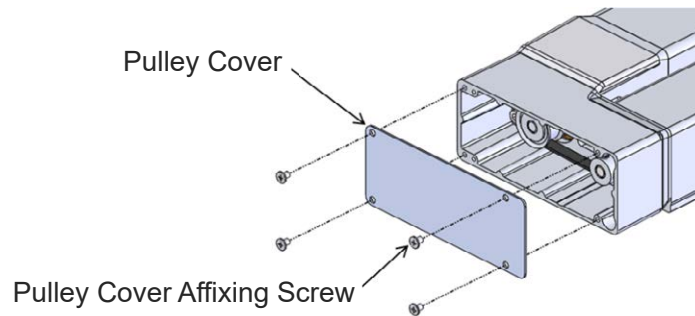
Tensile Force when Attaching Motor Unit

Type	Tension Force (N)
SA4R	20 to 25
SA6R	40 to 45
SA7R	70 to 80
SA8R	100 to 110

Motor Unit Affixing Screw Tightening Torque

Type	Tightening Torque (N·m)
SA4R, SA6R	3.2
SA7R, SA8R	6.3

- 6) Tighten up the pulley cover with the pulley cover affixing screws in the tightening torque shown in the table below.



Pulley Cover Affixing Screw Tightening Torque

Type	Type of Screws	Tightening Torque (N·m)
SA4R	Cross Recessed Slim-Head Screw (SUS) : M3	0.4
SA6R, SA7R, SA8R	Cross Recessed Flat-Head Screw (SUS) : M3	0.4

4.6.3 Motor unit replacement

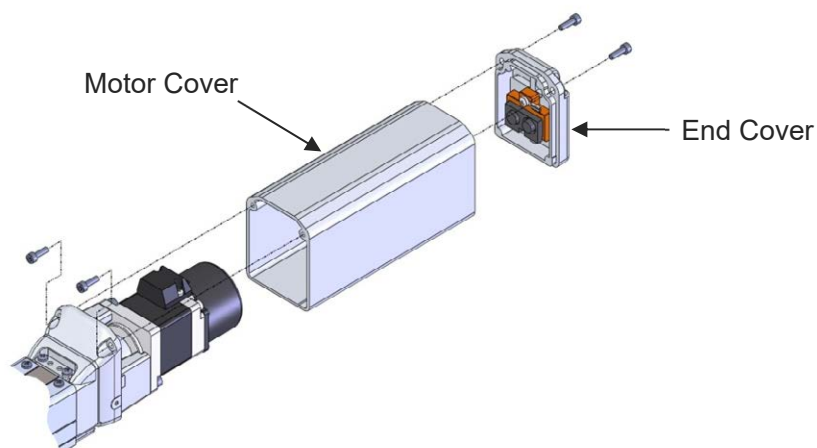
[Motor Straight Type]

Type	Motor Unit Model Code
RCS4-SA4C, RCS4CR-SA4C	RCS4-MUSRTA4C (-B)
RCS4-SA6C, RCS4CR-SA6C	RCS4-MUSRTA6C (-B)
RCS4-SA7C, RCS4CR-SA7C	RCS4-MUSRTA7C (-B)
RCS4-SA8C, RCS4CR-SA8C	RCS4-MUSRA8C (-B)

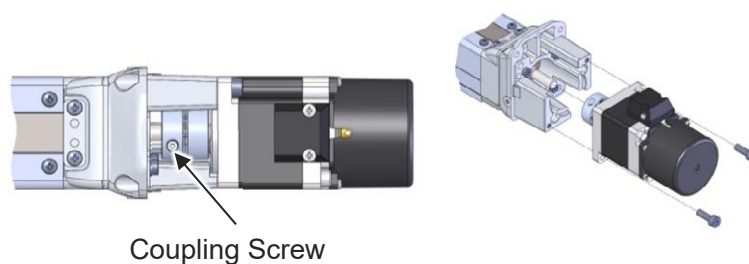
Selections when option in brackets () selected

B: Equipped with Brake

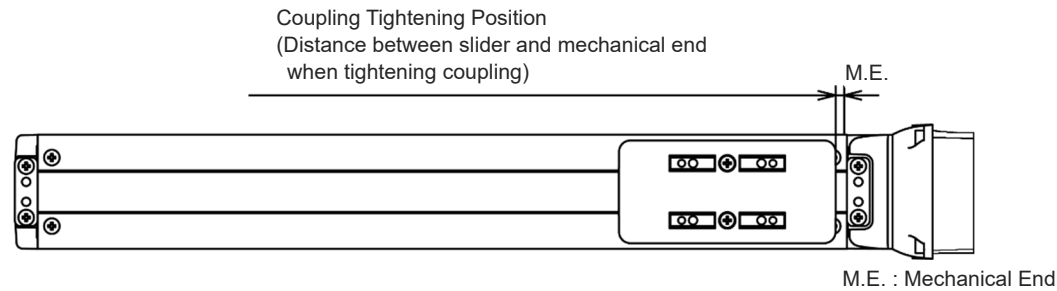
- 1) Detach the motor cover affixing screws.
- 2) Take off the end cover and motor cover.



- 3) Move the slider to the position where the coupling screw on the actuator side can be seen.
- 4) Loosen the coupling screw, detach the motor affixing screws and take off the motor unit.



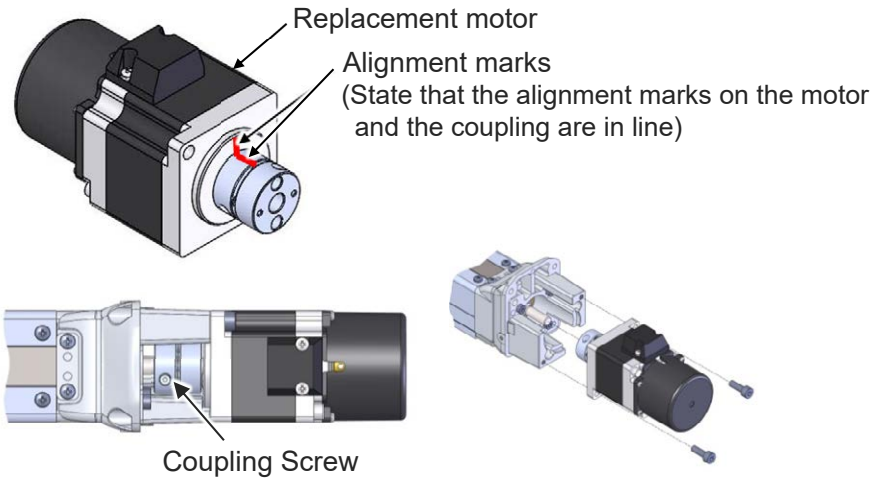
- 5) Keep the slider at the distance shown in the table below from the mechanical end.



Coupling Tightening Position

Type	Coupling Tightening Position (mm) (Distance between Slider and Mechanical End)
SA4C, SA6C, SA7C, SA8C	2

- 6) Confirm that the alignment marks on the motor and the coupling are in line with each other in the motor for replacement. Make the alignment lines in line if they are not. After confirming that the marks are aligned, loosely tighten the replacement motor with motor affixing bolts and screws, and then tighten up the coupling bolt with the torque shown in the table.



Coupling Screw Tightening Torque

Type	Tightening Torque (N·m)
SA4C	0.4
SA6C	0.9
SA7C, SA8C	1.5

- 7) Fully tighten the motor affixing screws in the tightening torque shown in the table.

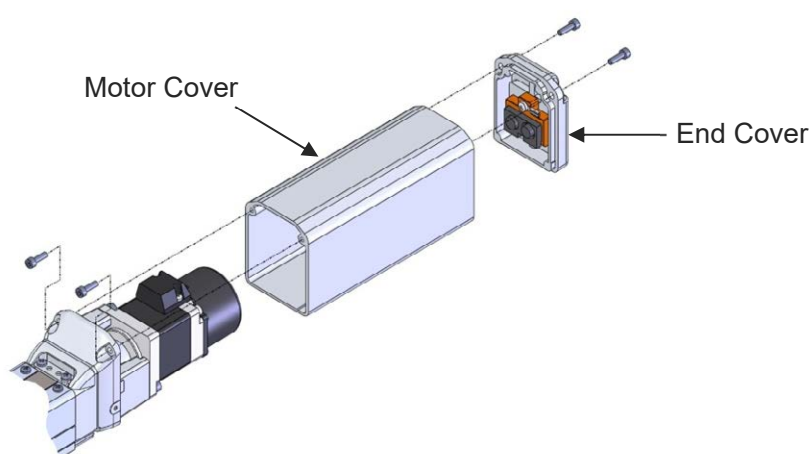
Motor Affixing Screw Tightening Torque

Type	Tightening Torque (N·m)
SA4C, SA6C	2.1
SA7C, SA8C	4.1

- 8) Attach the motor cover and end cover.

Tighten the motor cover affixing screws in the tightening torque shown in the table.

Pay attention not to get the cable pinched.



Motor Cover Affixing Screw Tightening Torque

Type	Type of Screws	Tightening Torque (N·m)
SA4C, SA6C	Hexagonal Socket Head Bolt : M3	0.9
SA7C, SA8C	Hexagonal Socket Head Bolt : M4	2.1

- * If there is a misalignment in the coupling tightened position (approx. 0.5mm) after the home-return operation, adjust the misalignment with the following parameters.

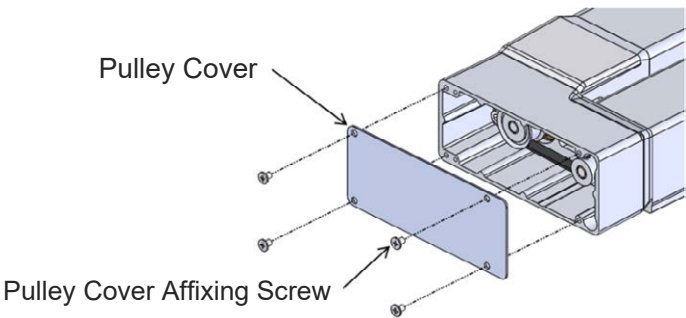
SCON	Parameter No. 22
MSCON	To be adjusted with offset movement amount
RCON-SC	(lead length or more not allowed)
RSEL	Driver Unit Parameter No. 22
	To be adjusted with offset movement amount
	(lead length or more not allowed)
XSEL	Axis-Specific Parameter No. 21
	To be adjusted with the home-return offset movement amount
	(lead length or more not allowed)

[Motor Reversing Type: When Replacing Motor Unit]

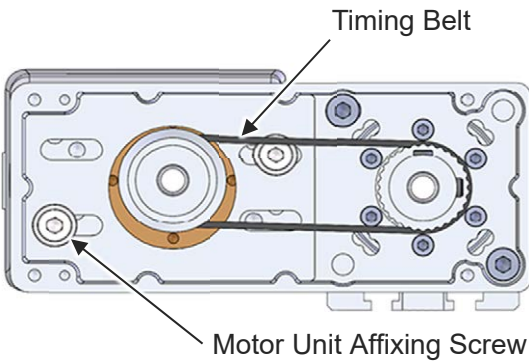
Type	Motor Unit Model Code
RCS4-SA4R	RCS4-MUSRTA4WSRA10R (-B) (-CJT/CJB/CJO) (-ML/MR) (-PU)
RCS4-SA6R	RCS4-MUSRTA6WSRA12R (-B) (-CJT/CJB/CJO) (-ML/MR) (-PU)
RCS4-SA7R	RCS4-MUSRTA7WSRA14R (-B) (-CJT/CJB/CJO) (-ML/MR) (-PU)
RCS4-SA8R	RCS4-MUSA8R (-B) (-CJT/CJB/CJO) (-ML/MR) (-PU)

Selections when option in brackets () selected
B: Equipped with Brake, CJT/CJB/CJO: Direction of Cable Ejection,
ML/MR: Direction of Motor Reversing * Mandatory to select, PU: Equipped with Pulley

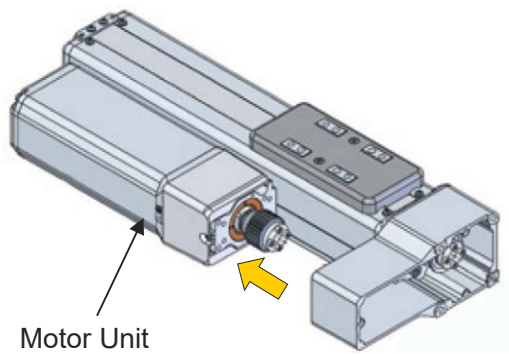
- 1) Detach the pulley cover affixing screws and take off the pulley cover.



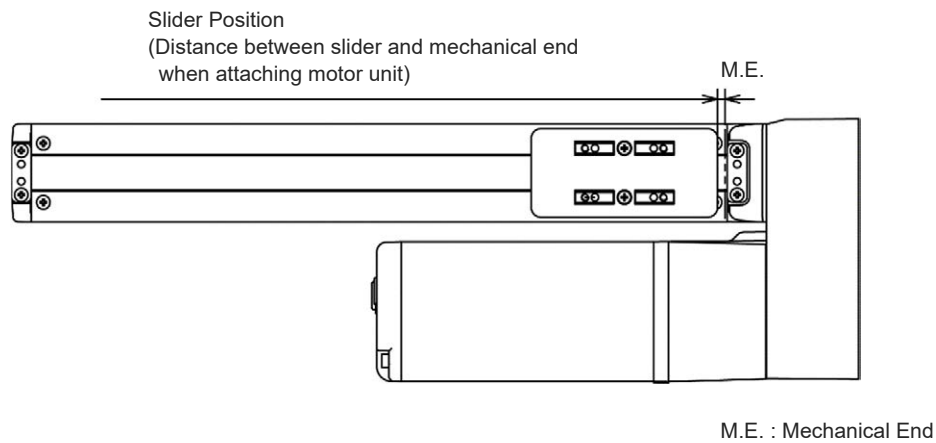
- 2) Detach the motor unit affixing screw and take off the timing belt.



- 3) Detach the motor unit.

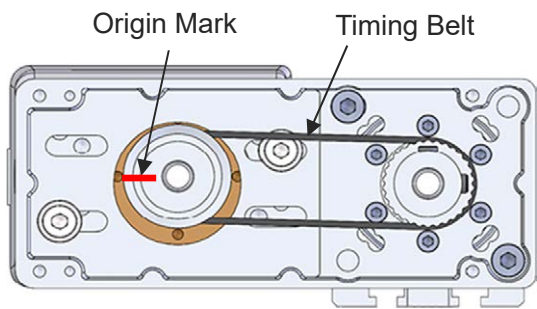


- 4) Attach the motor unit for replacement.
5) Keep the table at the distance shown in the table below from the mechanical end.

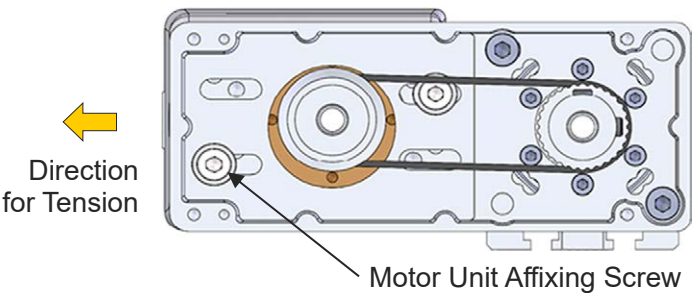


Slider Position	
Type	Slider Position (mm)
SA4R, SA6R, SA7R, SA8R	2

- 6) With the origin mark marked on the motor end pulley facing outwards, hang the timing belt on the pulleys.



- 7) Apply tension in the force shown in the table below to the motor unit, and tighten the motor unit affixing screw in the tightening torque shown in the table below.



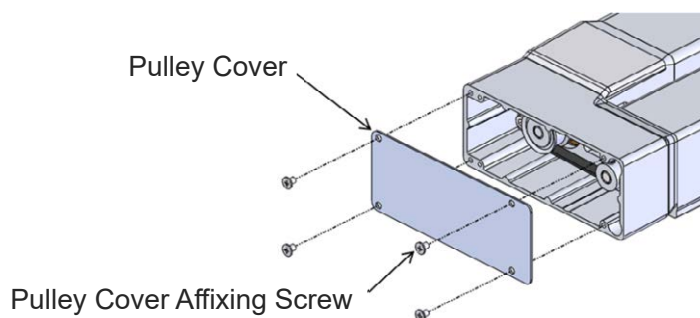
Tensile Force when Attaching Motor Unit

Type	Tension Force (N)
SA4R	20 to 25
SA6R	40 to 45
SA7R	70 to 80
SA8R	100 to 110

Motor Unit Affixing Screw Tightening Torque

Type	Tightening Torque (N·m)
SA4R, SA6R	3.2
SA7R, SA8R	6.3

- 8) Tighten up the pulley cover with the pulley cover affixing screws in the specified torque.



Pulley Cover Affixing Screw Tightening Torque

Type	Type of Screws	Tightening Torque (N·m)
SA4R	Cross Recessed Slim-Head Screw (SUS) : M3	0.4
SA6R, SA7R, SA8R	Cross Recessed Flat-Head Screw (SUS) : M3	0.4

- * If there is a misalignment in the coupling tightened position (approx. 0.5mm) after the home-return operation, adjust the misalignment with the following parameters.

SCON	Parameter No. 22
MSCON	To be adjusted with offset movement amount
RCON-SC	(lead length or more not allowed)
RSEL	Driver Unit Parameter No. 22
	To be adjusted with offset movement amount
	(lead length or more not allowed)
XSEL	Axis-Specific Parameter No. 21
	To be adjusted with the home-return offset movement amount
	(lead length or more not allowed)



Caution

- Make sure to hold the slider so it would not move in case of replacing a motor in vertical installation which is not equipped with a brake. It will be dangerous as the slider will be dropped, if it is not held, as soon as the motor gets taken off.



Notice

- Pay attention not to get the cable pinched when attaching the motor cover and end cover.
- When the actuator is not equipped with a brake, make sure that the motor is magnetized when attaching it so the shaft and the origin point get aligned.

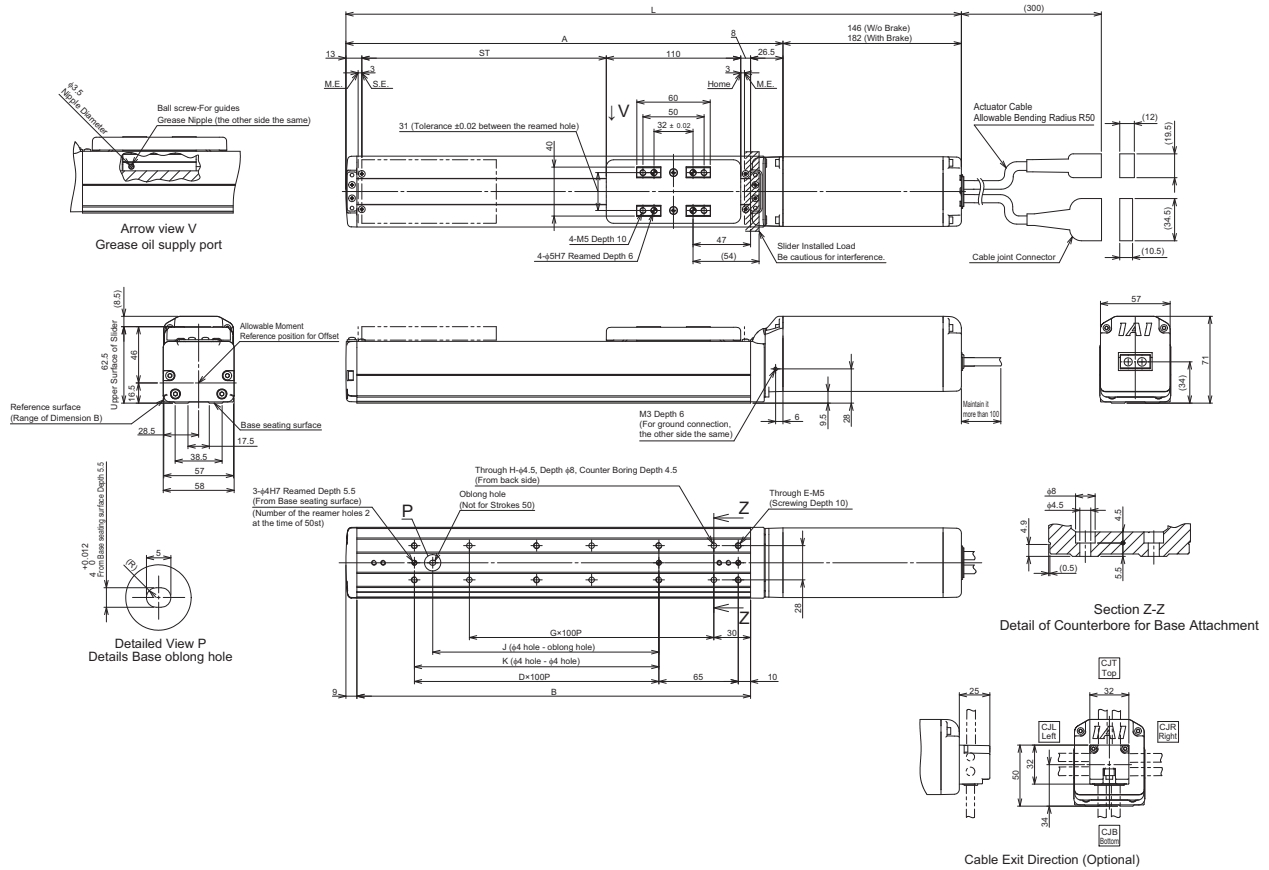
ROBO Cylinder

Chapter 5

External Dimensions

5.1	External Dimensions	5-1
5.1.1	RCS4-SA4C	5-1
5.1.2	RCS4-SA6C	5-2
5.1.3	RCS4-SA7C	5-3
5.1.4	RCS4-SA8C	5-4
5.1.5	RCS4-SA4R	5-5
5.1.6	RCS4-SA6R	5-7
5.1.7	RCS4-SA7R	5-8
5.1.8	RCS4-SA8R	5-10
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5.1.10	RCS4CR-SA6C	5-13
5.1.11	RCS4CR-SA7C	5-14
5.1.12	RCS4CR-SA8C	5-15

ST: Stroke, M.E.: Mechanical End, S.E.: Stroke End



■ Dimensions and Mass by Stroke

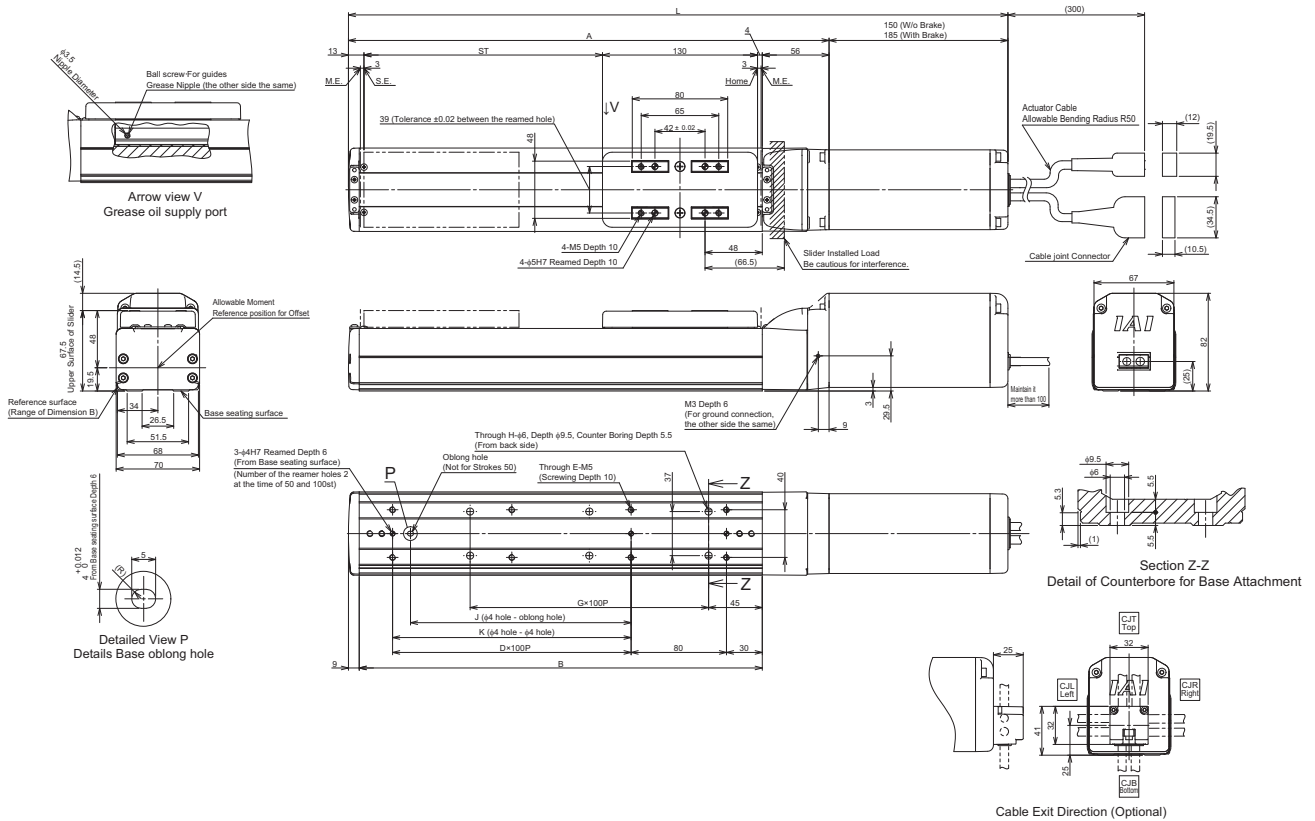
Unit: mm

Stroke	L		A	B	D	E	G	H	J	K	Mass [kg]	
	W/o Brake	With Brake									W/o Brake	With Brake
50	353.5	389.5	207.5	172	0	4	1	4	0	0	2.0	2.3
100	403.5	439.5	257.5	222	1	6	1	4	85	100	2.2	2.5
150	453.5	489.5	307.5	272	1	6	2	6	85	100	2.4	2.7
200	503.5	539.5	357.5	322	2	8	2	6	185	200	2.6	2.9
250	553.5	589.5	407.5	372	2	8	3	8	185	200	2.7	3.0
300	603.5	639.5	457.5	422	3	10	3	8	285	300	2.9	3.2
350	653.5	689.5	507.5	472	3	10	4	10	285	300	3.1	3.4
400	703.5	739.5	557.5	522	4	12	4	10	385	400	3.3	3.6
450	753.5	789.5	607.5	572	4	12	5	12	385	400	3.4	3.7
500	803.5	839.5	657.5	622	5	14	5	12	485	500	3.6	3.9
550	853.5	889.5	707.5	672	5	14	6	14	485	500	3.8	4.1
600	903.5	939.5	757.5	722	6	16	6	14	585	600	4.0	4.3
650	953.5	989.5	807.5	772	6	16	7	16	585	600	4.2	4.5
700	1003.5	1039.5	857.5	822	7	18	7	16	685	700	4.3	4.6
750	1053.5	1089.5	907.5	872	7	18	8	18	685	700	4.5	4.8
800	1103.5	1139.5	957.5	922	8	20	8	18	785	800	4.7	5.0

5.1 External Dimensions

5.1.3 RCS4-SA7C

ST: Stroke, M.E.: Mechanical End, S.E.: Stroke End

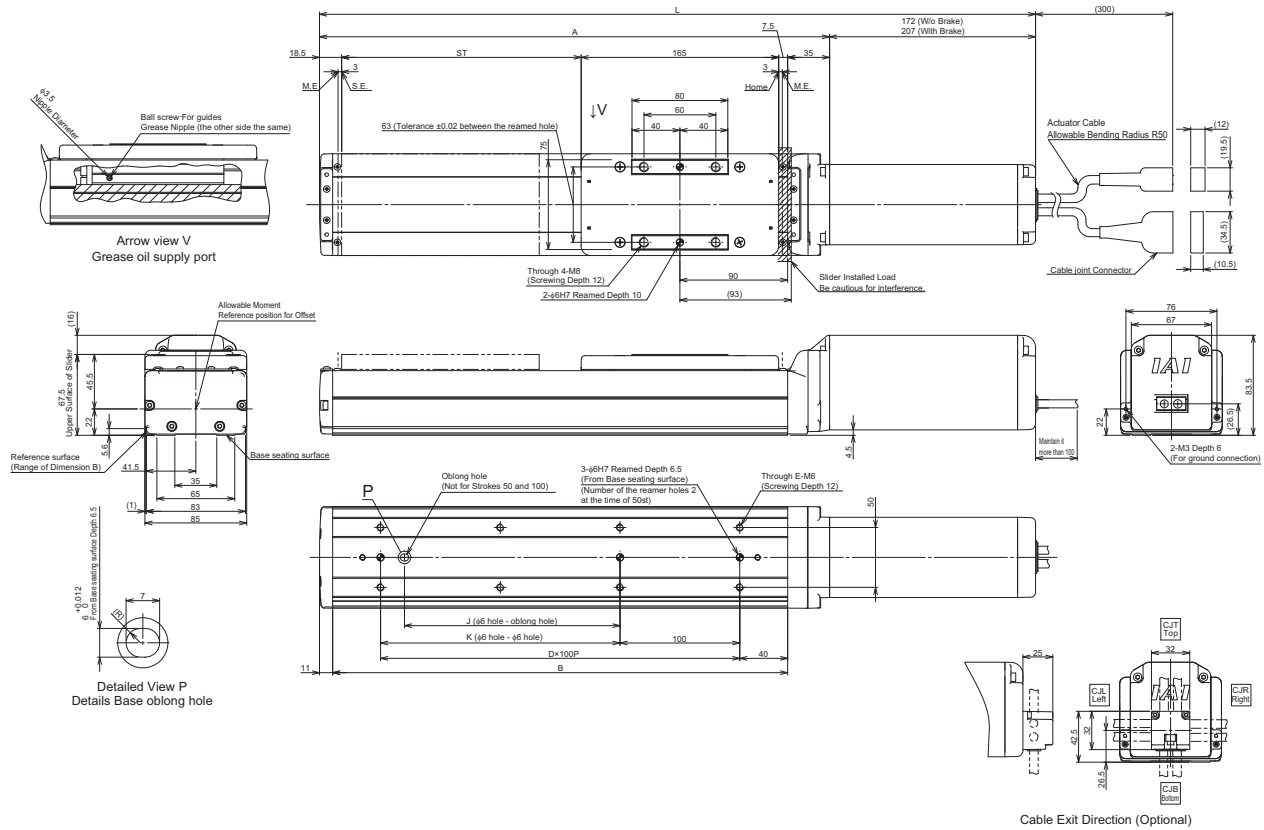


■ Dimensions and Mass by Stroke

Unit: mm

Stroke	L		A	B	D	E	G	H	J	K	Mass [kg]	
	W/o Brake	With Brake									W/o Brake	With Brake
50	403	438	253	188	0	4	1	4	0	0	3.5	4.0
100	453	488	303	238	1	6	1	4	85	0	3.8	4.3
150	503	538	353	288	1	6	2	6	85	100	4.0	4.5
200	553	588	403	338	2	8	2	6	185	200	4.3	4.8
250	603	638	453	388	2	8	3	8	185	200	4.5	5.0
300	653	688	503	438	3	10	3	8	285	300	4.7	5.2
350	703	738	553	488	3	10	4	10	285	300	5.0	5.5
400	753	788	603	538	4	12	4	10	385	400	5.2	5.7
450	803	838	653	588	4	12	5	12	385	400	5.4	5.9
500	853	888	703	638	5	14	5	12	485	500	5.7	6.2
550	903	938	753	688	5	14	6	14	485	500	5.9	6.4
600	953	988	803	738	6	16	6	14	585	600	6.1	5.6
650	1003	1038	853	788	6	16	7	16	585	600	6.4	6.9
700	1053	1088	903	838	7	18	7	16	685	700	6.6	7.1
750	1103	1138	953	888	7	18	8	18	685	700	6.8	7.3
800	1153	1188	1003	938	8	20	8	18	785	800	7.1	7.6

ST: Stroke, M.E.: Mechanical End, S.E.: Stroke End

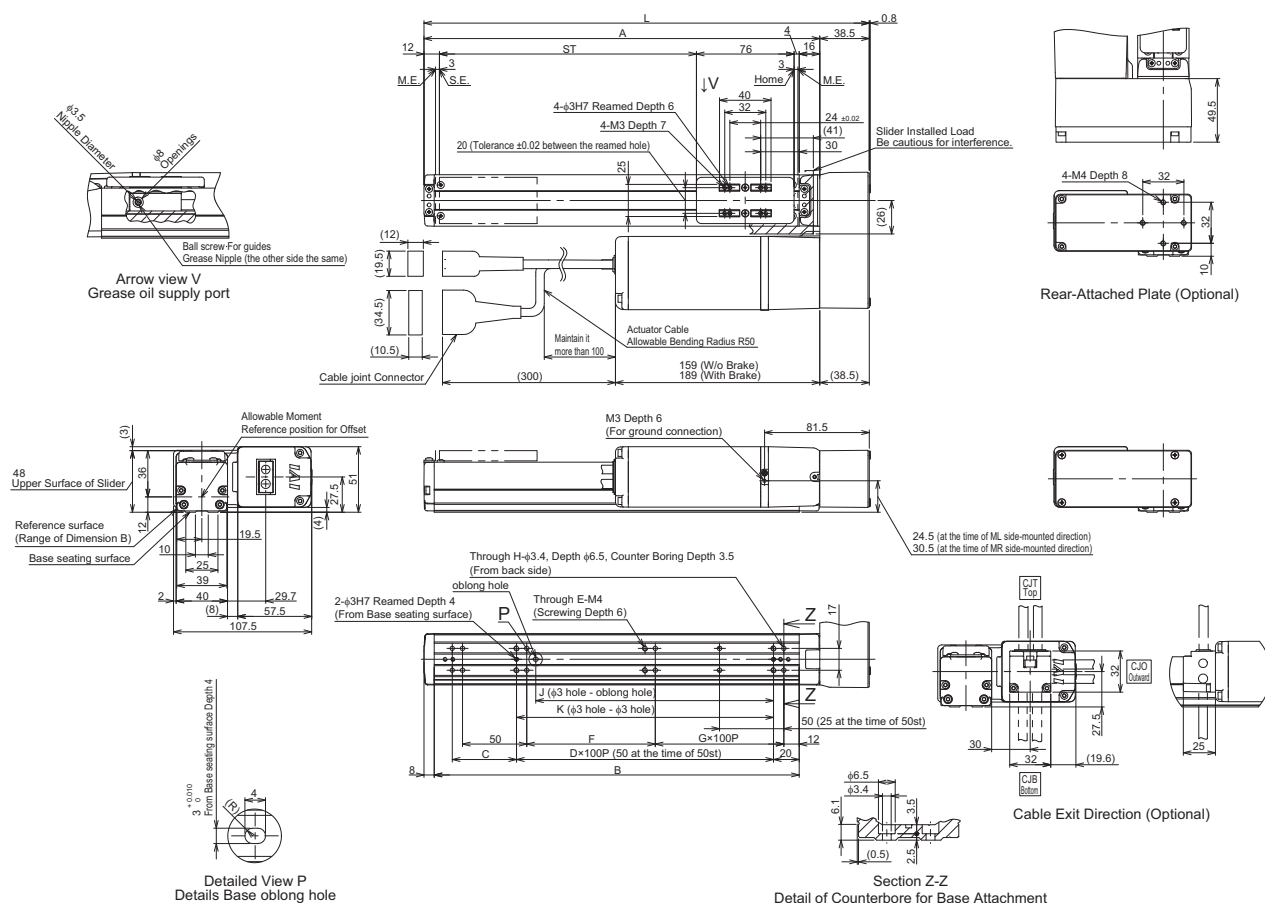


■ Dimensions and Mass by Stroke

Unit: mm

Stroke	L		A	B	D	E	J	K	Mass [kg]	
	W/o Brake	With Brake							W/o Brake	With Brake
50	448	483	276	230	1	4	0	0	4.7	5.2
100	498	533	326	280	2	6	0	100	5.0	5.5
150	548	583	376	330	2	6	80	100	5.3	5.8
200	598	633	426	380	3	8	180	200	5.6	6.1
250	648	683	476	430	3	8	180	200	5.9	6.4
300	698	733	526	480	4	10	280	300	6.2	6.7
350	748	783	576	530	4	10	280	300	6.5	7.0
400	798	833	626	580	5	12	380	400	6.8	7.3
450	848	883	676	630	5	12	380	400	7.1	7.6
500	898	933	726	680	6	14	480	500	7.4	7.9
550	948	983	776	730	6	14	480	500	7.6	8.1
600	998	1033	826	780	7	16	580	600	7.9	8.4
650	1048	1083	876	830	7	16	580	600	8.2	8.7
700	1098	1133	926	880	8	18	680	700	8.5	9.0
750	1148	1183	976	930	8	18	680	700	8.8	9.3
800	1198	1233	1026	980	9	20	780	800	9.1	9.6
850	1248	1283	1076	1030	9	20	780	800	9.4	9.9
900	1298	1333	1126	1080	10	22	880	900	9.7	10.2
950	1348	1383	1176	1130	10	22	880	900	10.0	10.5
1000	1398	1433	1226	1180	11	24	980	1000	10.3	10.8
1050	1448	1483	1276	1230	11	24	980	1000	10.6	11.1
1100	1498	1533	1326	1280	12	26	1080	1100	10.8	11.3

ST: Stroke, M.E.: Mechanical End, S.E.: Stroke End

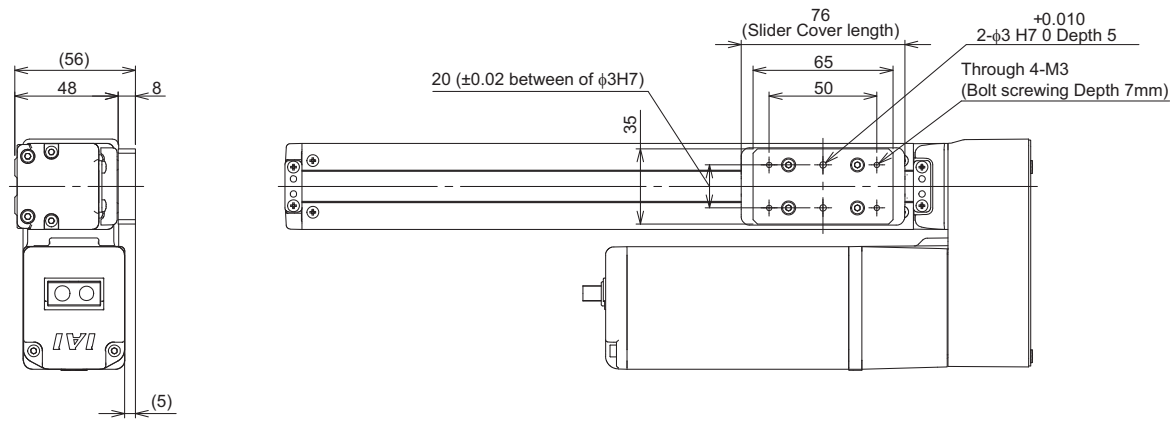


■ Dimensions and Mass by Stroke

Unit: mm

Stroke	L	A	B	C	D	E	F	G	H	J	K	Mass [kg]	
												W/o Brake	With Brake
50	196.5	158	134	50	-	6	50	0	8	35	50	1.5	1.7
100	246.5	208	184	50	1	6	100	0	8	85	100	1.6	1.8
150	296.5	258	234	100	1	6	50	1	10	85	100	1.7	1.9
200	346.5	308	284	50	2	8	100	1	10	185	200	1.7	1.9
250	396.5	358	334	100	2	8	50	2	12	185	200	1.8	2.0
300	446.5	408	384	50	3	10	100	2	12	285	300	1.9	2.1
350	496.5	458	434	100	3	10	50	3	14	285	300	2.0	2.2
400	546.5	508	484	50	4	12	100	3	14	385	400	2.1	2.3
450	596.5	558	534	100	4	12	50	4	16	385	400	2.2	2.4
500	646.5	608	584	50	5	14	100	4	16	485	500	2.3	2.5

Slider Spacer (Optional Model: SS)

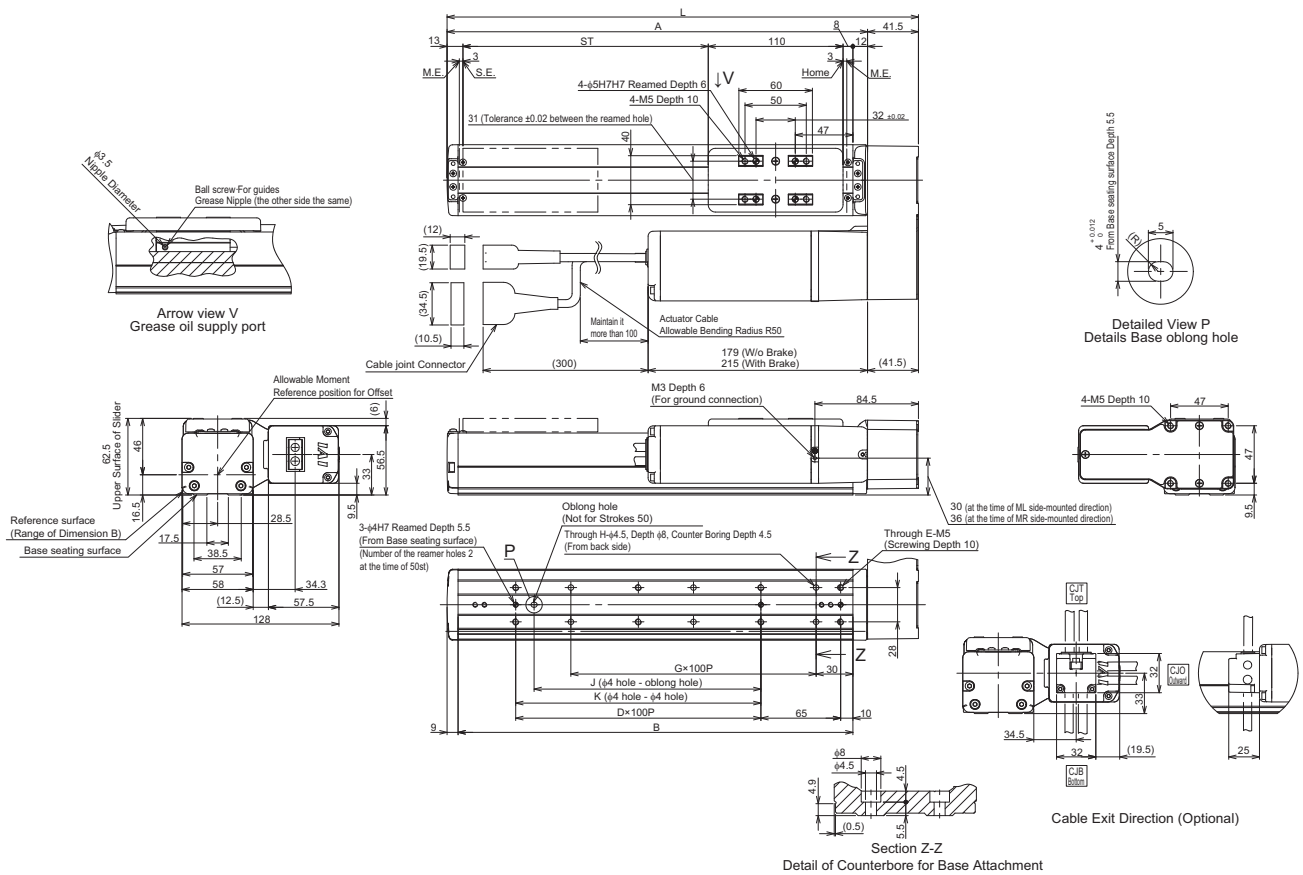


5.1 External Dimensions

5.1.6 RCS4-SA6R

ST: Stroke, M.E.: Mechanical End, S.E.: Stroke End

5. External Dimensions



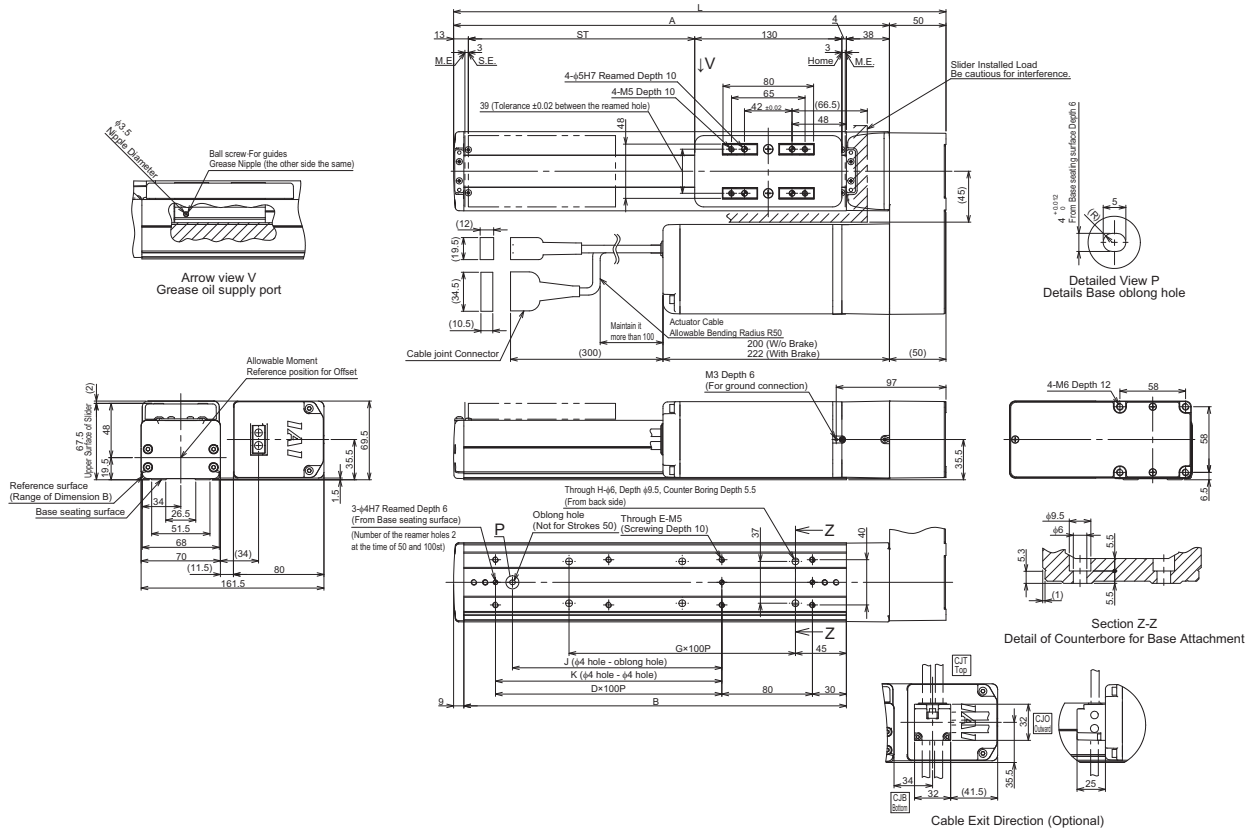
■ Dimensions and Mass by Stroke

Unit: mm

Stroke	L	A	B	D	E	G	H	J	K	Mass [kg]	
										W/o Brake	With Brake
50	234.5	193	172	0	4	1	4	0	0	2.4	2.7
100	284.5	243	222	1	6	1	4	85	100	2.5	2.8
150	334.5	293	272	1	6	2	6	85	100	2.7	3.0
200	384.5	343	322	2	8	2	6	185	200	2.9	3.2
250	434.5	393	372	2	8	3	8	185	200	3.1	3.4
300	484.5	443	422	3	10	3	8	285	300	3.2	3.5
350	534.5	493	472	3	10	4	10	285	300	3.4	3.7
400	584.5	543	522	4	12	4	10	385	400	3.6	3.9
450	634.5	593	572	4	12	5	12	385	400	3.8	4.1
500	684.5	643	622	5	14	5	12	485	500	4.0	4.3
550	734.5	693	672	5	14	6	14	485	500	4.1	4.4
600	784.5	743	722	6	16	6	14	585	600	4.3	4.6
650	834.5	793	772	6	16	7	16	585	600	4.5	4.8
700	884.5	843	822	7	18	7	16	685	700	4.7	5.0
750	934.5	893	872	7	18	8	18	685	700	4.9	5.2
800	984.5	943	922	8	20	8	18	785	800	5.0	5.3

5.1.7 RCS4-SA7R

ST: Stroke, M.E.: Mechanical End, S.E.: Stroke End

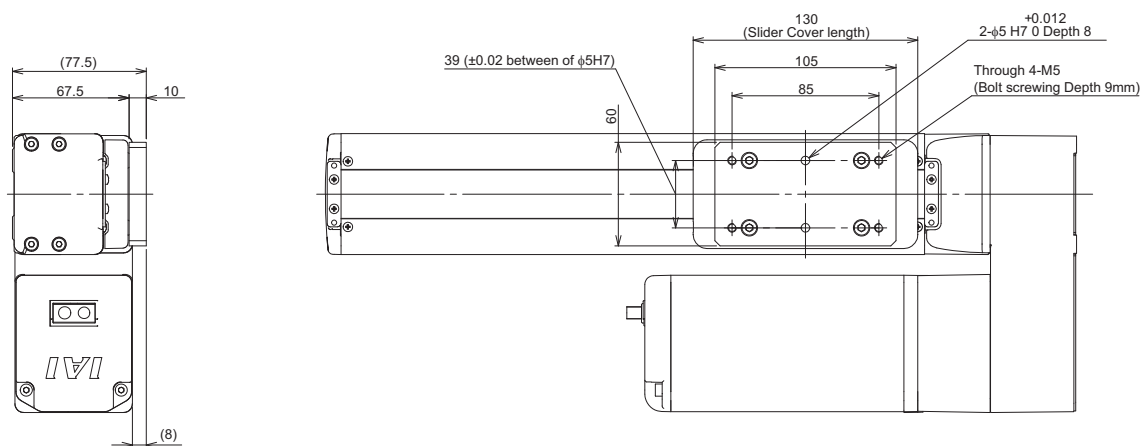


■ Dimensions and Mass by Stroke

Unit: mm

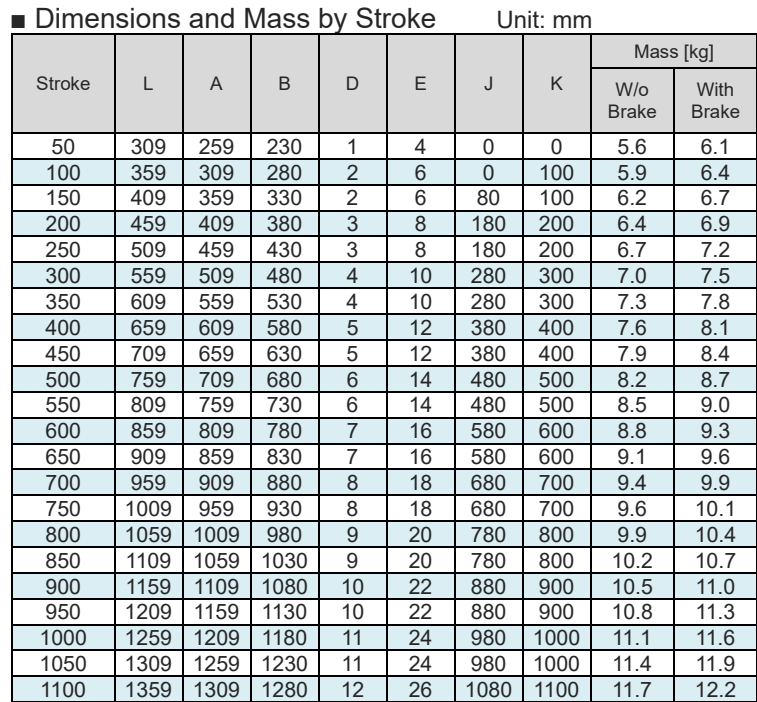
Stroke	L	A	B	D	E	G	H	J	K	Mass [kg]	
										W/o Brake	With Brake
50	285	235	188	0	4	1	4	0	0	4.3	4.8
100	335	285	238	1	6	1	4	85	0	4.6	5.1
150	385	335	288	1	6	2	6	85	100	4.8	5.3
200	435	385	338	2	8	2	6	185	200	5.0	5.5
250	485	435	388	2	8	3	8	185	200	5.3	5.8
300	535	485	438	3	10	3	8	285	300	5.5	6.0
350	585	535	488	3	10	4	10	285	300	5.8	6.3
400	635	585	538	4	12	4	10	385	400	6.0	6.5
450	685	635	588	4	12	5	12	385	400	6.2	6.7
500	735	685	638	5	14	5	12	485	500	6.5	7.0
550	785	735	688	5	14	6	14	485	500	6.7	7.2
600	835	785	738	6	16	6	14	585	600	6.9	7.4
650	885	835	788	6	16	7	16	585	600	7.2	7.7
700	935	885	838	7	18	7	16	685	700	7.4	7.9
750	985	935	888	7	18	8	18	685	700	7.6	8.1
800	1035	985	938	8	20	8	18	785	800	7.9	8.4

Slider Spacer (Optional Model: SS)



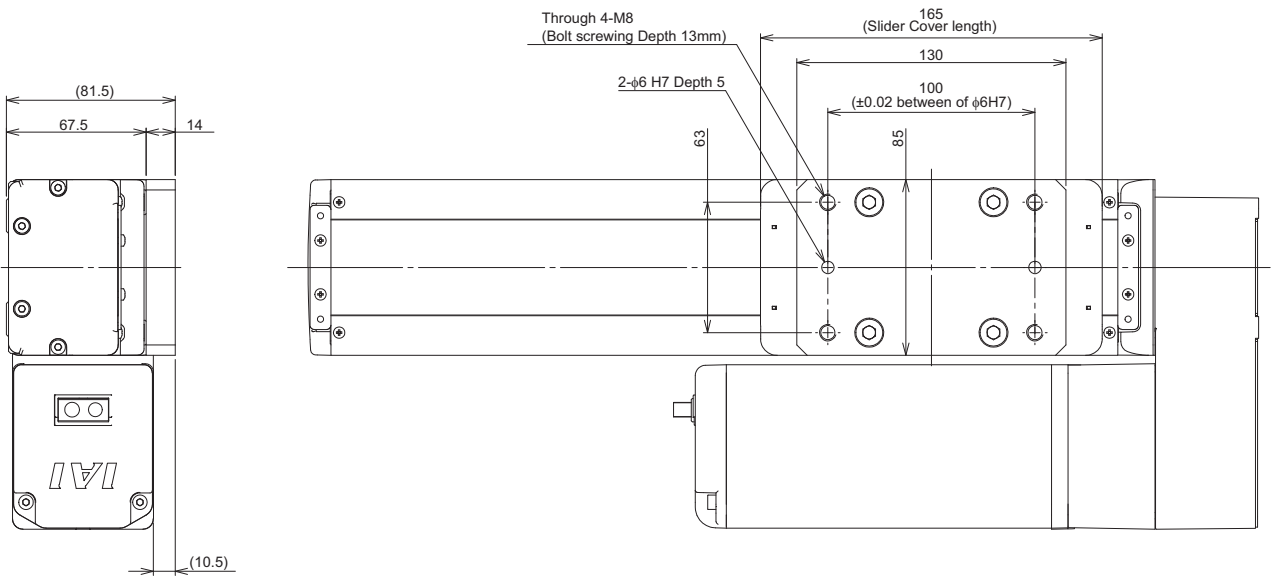
ST: Stroke, M.E.: Mechanical End, S.E.: Stroke End

5. External Dimensions



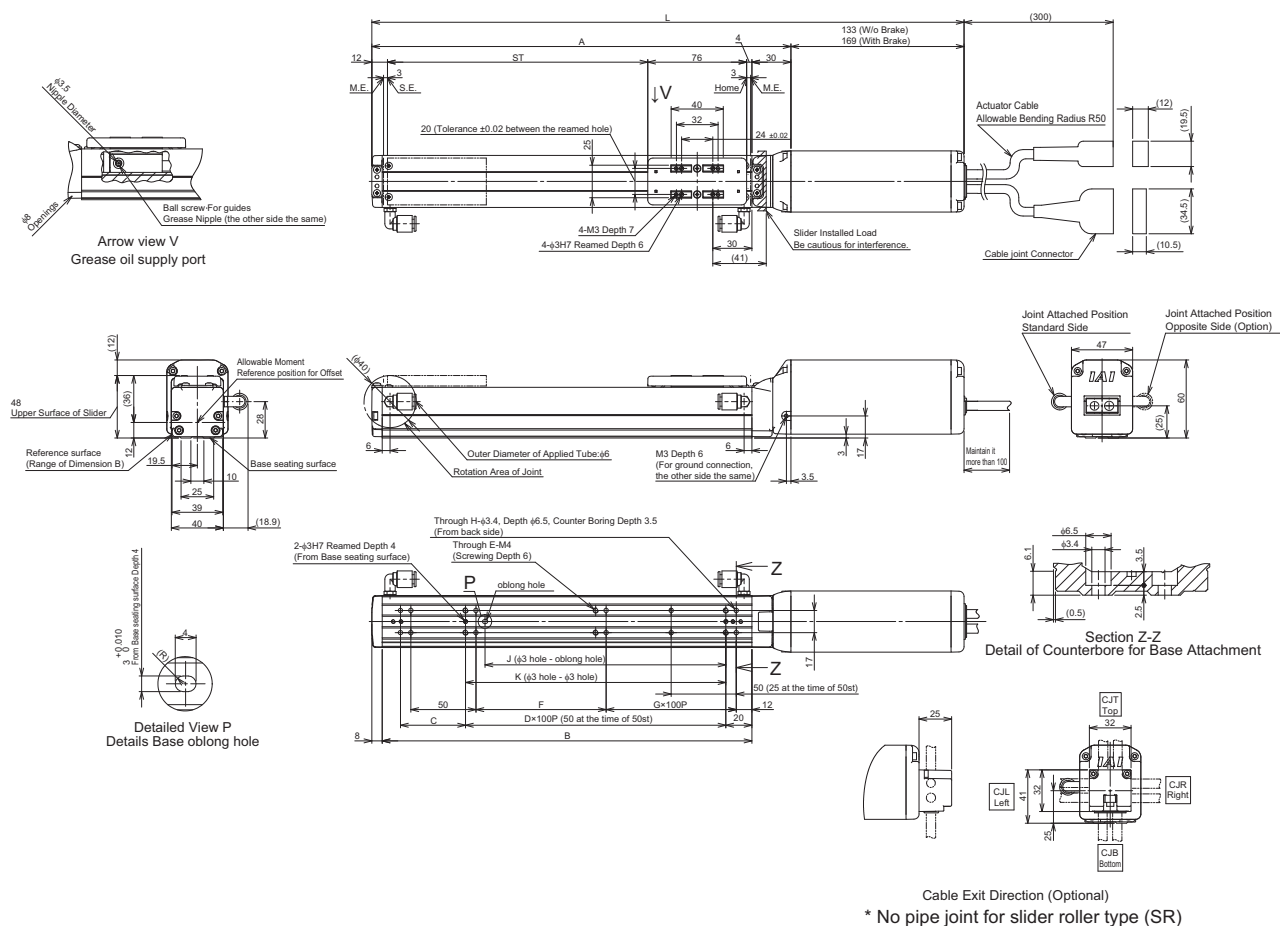
5.1 External Dimensions

Slider Spacer (Optional Model: SS)



5.1.9 RCS4CR-SA4C

ST: Stroke, M.E.: Mechanical End, S.E.: Stroke End

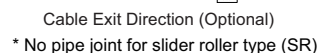


■ Dimensions and Mass by Stroke

Unit: mm

Stroke	L		A	B	C	D	E	F	G	H	J	K	Mass [kg]	
	W/o Brake	With Brake											W/o Brake	With Brake
50	305	341	172	134	50	-	6	50	0	8	35	50	1.2	1.4
100	355	391	222	184	50	1	6	100	0	8	85	100	1.3	1.5
150	405	441	272	234	100	1	6	50	1	10	85	100	1.4	1.6
200	455	491	322	284	50	2	8	100	1	10	185	200	1.5	1.7
250	505	541	372	334	100	2	8	50	2	12	185	200	1.5	1.7
300	555	591	422	384	50	3	10	100	2	12	285	300	1.6	1.8
350	605	641	472	434	100	3	10	50	3	14	285	300	1.7	1.9
400	655	691	522	484	50	4	12	100	3	14	385	400	1.8	2.0
450	705	741	572	534	100	4	12	50	4	16	385	400	1.9	2.1
500	755	791	622	584	50	5	14	100	4	16	485	500	2.0	2.2

ST: Stroke, M.E.: Mechanical End, S.E.: Stroke End

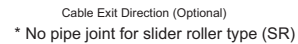


Unit: mm

5-13

ST: Stroke, M.E.: Mechanical End, S.E.: Stroke End

5. External Dimensions

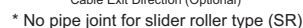


Unit: mm

ME3769-1H

5. External Dimensions

ST: Stroke, M.E.: Mechanical End, S.E.: Stroke End



Unit: mm

5-15

ROBO Cylinder

Chapter 6

Life

6.1	Concept of life for slider type	6-1
6.1.1	How to calculate operation life	6-1
6.1.2	Relation between operation life and moment	6-3

6.1 Concept of life for slider type

The mechanical life of the slider type is represented by that of the linear guide receiving the greatest moment load.

The operation life of the linear guide is to be determined by the total driving distance reachable without flaking (peeling on rail surface) in 90% of a group of products operated under the same conditions.

Operation life can be calculated with the method shown below.

6.1.1 How to calculate operation life

For the operation life of the linear guide, use the dynamic allowable moment stated in "1.2 specifications", and calculate with the formula below.

$$L = \left(\frac{C_M}{M} \right)^3 \times 5,000\text{km}$$

L : Operation life (km)
C_M : Dynamic allowable moment (N·m)
M : Moment acting (N·m)
5,000km : Standard rated life of ROBO Cylinder

In addition, calculate with the formula below if the life may be shortened due to vibration or mounting status.

$$L = \left(\frac{C_M}{M} \cdot \frac{f_{ws}}{f_w} \cdot \frac{1}{f_\alpha} \right)^3 \times 5,000\text{km}$$

L : Operation life (km)
C_M : Dynamic allowable moment (N·m)
M : Moment acting (N·m)
5,000km : Standard rated life of ROBO Cylinder
f_{ws} : Standard load coefficient
f_w : Load coefficient
f_α : Mounting coefficient

For "Standard load coefficient f_{ws} ", "Load coefficient f_w " and "Mounting coefficient f_a ", refer to the contents below for configuration and selection.

[Standard load coefficient f_{ws}]

For ROBO Cylinders described in this manual, calculate with $f_{ws} = 1.2$.

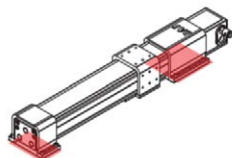
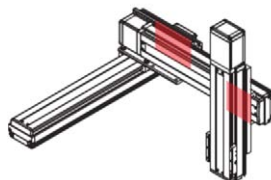
[Load coefficient f_w]

This coefficient allows consideration of the effects on life due to operating conditions.

Load coefficient f_w	Operating conditions	Guideline for acceleration/deceleration
1.0 to 1.5	Low vibration/impact, slow operation	1.0G or less

[Mounting coefficient f_a]

This coefficient allows consideration of the effects on life due to ROBO Cylinder mounting status.

Mounting coefficient f_a	1.0	1.2	1.5
Mounting status	Fixed on entire surface	Fixed at both ends	Fixed locally
			

- The actuator used in the explanatory figures above is not the RCS4 ROBO Cylinder. Replace with ROBO Cylinder and select the mounting coefficient.
- Even when seated over the total length of the product, select 1.2 or 1.5 for the mounting coefficient depending on the position of screw fixing and not "fixing over the entire surface".
- It should be judged as "fixing over the entire surface" only when all the screw holes (counterbored holes) equipped on the seat surface are used for fixation.

6.1.2 Relation between operation life and moment

The operation life depends on the moment acting on the slider.

With a light load, it may be longer than 5,000km, the standard rated life.

With no consideration of vibration and mounting conditions, the operation life is 40,000km according to the calculation with the formula on the previous page, assuming that $0.5 C_M$ (half of dynamic allowable moment) of moment is applied.

We see that it can be 8 times longer than the standard rated life, which is 5,000km.

ROBO Cylinder

Chapter 7

Warranty

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7.1 Warranty period

Whichever of the following periods is shorter:

- 18 months after shipment from IAI
- 12 months after delivery to the specified location
- 2,500 hours of operation

7.2 Scope of the warranty

Our products are covered by warranty when all of the following conditions are met.

Faulty products covered by warranty will be replaced or repaired free of charge:

- (1) The breakdown or malfunction in question pertains to our product as delivered by IAI or our authorized dealer.
- (2) The breakdown or malfunction in question occurred during the warranty period.
- (3) The breakdown or malfunction in question occurred while the product was in use for an appropriate purpose under the operating conditions and operating environment specified in the instruction manual and catalog.
- (4) The breakdown or malfunction in question was caused by a specification defect, malfunction, or poor product quality.

Note that breakdowns due to any of the following reasons are excluded from the scope of warranty:

- (a) Anything other than our product
- (b) Modification or repair performed by a party other than IAI (unless approved by IAI)
- (c) Anything that could not be easily predicted with the level of science and technology available at the time of shipment from IAI
- (d) Natural disaster, unnatural disaster, incident or accident for which we are not liable
- (e) Natural fading of paint or other symptoms of aging
- (f) Wear, depletion or other expected result of use
- (g) Operation noise, vibration or other subjective sensations not affecting function or maintenance

Note that the warranty only covers our product as delivered and that any secondary loss arising from a breakdown of our product is excluded from the scope of warranty.

7.3 Honoring the warranty

As a rule, the product must be consigned to IAI for repair under warranty.

7.4 Limited liability

- (1) We assume no liability for any special damage, consequential loss or passive loss such as a loss of expected profit arising from or in connection with our product.
- (2) We assume no liability for any program or control method created by the customer to operate our product or for the results of any such program or control method.

7.5 Conformance with applicable standards/regulations, etc., and application conditions

- (1) If our product is combined with another product or any system, equipment, etc., used by the customer, the customer must first check the applicable standards, regulations and/or rules. The customer is also responsible for confirming that such combination with our product conforms to the applicable standards, etc.
In such a case we assume no liability for the conformance of our product with the applicable standards, etc.
- (2) Our product is for general industrial use. It is not intended or designed for the applications specified below, which require a high level of safety. Accordingly, as a rule our product cannot be used in these applications.

Contact IAI if you must use our product for any of these applications:

- (a) Medical equipment used to maintain, control or otherwise affect human life or physical health
- (b) Mechanisms and machinery designed for the purpose of moving or transporting people (vehicles, railway facilities, aviation facilities etc.)
- (c) Machinery components essential for safety (safety devices etc.)
- (d) Equipment used to handle cultural assets, art or other irreplaceable items
- (3) Contact IAI in advance if our product is to be used in any condition or environment that differs from that specified in the catalog or instruction manual.

7.6 Other Items excluded from warranty

The price of the product delivered to you does not include expenses associated with programming, the dispatch of engineers, etc. Accordingly, a separate fee will be charged in the following cases even during the warranty period:

- (1) Guidance for mounting/adjustment and witnessing of test operation
- (2) Maintenance and inspection
- (3) Technical guidance and education on operating/wiring methods, etc.
- (4) Technical guidance and education on programming and other items related to programs



7. Warranty

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Revision history

Revision date	Revised content
2017.10	First Edition
2018.02	1B edition Pg. 1-14, 1-26 Lead 48 0.7G changed from 8 to 6
2018.09	1C edition Pg. 4-8 Grease changed for leads other than Lead 48 in SA8
2019.01	1D edition Pg. 4-25 Note added for confirmation of alignment marks on replacement motor
2020.07	1E edition Pg. 1-5, 1-8, 1-11, 1-14, 1-17, 1-20, 1-23, 1-26 Contents added for payload and acceleration/deceleration at offboard tuning Pg. 4-22, 4-29 Motor unit affixing screw tightening torque changed SA4R, SA6R 2.1 → 3.23 SA7R, SA8R 4.1 → 6.31
2020.10	1F edition Pg. 2-5 Description added stating installation with motor on the bottom not recommended
2020.12	1G edition Pg. 2-5 Figure for vertical installation changed to motor reversed type Pg. 4-15, 4-18 Note added for caution in stainless steel sheet replacement work
2022.02	1H edition • Complied global standard changed to "RoHS3 Directive" • Applicable controller T4 added in How to Read Model Number • RCON and RSEL added in Connecting with the Controller • SA4 added and correction made in caution in Greasing Method • Correction made in (3) Image in Timing Belt Replacement • Correction made in each of (5) Image in Motor Unit Replacement for Straight Type / Reversing Type • Correction made and layout revised



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